

# neUROn Design Manifest

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# Chapter 1

## Introduction and History

The original neUroN was coded in C in 1992 when I was at Baylor College of Medicine. It was named “neural computational for UROlogical numericals,” a somewhat mercurial reference to its application domain, the medical field of Urology. In actuality, neUroN is a general purpose neural computational modeling environment. neUroN has been, and will continue to be used to model many problems beyond the Urological domain.

Between 1992 and 2000, neUroN evolved in fundamental ways. In 1996, Young Hong migrated neUroN to C++, and the resulting environment was named “neUroN++”. Many have coded these fundamental changes, including Vinod Kutty, Yuan Qin, Luke Cho, Joe Jovero, and Kai Liu. In general, beyond migrating neUroN to object-oriented programming, these fundamental changes have taken 2 forms: programming novel neural computational training algorithms and programming statistical analysis of training behavior. Throughout this project, Richard Golden Ph.D. at the University of Texas at Dallas has suggested algorithms which the team has implemented.

The impetus for programming novel training algorithms should be obvious. Neural computational algorithms are iterative, and problems modeled by the neural computational approach can easily become intractable, even with modern computational hardware. A single training session may require several hours, and multiple training sessions may require days or even weeks. There are 3 basic ways in which training sessions may become shorter:

1. Faster hardware may be used. This is clearly the way in which the programmer has the least control.
2. More efficient code may be written. The importance of this cannot be stressed enough, and it will serve as a basic design principle for the neUroN project. In the original neUroN code, for example, arrays were expressed entirely as pointer operations. It was not until the compiler benchmarks demonstrated similar outcomes for array constructs that arrays were used in the code.

3. Faster training algorithms may be encoded. A substantial part of the neUROn project is devoted to investigating novel training algorithms, and this has paid handsomely. Investigated algorithms have been demonstrated to speed training by 2 orders of magnitude or more.

The impetus for programming statistical analysis of network behavior was less obvious, but perhaps more significant. When we began reporting results to the Urological community, we were surprised to find a universal response was, “so the network accurately predicts  $X$ , but what were the important variables?” In retrospect, this response is unsurprising. The majority of our audience, our potential users, were physicians or scientists involved in the medical field. It is not enough to accurately model an outcome for these users, it is critical to find the significant input features so that these features may be altered by the user in the clinical environment. For example, if it is determined that medication  $X$  is statistically significant in the model  $Y$  that predicts cancer recurrence, the user will put that information to use in the clinical domain.

Beginning in 2000, neUROn was entirely redesigned using this document as a specification, and taking full advantage of the object-oriented nature of C++. Whereas the migration from 1996 to 2000 was an incorporation of C code into the C++ paradigm, the hybrid code which resulted was brittle, with previous features breaking as new features were added. In 2000, the code was redesigned from the “ground up”. This document is intended to be a description of neUROn programming, past, present, and future. As such, the document is a plan that is expected to evolve as neUROn is redesigned and recoded in the future.

The current neuron 3.0 work resumes that design discipline around a modern C++ engine and a loopback web interface. It preserves the original numerical, model, and statistical object hierarchy while adding reproducible scripted sessions; a GUI and REST API with command-line parity; asynchronous training and cancellation; explicit convergence semantics; automatic optimizer probing; plateau stopping; validation-based OBD architecture selection; shared, stratified, and group-disjoint cross-validation plans; untouched locked-test evaluation; and machine-readable prediction, metric, and design artifacts.

Statistical reporting has likewise been made explicit about its sampling unit. Independent-row locked tests use paired DeLong covariance; clustered locked tests use Obuchowski’s nonparametric clustered covariance. Both share the same Mann–Whitney point AUC, placements, tie handling, interval algebra, and paired contrast rules. Group-aware partitioning and clustered inference remain separate mechanisms: the first prevents leakage, while the second estimates uncertainty when rows within a sampling unit are correlated.

Part I describes these current interfaces and workflows. Part II retains the mathematical derivations, documentation rules, and object specifications which make the manifest a programming plan rather than a user manual. Chapter 12 now connects the established objects with the newer partition, training-control, selection, covariance, cross-validation, and reporting services.

## Part I

# Current neuron 3.0

## Chapter 2

# neuron 3.0 Architecture

The 2026 reanimation preserves the object design described in this Manifest while giving it three contemporary ways to be used: the original menu interface, a local graphical interface, and a loopback HTTP interface. The same C++ model and statistical classes serve all three. Python tools prepare data and export trained models, but do not duplicate the engine.

The central architectural rule is that every mechanism has one authoritative implementation in the class that conceptually owns it. The graphical interface may call that implementation, and a cross-validation run may call it repeatedly, but neither copies its mathematics. Dependencies point down through the layers:

|                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------|
| Human and programmatic interfaces: GUI, REST API, frozen CLI menus                                                  |
| Experiment orchestration: cross-validation, OBD sizing, reports, automatic optimizer selection                      |
| Model and data interfaces: Model, DataSet, TwoSet, split plans, model factory and clone helpers                     |
| Training and concrete methods: Iterative, Network, Logistic, SimpleProp, BareProp, BackProp, LDFA, QDFA, RegressNet |
| Numerical foundation: Matrix, vector operations, statistics and utility functions                                   |

A lower layer never learns a higher concern. Matrix can gather rows but does not know what a fold is. DataSet can materialize a fold but does not choose a model. A model fits itself but does not run a cross-validation loop. OBD chooses a hidden-layer size but does not own final evaluation. Cross-validation owns repetition, not training. This separation keeps the matrix notation, the implementation, and the statistical report traceable to one another.

## 2.1 Legibility and speed as joint requirements

Two constraints govern every line of engine code, and neither may be traded for the other or for a smaller source file.

The first is legibility. The computational core was built on Matrix, the vector operations and Population so that a reader can move between a published equation and its implementation in either direction. A gradient in this Manifest and the statement that computes it should be recognizably the same object. An abstraction that hides a published formula has failed even when it removes duplication: the biased and unbiased forms of a gradient are two equations, and keeping both visible is worth the repetition that a flag would remove.

The second is speed. Neural computation is compute-bound, and the engine is compiled C++ for that reason. The numerical classes therefore offer each operation twice: a form that fills a destination the caller supplies, and a form that returns a new object. That pair is deliberate and is not duplication. The returning form is for formulae written once; the destination form is what belongs inside a training loop, where a fresh allocation per exemplar per iteration is the difference between a run and a wait. The compound operators and the range overloads exist for the same reason, the latter also because a range expresses a mathematical domain – every hidden unit except the bias slot – more directly than computing the whole vector and repairing it afterwards.

It follows that removing duplicated source is not, by itself, an improvement. Indirection through a function object, a virtual call, or a copy introduced into a per-exemplar path to unify two loops trades a measurable cost for a cosmetic gain. In cold paths – reporting, file writing, one-time setup – those tools are unobjectionable. A suspected performance problem is measured before an algorithm changes in response to it.

The operating form of this section, with the specific idioms and the episode that prompted writing it down, is standing rule 7 in `CLAUDE.md`; rule 4 states the legibility half and rule 6 the ownership half.

## 2.2 Interfaces

### 2.2.1 Command line

The interactive menus remain the authoritative feature inventory and are intentionally frozen. Reproducible command-line runs feed a complete answer file to the engine and always provide a seed:

```
./build/neuron --seed 42 < session.in > session.out
```

The transcript is the permanent record of the run. The program is not intended to be driven by keystrokes in an automated experiment.

### 2.2.2 Graphical interface

```
./build/neuron --gui
```

This starts an embedded `cpp-httpd` server on `127.0.0.1` at an operating-system-assigned port and opens a single local web page. The page provides the full load, model, train, analysis, report, and save workflow. It includes realtime training progress and ROC plots. The server is loopback-only: it is a local application, not a network service.

GUI and CLI parity is a contract. Every CLI capability must have both a page control and an HTTP parameter. The maintained option-by-option map is `docs/gui_cli_parity.md`. Features added beyond the frozen CLI, including automatic optimizer selection, OBD sizing, and cross-validation, are available through both the GUI and the HTTP interface.

### 2.2.3 Python tools

`tools/mkdataset.py` turns raw CSV data into the numeric, outcome-last format used by the engine and writes the key and input-group files needed to interpret the encoded columns. `tools/neuron2web.py` combines a saved network, scaling factors, and a human-readable label specification into a self-contained HTML calculator. The deployed calculator performs inference entirely in the browser.

## 2.3 Data ownership and splitting

`DataSet` owns loaded and normalized observations. `TwoSet` owns classification and ROC reporting for a modeled set. The dedicated `split` layer owns index plans and provides:

- outcome-stratified train and test splits;
- optional covariate stratification with quantile cells and Hamilton apportionment;
- group-aware splits that keep an entire cluster on one side;
- three-way train, validation, and untouched test splits;
- outcome-stratified and outcome-by-covariate folds for cross-validation; and
- outcome-balanced group folds that never divide a cluster.

These mechanisms solve different problems. Covariate stratification makes two samples more alike and is useful for subgroup coverage. Grouping makes evaluation harder by preventing site or cluster leakage. In a grouped comparison, one group key governs the outer folds, the locked holdout, and nested OBD's inner validation split. No model-selection stage can therefore see rows from a cluster that it later treats as unseen. A validation set permits model selection without repeatedly looking at the final test set.

## 2.4 Models and training

**Model** defines the common modeling interface. **Network** adds connected layers and weights. **Iterative** owns the training loop, stopping conditions, progress callbacks, and convergence state. Concrete model classes own their distinct mathematics:

**SimpleProp** One hidden layer with bias nodes.

**BareProp** One hidden layer without bias nodes. **SimpleProp** and **BareProp** share their state, training, and forward equations through **OneHiddenNet** and remain distinct concrete types; only the biased model establishes a bias slot.

**BackProp** Multiple hidden layers and multi-output networks.

**Logistic** Binary logistic regression, including Wald tests and the design-matrix condition number.

**LDFA and QDFA** Linear and quadratic discriminant analysis.

**RegressNet** Forward and reverse stepwise variable selection using the Wilks likelihood-ratio test.

Training supports canonical gradient descent, conjugate gradient descent, and Shanno’s method. The automatic selector probes all three from cloned starting weights for a short wall-clock budget, adopts the lowest-error probe, and continues from the progress it made.

Stopping conditions are evaluated every iteration, independently of how often an iteration row is printed. Reaching the iteration ceiling means that the requested operation ran, but the fit did not converge. The weights remain available for continuation; an unfinished fit is not eligible for model selection or cross-validation scoring.

## 2.5 Model selection and evaluation

### 2.5.1 OBD hidden-layer sizing

The **obd** layer grows a **SimpleProp** hidden layer while held-out error improves, then prunes by saliency. Each candidate size is stopped at its held-out minimum or by a valid training stopping condition. With a three-way split, the search monitors validation and leaves test untouched. A successful standalone OBD run replaces the current model with the selected, trained network. The published saliency idea, neuron’s validation-driven adaptation, and the complete object contract are explained in section [12.8](#).

### 2.5.2 Cross-validation

`crossval` applies logistic regression, LDFA, QDFA, a fixed neural architecture, or nested OBD to one shared fold plan. `cvadapters` connects those procedures to their existing model implementations. `cvreport` renders a three-tier result:

1. a headline comparison suitable for decisions;
2. a detailed human-readable audit; and
3. machine-readable prediction, metric, and run files.

Nested OBD repeats architecture selection inside each training fold. It therefore estimates the performance of the selection procedure, not the performance of an architecture chosen using all observations. Cross-validation is a standalone analysis and does not replace the current model. Section 12.9 explains why selection must be nested inside each assessment fold and cites the supporting model-selection evidence.

The default plan is outcome-stratified. A request may instead balance folds on outcome-by-covariate cells or assign whole groups with a seeded, outcome-balanced bin-packing plan. Covariate stratification and grouping are alternatives: no untested joint objective is silently invented. The completed design records achieved rows, events, and groups per fold, its largest group and imbalance, and a zero-leakage check in the Tier 3 run artifact.

### 2.5.3 Locked-test inference

A locked test set may be removed before cross-validation. Procedures are compared on the development rows, refitted by their own rules, and scored once on the locked rows. The evaluation design is structured by `evaldesign`: the declared sampling unit and achieved partition select the only permitted covariance estimator.

For genuinely independent rows, `delong` supplies ordinary paired AUC intervals and the prespecified contrast. For rows clustered by the request's group key, `clustered_auc` supplies Obuchowski's non-parametric clustered ROC covariance. Both use the same `aucov` placements, mid-ranks, point AUC, and contrast algebra. They can therefore differ only in the covariance appropriate to the sampling unit. Grouping prevents leakage but does not make rows independent; an independent-row declaration over a group-disjoint design is refused rather than passed to ordinary DeLong. Section 12.6 derives the distinction, defines the paired contrast, and cites both estimators' primary papers.

The point AUC remains the row-level Mann–Whitney probability under both methods. Clustering changes uncertainty, not the point estimand. The effect has no fixed direction: ordinary row-based standard errors can be too small or too large, depending on the within-cluster covariance.

## 2.6 Statistical reporting

Binary models report classification accuracy, sensitivity, specificity, and two complementary ROC estimates. The trapezoidal area is the exact non-parametric AUC over every distinct operating point. The primary binormal  $A_z$  is fitted to the z-ROC curve and is reported with a stratified bootstrap confidence interval. Reports state the number of fitted operating points, bootstrap resamples, and failures. Logistic models additionally report coefficient-level Wald tests and the condition number.

## 2.7 Run artifacts and audit trail

The directory of the first loaded dataset is the run directory. Networks, scaling factors, normalized sets, guesses, reports, `neuron.log`, and `neuron.actions.log` are written there. The latter records every GUI or API action with a timestamp and its parameter values. Long operations publish structured progress through one shared status channel and honor graceful cancellation at an iteration boundary.

## 2.8 Source map for the modern layers

| Files                                                                               | Responsibility                                                                                                               |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <code>gui.*</code> ,<br><code>gui_page.html</code>                                  | Local server, REST handlers, and browser interface.                                                                          |
| <code>asyncjob.*</code> ,<br><code>procgaurd.*</code>                               | The asynchronous-job lifecycle every long operation shares, and the command-line program's top-level exception boundary.     |
| <code>split.*</code>                                                                | Stratified, group-aware, three-way, arbitrary-stratum k-fold, and whole-group k-fold index plans.                            |
| <code>crossval.*</code> ,<br><code>cvadapters.*</code> ,<br><code>cvreport.*</code> | Procedure comparison, connections to model implementations, and human plus machine-readable reports.                         |
| <code>obd.*</code>                                                                  | Grow-and-prune hidden-layer selection.                                                                                       |
| <code>autoalgo.*</code> ,<br><code>plateau.h</code>                                 | Optimizer probing and plateau detection shared by training workflows.                                                        |
| <code>evaldesign.*</code>                                                           | Typed partition, sampling-unit, and inference policy metadata.                                                               |
| <code>auccov.*</code> ,<br><code>delong.*</code> ,<br><code>clustered_auc.*</code>  | Shared AUC placements and contrast algebra, ordinary independent-row DeLong covariance, and Obuchowski clustered covariance. |
| <code>modelfactory.*</code> ,<br><code>netclone.*</code>                            | Construction and faithful cloning through the common model interfaces.                                                       |

| <b>Files</b>                          | <b>Responsibility</b>                                                                    |
|---------------------------------------|------------------------------------------------------------------------------------------|
| dataset.*,<br>twoset.*                | Observation ownership, normalization,<br>classification, ROC, and statistical reporting. |
| model.*,<br>network.*,<br>iterative.* | Core model contract, network structure, and<br>authoritative training loop.              |
| onehidden.*                           | State, training, and forward equations shared by<br>the two one-hidden-layer models.     |
| matrix.*,<br>vector_ops.*,<br>stats.* | Bounds-checked numerical and statistical<br>foundation.                                  |

## Chapter 3

# Command-line interface

The command-line menus remain fully supported and are the authoritative feature inventory, but they are frozen. New interactive work normally uses the GUI described in Chapter 2; new programmatic work uses the REST API in Chapter 4.

### 3.1 Reproducible sessions

Automated runs write every menu answer to a text file, one answer per line, and feed that file through standard input. Always supply a seed and capture standard output:

```
./build/neuron --seed 42 < session.in > session.out
```

Read the completed transcript and verify that it ends after the quit confirmation, not at an error or unanswered prompt. Do not automate the menus by typing into a live terminal.

### 3.2 Menu organization

**Specify dataset** Set input and output counts; load raw data or a normalized training and test pair; create a seeded train and test split; set outcome type, normalization bounds, and ROC reporting; save normalized sets and scaling factors.

**Specify model** Create a feed-forward network or binary logistic model; set bias nodes, hidden layers, and error function; load a saved network; set logging.

**Use model** Randomize weights; configure learning rate, weight decay, batch or epoch behavior, printing, and stopping conditions; train with canonical gradient descent, conjugate gradient descent, or Shanno's method; run stepwise selection; save the network and predictions.

**Discriminant function analysis** Run linear or quadratic discriminant analysis and save its predictions.

The exact menu-to-GUI-to-API mapping is maintained in `docs/gui_cli_parity.md`. That file is the current option-level reference and is updated whenever an interface changes.

### 3.3 Dataset and run files

Numeric data place the outcome in the final column. Binary outcomes are encoded as 0 and 1. The first loaded dataset establishes the run directory, where the engine writes networks, scaling factors, reports, predictions, `neuron.log`, and `neuron.actions.log`.

For raw CSV data, use `tools/mkdataset.py` as described in Chapter 5. When an intercept-bearing model will be trained, reference-category coding avoids collinearity between a full one-hot group and the intercept.

### 3.4 Convergence and reporting

The iteration ceiling is a safety limit, not evidence of convergence. If training reaches it, the current weights can be continued, but the fit is unfinished. Model-selection and cross-validation procedures refuse unfinished candidate fits.

For binary outcomes, interpret held-out performance through classification measures and ROC areas, not accuracy alone. The primary fitted binormal  $A_z$  is reported with its bootstrap 95% confidence interval, operating-point count, resample count, and any failures. The exact empirical AUC and Hanley–McNeil interval are an independent cross-check. Logistic reports additionally provide Wald tests and a condition number.

### 3.5 Current command-line options

Usage: `neuron [--seed N] [--gui [--no-browser]] [--version]`

`--gui` starts the local browser interface. `--no-browser` leaves it running without opening a browser. `--version` prints the package version and exits.

## Chapter 4

# Loopback REST API

The graphical interface is a client of the same HTTP endpoints that are available to scripts. Start the service without opening a browser:

```
./build/neuron --gui --no-browser
```

The engine prints a URL such as `http://127.0.0.1:54321/`. The port is assigned at runtime, so clients should read it from standard output. Requests use ordinary form fields or multipart file uploads. Successful JSON responses contain `"ok":true`; failures contain `"ok":false` and a human-readable `message`.

The server deliberately binds only to loopback and has no authentication. It is intended for a user or local agent on the same computer. It must not be exposed as a public web service.

### 4.1 A complete minimal session

```
URL=http://127.0.0.1:54321
```

```
curl -s -X POST "$URL/api/load" \  
-d "mode=raw&path=data.txt&fraction=0.25&seed=42"
```

```
curl -s -X POST "$URL/api/model" \  
-d "type=logistic"
```

```
curl -s -X POST "$URL/api/train" \  
-d "algorithm=1&maxiter=20000&seed=42"
```

```
curl -s "$URL/api/stats"  
curl -s "$URL/api/save/network" -o network.txt  
curl -s "$URL/api/save/scales" -o scales.txt
```

File fields accept either a path for a local script or an uploaded multipart part for a browser. Uploaded files are copied into the run directory using the safe final component of their filename.

## 4.2 Endpoint summary

| Verb | Path              | Purpose                                                                       |
|------|-------------------|-------------------------------------------------------------------------------|
| GET  | /api/version      | Engine package and version string.                                            |
| POST | /api/load         | Load, normalize, and optionally split a dataset.                              |
| POST | /api/model        | Create a model or load a saved network.                                       |
| POST | /api/randomize    | Reset the current model's weights from a seed.                                |
| POST | /api/train        | Train synchronously or start an asynchronous run.                             |
| GET  | /api/train/status | Read progress and the completed result of the active or most recent long job. |
| POST | /api/train/stop   | Request graceful cancellation at an iteration boundary.                       |
| GET  | /api/stats        | Return the current trained model's ROC and classification statistics.         |
| POST | /api/regress      | Run forward or reverse stepwise regression.                                   |
| POST | /api/dfa          | Run linear or quadratic discriminant analysis.                                |
| POST | /api/obd          | Start asynchronous hidden-layer sizing.                                       |
| POST | /api/cv           | Start asynchronous procedure comparison and optional locked-test inference.   |
| GET  | /api/save/:what   | Download a session artifact.                                                  |

## 4.3 Dataset loading: POST /api/load

**mode** `raw` to normalize and split one file; `train` to load an already normalized training set and optional matched test set.

**path** or **file** Dataset path or multipart upload.

**testpath** or **testfile** Optional matched test set in `train` mode.

**inputs, outputs** Optional column-count overrides. Outputs default to 1; multiple outputs select BackProp and accuracy-only reporting.

**discrete** 1 for classification or 0 for a continuous outcome.

**fraction, test\_n** Test share or exact test count. Use one form. A value of 0 keeps all rows in training.

**val\_fraction, val\_n** Optional validation share or count for a three-way split. The count form accompanies **test\_n**.

**seed** Reproducible split seed.

**strata, strata\_bins** Comma-separated one-based input columns for covariate stratification and the number of quantile bins for continuous columns.

**group** Comma-separated one-based input columns whose joint values define indivisible clusters. Grouping takes precedence over covariate stratification.

**threshold** Classification threshold between 0 and 1.

**in\_lower, in\_upper** Input normalization bounds.

**out\_lower, out\_upper** Output normalization bounds for a continuous outcome.

**roc\_report** **both** or **either**, defining the class-availability rule for ROC reporting.

**roc\_min** Minimum observations needed for statistical ROC reporting.

**history** 1 or 0 for dataset-operation logging.

Outcome stratification is always applied to ordinary random splits. Covariate stratification and grouping are deliberate alternatives. A three-way split cannot currently be combined with either of them. The response includes achieved row counts and, when appropriate, a representativeness or zero-leakage diagnostic.

## 4.4 Model construction: POST /api/model

**type** **logistic** or **simpleprop**.

**hidden** A comma-separated layer list such as 5 or 8,4. Several layers create BackProp.

**bias** 0 creates BareProp for a one-hidden-layer network; 1 keeps bias nodes.

**errfunc** **auto**, **xentropy**, or **lms**.

**mode** **load** reads the network type and architecture from a saved network.

**path or file** Saved network path or multipart upload when **mode=load**.

**log\_lastop, log\_history** 1 or 0 logging controls.

Model construction requires a loaded dataset because the input and output dimensions are data-dependent. A loaded network is checked against those dimensions before it becomes current.

## 4.5 Weight reset: POST `/api/randomize`

The optional **seed** field resets the current trainable model to a reproducible starting point. Training itself continues from current weights; repeated calls to `/api/train` do not randomize implicitly.

## 4.6 Training: POST `/api/train`

**algorithm** 1 for canonical gradient descent, 2 for conjugate gradient, 3 for Shanno, or **auto** to probe and choose.

**maxiter** Required positive iteration ceiling.

**seed** Seed used by training and automatic selection.

**async** 1 returns immediately and publishes progress through `/api/train/status`.

**eta, autostep** Manual learning rate and automatic-step toggle.

**batch\_epoch** Batch or epoch mode. Logistic regression requires it to remain on.

**weight\_decay, decay** Weight-decay toggle and lambda.

**minerr, change, errwindow, gradmax** Classic stopping conditions. Omitting a field keeps its current value; sending an empty field disables that condition.

**autostop, autostop\_tol, autostop\_window** Plateau detector and its tolerance and moving-window width.

**logprint, printcount** Iteration-table presentation only. These fields never change the fitted endpoint.

A blocking call returns the complete report, ROC series, and structured statistics. An asynchronous call returns acceptance immediately; the same complete result later appears in status. The response separates operation success from fit validity with fields including **converged**, **ceilingExhausted**, and **stopReason**.

## 4.7 Long-job control

### 4.7.1 GET /api/train/status

Training, stepwise regression, OBD, and cross-validation share one job channel. Status returns **running**, the current phase, decimated training and held-out error series, and procedure-specific progress. OBD adds its phase and hidden-node count. Stepwise adds the current pass and candidate. Cross-validation adds stage, procedure, run, fold, and nested-search detail when present. When the job finishes, **result** contains the same object its blocking form would have returned.

### 4.7.2 POST /api/train/stop

Cancellation is graceful: the engine finishes the current iteration and produces the appropriate partial audit. Bodyless POST requests must still carry a zero-length body:

```
curl -s -X POST -d "" "$URL/api/train/stop"
```

A bare `curl -X POST` sends no Content-Length header and can make the server wait for its read timeout before dispatching.

Only status and stop remain available while a job owns the engine. Other engine-touching endpoints return HTTP 409 with **"busy":true**; there is no hidden request queue.

## 4.8 Standalone analyses

### 4.8.1 POST /api/dfa

`type=linear` runs LDFA and `type=quadratic` runs QDFA. Optional `log_lastop` and `log_history` fields control logging. The response includes a captured report and, for a discrete single-output dataset, ROC series and structured statistics. DFA does not replace the current model.

### 4.8.2 POST /api/regress

**structure** Semicolon-separated conceptual variables; commas join nodes and hyphens denote inclusive ranges, for example `0;1-4;5;6,7`.

**direction** `forward` or `reverse`.

**threshold** Selection p-value threshold.

**async** `1` for status, progress, and Stop.

Stepwise requires a trained cross-entropy model. Candidate subnetworks are fitted on clones, never mutate the current model, and must converge before entering a Wilks comparison. The result contains the selection path, every candidate audit, the final variables settled by completed passes, and a **complete** flag.

### 4.8.3 POST /api/obd

**hidden\_start, hidden\_max** Starting size and safety ceiling.

**iter\_budget, sample\_every** Per-size ceiling and held-out sampling cadence.

**early\_stop\_tol, early\_stop\_patience** Within-size held-out reversal rule.

**grow\_patience, prune\_tol** Growth and pruning rules.

**autostop\_tol, autostop\_window** Per-size training plateau backstop.

**algorithm** 1, 2, 3, or auto.

**seed** Reproducibility seed.

OBD is asynchronous-only and requires a discrete, single-output dataset with a held-out monitoring set. Its result names that set as **validation** or **test**, records every size trial, and replaces the current model only after a successful search.

### 4.8.4 POST /api/cv

**folds, seed** Fold count and reproducibility seed.

**logistic, ldffa, qdffa, neural** Procedure-selection toggles.

**neural\_obd, neural\_hidden** Nested OBD or a fixed neural size.

**maxiter, hidden\_max, iter\_budget** Ordinary-fit and nested-search ceilings.

**algorithm** Nested OBD optimizer, including **auto**.

**autostop\_tol, autostop\_window, inner\_val** Nested-fit plateau rule and inner validation share.

**strata, strata\_bins** Optional one-based input columns for outcome-by-covariate fold balance and the number of quantile bins for continuous columns.

**group** Optional one-based input columns whose joint values define indivisible sampling groups. The same key governs the outer folds, locked holdout, and nested OBD inner validation split.

**locked\_fraction, locked\_n** Optional locked test share or count. Use one form.

**primary, reference** Prespecified procedure tokens for the signed AUC contrast.

**independence rows** opts into ordinary paired DeLong inference for a row-wise design. **cluster** selects Obuchowski clustered covariance and requires **group** plus a locked test. Omit the field for point estimates only.

Cross-validation is asynchronous-only, requires raw discrete single-output data, and does not alter the current model. The result contains headline and detailed reports plus paths to Tier 3 CSV and JSON artifacts.

Without **strata** or **group**, folds are outcome-stratified. **strata** balances outcome-by-covariate cells; **group** makes whole-cluster folds for evaluation on unseen groups. The two cannot currently be combined. These fields belong to the CV request and are never inherited from **/api/load**. A grouped request also makes the locked holdout and nested inner validation group-disjoint, with no row-wise fallback when a whole-group partition is infeasible.

A locked test adds original-row predictions and a signed, prespecified AUC contrast. Independent-row DeLong and clustered Obuchowski inference are explicit alternatives selected by the declared sampling unit; neither ever falls back to the other. The clustered preflight requires at least two locked-test clusters carrying each outcome class. Tier 1 names the estimator and the number of independent clusters. **cv\_run.json** records the typed fold and locked-test designs, including per-fold rows, events, groups, imbalance, largest group, and leakage. The two prediction CSV files use the original raw-row identity and, for grouped designs, stable cluster identifiers, so they join without reconstructing either key.

## 4.9 Statistics and downloads

### 4.9.1 GET /api/stats

Returns structured train and optional test statistics for the current trained classification model. Each set includes classification measures, ROC estimates, confidence intervals, and curve data. The training endpoint also embeds these values with the captured human-readable report.

### 4.9.2 GET /api/save/:what

Replace **:what** with one of:

|                                            |                                                  |
|--------------------------------------------|--------------------------------------------------|
| <b>network</b>                             | saved model weights and architecture             |
| <b>scales</b>                              | natural-to-normalized scaling factors            |
| <b>train_set, test_set</b>                 | normalized datasets                              |
| <b>train_guesses, test_guesses</b>         | model predictions                                |
| <b>dfa_train_guesses, dfa_test_guesses</b> | DFA predictions                                  |
| <b>report</b>                              | captured report from the last training operation |

The engine writes the artifact in the run directory and returns the same bytes as a download. Missing prerequisites return HTTP 400 with a JSON explanation.

## 4.10 Response and audit conventions

The API returns machine-readable values together with the engine's captured report, rather than asking clients to reconstruct statistical meaning from a few scalars. JSON numeric fields that cannot be calculated are null, not invented sentinel estimates.

Every API action, including save and stop, is timestamped in `neuron.actions.log`. Engine operations and self-describing training headers are written to `neuron.log`. Together with the seed and saved artifacts, these logs make a local GUI or scripted session reproducible and auditable.

## Chapter 5

# Data preparation and deployment tools

The current support tools are standard-library Python programs in `tools/`. They require Python 3 but no installed packages. The Perl, Palm OS, and early iPhone exporters from neUROn2++ are not part of neuron 3.0.

| Program                    | Purpose                                                                                        |
|----------------------------|------------------------------------------------------------------------------------------------|
| <code>mkdataset.py</code>  | Convert a delimited table into a numeric neuron dataset plus interpretation files.             |
| <code>neuron2web.py</code> | Convert a saved model and its natural-unit interface into a self-contained browser calculator. |

The maintained command reference is `tools/README.md`. Both programs are exercised against committed expected outputs by `tests/tools/run_tools.sh`.

### 5.1 Preparing a dataset

`mkdataset.py` converts an ordinary delimited file into the numeric, outcome-last form used by the engine. It can change delimiters, one-hot encode categorical inputs, map a two-valued text outcome to 0 and 1, create missing-value indicator pairs, and omit one reference category from each categorical group.

```
python3 tools/mkdataset.py --onehot --refcat \  
  --key key.txt --inputs inputs.txt \  
  -o data.txt raw.csv
```

The key maps encoded columns back to human variables. `inputs.txt` groups nodes belonging to the same conceptual variable for stepwise analysis. The program reports the resulting column count; the number of input nodes is one less than that count for a single outcome.

### 5.1.1 Current grooming behavior

- The first row is treated as column labels unless `--noheader` is supplied.
- Python's CSV parser preserves quoted fields containing embedded delimiters. `--delimiter` accepts any single character, including `\t` for a tab.
- Rows with a blank outcome are dropped with a warning. `--noelim` retains them when that behavior is intentional.
- A numeric input containing blanks becomes a two-node indicator pair: presence followed by value. A blank is represented by `0,0`; a present value by `1,value`.
- `--onehot` with no column list detects non-numeric columns automatically. A comma-separated list of one-based column numbers limits conversion to selected columns.
- A two-level text outcome becomes one binary output. Text input levels become grouped indicator nodes; a blank input activates none.
- `--refcat` omits the alphabetically first input level and records it as the reference. This is the appropriate form for logistic regression and other intercept-bearing models because it avoids a perfectly collinear full indicator group.
- `--key` records every emitted column and its source label. `--inputs` records conceptual-variable groups in the syntax accepted by stepwise regression.
- `--quiet` suppresses the narrative report without changing the output files.

## 5.2 Creating a browser calculator

`neuron2web.py` combines three artifacts:

1. a saved network containing architecture and weights;
2. scaling factors mapping natural values to model inputs; and
3. a label specification describing fields, units, categories, and outcome text.

```
python3 tools/neuron2web.py \  
--network network.txt --scales scales.txt \  
--spec spec.txt -o calculator.html
```

The resulting HTML file is self-contained. Its JavaScript, model, scaling rules, and form are embedded, so it can be opened from disk, emailed, or served by any static web host. Inference remains in the visitor's browser.

### 5.2.1 Supported models and form fields

The exporter currently accepts saved SimpleProp and binary logistic models. It refuses other network types rather than approximating their architecture. The label specification supports:

- N A numeric field with label and optional units.
- C A categorical menu whose levels expand to indicator columns. An asterisk marks a reference level omitted by `--refcat`.
- B A binary choice.
- K A missingness indicator paired with a numeric field.
- O Labels for outcomes 0 and 1.
- R Optional text for the reported odds.

The complete label grammar and validation contract are in `docs/deploy.md`. The tool checks the specification against the saved network's input count and the scaling file before it writes the calculator.

### 5.2.2 Verification and preview

Before deployment, use `--eval` with a known natural-unit row to verify the complete scaling and forward-pass path. A dataset that arrived already normalized needs the original natural-to-normalized scales; scales saved from a later training run over normalized values would merely reproduce that normalized interface.

`--serve` writes the calculator, serves it on loopback at an operating-system-assigned port, and opens a preview in the browser. `--title` supplies the page title. `--quiet` suppresses informational output. Neither preview nor deployment requires an external JavaScript package or web service.

## 5.3 Worked examples

The illustrated Civic Choice walkthrough at `docs/datasets/civic-choice/WALKTHROUGH.md` is the current end-to-end guide. The dataset-specific READMEs under `docs/datasets/` provide further preparation and deployment examples.

## Part II

# Foundational design record

The chapters in this part preserve the mathematical derivations, documentation philosophy, numerical layer, and object design from the neURON2++ redesign. They explain why the engine is organized as it is, but their method inventories and code examples are not the current public interface specification. For present architecture, build, menus, and HTTP behavior, use Part I and the source tree.

## Chapter 6

# Methodology

The following chapter outlines algorithms currently implemented in neURON, as well as many intended to be implemented in the future. As usual, we are indebted to Dr. Golden. The terminology ANN is used to refer to “Artificial Neural Network”.

1. **Error Gradient Calculation Algorithm.** The gradient of the objective function is computed by calling a *forward propagation algorithm* followed by a *backward error propagation algorithm*.
2. **Parameter Estimation Algorithm.** The gradient computed by the *Error Gradient Calculation Algorithm* is then used by the parameter estimation algorithm for the purposes of learning.
3. **Data Analysis Algorithm.** If this algorithm decides that reliable statistical inference is possible, this algorithm indicates which input nodes in the network are making statistically significant contributions to the network’s performance. Otherwise, this algorithm makes recommendations to the user regarding possible manipulations designed to increase statistical inference reliability. Future work will involve computing confidence intervals on the predictions and relaxing the “Goodness-of-Fit” assumption which is assumed throughout this document.

Note that math calculations for the following algorithms such as matrix inverse, eigenvalues, “line-search”, cumulative distribution functions for the chi-square random variable for statistical test result reporting can be done using the routines in *Numerical Recipes in C*. In fact, *Numerical Recipes* has a conjugate gradient routine which could be used to develop a learning algorithm for the ANN system as well.

## 6.1 Error Gradient Calculation Algorithm

### 6.1.1 Data Records

Note that the data is assumed to have the following format:  $(\mathbf{s}_k, \mathbf{t}_k)$  where  $\mathbf{s}_k$  is the  $d$ -dimensional input vector for the  $k$ th data record and  $d$ -dimensional vector  $\mathbf{t}_k$  is a list of “targets” for the  $d$  output units. If the training data consists of  $N$  data records, then we have  $k = 1, \dots, N$ .

### 6.1.2 Objective Function

The *error gradient* is defined as the gradient (i.e., vector first derivative) of a sufficiently smooth *error function* with respect to the “weights” (i.e., parameters) of the ANN. In this document version, only feedforward  $m$ -layer ANN systems will be considered where the number of “layers” is defined by “connections”. Thus, a 1-layer ANN is essentially a linear feedforward ANN with one set of input units and one set of output units and no hidden units. Let  $\mathbf{y}$  denote a vector containing all weights of an  $m$ -layer ANN. The activation levels of the units in the output layer of an  $m$ -layer ANN generated in response to training stimulus  $k$  are denoted by the *activation vector*,  $\mathbf{a}_k^{(m)}$ . Note that  $\mathbf{a}_k^{(m)}$  is a function of both  $\mathbf{y}$  and the input vector  $\mathbf{s}_k$ .

The *error function*,  $E_k$ , for the  $k$ th training pattern is defined by the formula:

$$E_k(\mathbf{y}) = e(t^k, \mathbf{a}_k^{(m)}) + \frac{1}{2\sigma_w^2} \sum_{i=1}^m |\mathbf{y}|^2. \quad (6.1)$$

The second term on the right hand side of (6.1) is required to guarantee reliable statistical inference. It basically “stabilizes” the learning process by preventing the weights in the network from becoming too large or too small. In practice, a connection weight is expected to lie between  $-1.96\sigma_w$  and  $+1.96\sigma_w$  with probability equal to about 95%. Adjustment of this parameter can “change results” of the statistical analysis so it is best to fix  $\sigma_w$  to be some constant number and try not to change its value. A reasonable choice for  $\sigma_w$  would be  $\sigma_w = 100$ . This basically constrains the “learning process” to find connection strengths for the ANN such that the connection strength weight ranges between  $-200$  and  $+200$ . The choice of  $\sigma_w$  should be reported in conjunction with all statistical analysis results. Note that as  $\sigma_w \rightarrow \infty$ , the effect of the summation term becomes negligible.

The first term on the right hand side of (6.1) is called the *prediction error function* term. The prediction error function  $e$  must have continuous second partial derivatives with respect to the weights. For example, the classic choice of the prediction error function  $e$  is to choose

$$e(\mathbf{t}^k, \mathbf{a}_k^{(m)}) = (1/2)|\mathbf{t}^k - \mathbf{a}_k^{(m)}|^2 \quad (6.2)$$

to be the *least squares prediction error function*. This is a good choice when the target vector  $\mathbf{t}^k$  is a continuous (as opposed to a discrete) variable.

When the target vector consists of a set of binary variables taking on the values of zero or one, then a good choice is the *cross-entropy prediction error function* defined by the formula:

$$e(\mathbf{t}^k, \mathbf{a}^{(m)}) = -\mathbf{t}^k \cdot \mathbf{*} \log[\mathbf{a}_k^{(m)}] - (1 - \mathbf{t}^k) \cdot \mathbf{*} \log[1 - \mathbf{a}_k^{(m)}]. \quad (6.3)$$

Note that the notation  $\cdot \mathbf{*}$  denotes element by element matrix multiplication, the notation  $\mathbf{1}$  denotes a column vector of ones, and the notation  $\mathbf{log}$  denotes a vector-valued function defined such that the  $\mathbf{y} = \mathbf{log}(\mathbf{x})$  whenever the  $i$ th element of the vector  $\mathbf{y}$ ,  $y_i$  is equal to the natural log (log base  $e$ ) of the  $i$ th element of the vector  $\mathbf{x}$ . When the cross-entropy prediction error function is used, the transfer function for all output units should be the sigmoidal logistic transfer function which is defined formally in the next section of this report.

### 6.1.3 Forward Propagation of Activations

The response of the artificial neuron units in layer  $j$  is computed from the activation levels of the artificial neuron units in layer  $j - 1$  using the *forward propagation algorithm* formulas. These formulas are defined by an update formula for computing the *netinput*,  $\mathbf{u}_k^{(j)}$ , to a unit in layer  $j$  for the  $k$ th training stimulus:

$$\mathbf{u}_k^{(j)} = \mathbf{W}^{(j)} \mathbf{a}_k^{(j)} + \mathbf{b}^{(j)} \quad (6.4)$$

and the *output activation level* for the  $k$ th training stimulus in layer  $j + 1$ :

$$\mathbf{a}_k^{(j)} = \mathbf{f}^{(j)}(\mathbf{u}_k^{(j)}) \quad (6.5)$$

using the *layer transfer function*,  $\mathbf{f}^{(j)}$ . Note that  $\mathbf{a}_k^{(0)} = \mathbf{s}_k$  by definition.

The layer transfer function,  $\mathbf{f}^{(j)}$ , is the chosen vector-valued transfer function for layer  $j$ ,  $\mathbf{W}^{(j)}$  is the connection matrix from the units in layer  $j - 1$  to the units in layer  $j$ , and  $\mathbf{b}^{(j)}$  is the bias vector for the units in layer  $j$ . The subscript  $k$  refers to the  $k$ th training stimulus. The function  $\mathbf{f}^{(j)}$  must have continuous second partial derivatives.

For example, let  $\mathbf{u} = [u_1, u_2, u_3, u_4, u_5]$ . A function form for  $\mathbf{f}^{(j)}$  might be:

$$\mathbf{f}^{(j)}(\mathbf{u}) = [u_1, u_2, \mathcal{S}(u_3), \exp(u_4^2), 2\mathcal{S}(u_5) - 1] \quad (6.6)$$

where  $\mathcal{S}(u_3) = 1/(1 + \exp(-u_3))$ . The layer transfer function in (6.6) illustrates a situation where the first two elements are *linear transfer functions*, the third element is a *logistic sigmoidal* or just *sigmoidal* function, the fourth element is an example of one type of *radial basis* transfer function, and the fifth element is an example of one type of hyperbolic tangent sigmoidal function (i.e., the asymptotes are -1 and +1 unlike the 0 and 1 asymptotes of the logistic sigmoid).

### 6.1.4 Backward Propagation of Errors

Let the vector  $\delta_k^{(j)} = dE_k/d\mathbf{a}_k^{(j)}$  (i.e., the derivative of the error with respect to the activation pattern vector in layer  $j$  evaluated at the current set of weights

and the  $k$ th training vector). The vector  $\delta_k^{(j)}$  is called the *error signal* for the  $j$ th layer of units whose activation levels are defined by the vector  $\mathbf{u}^{(j)}$  (note this notation may slightly differ from the Rumelhart et al., 1986, paper). Let the matrix  $\mathbf{Df}_k^{(j)}$  be the vector first derivative (i.e., the “jacobian”) of the vector transfer function  $\mathbf{f}_k^{(j)}$  and  $k$  refers to the  $k$ th training stimulus.

Consider a  $m$ -layer feedforward network consisting of layers  $1, \dots, m$ . The *backward error propagation algorithm* formula for propagating an error signal for the output units of layer  $j$  to the output units of layer  $j - 1$  (i.e., the input units of layer  $j$ ) using the formula:

$$\delta_k^{(j-1)} = [\mathbf{W}^{(j)}]^T \mathbf{Df}_k^{(j)} \delta_k^{(j)} \quad (6.7)$$

for  $j = 1, \dots, m - 1$ . The error signal for the  $m$ th layer is the vector  $\delta_k^{(m)} = de_k/d\mathbf{a}_k^{(m)}$  where  $e_k$  is defined as in (6.1).

For example, suppose  $\mathbf{f}$  is defined such that  $\mathbf{u} = \mathbf{f}(\mathbf{u})$  (i.e., a “linear transfer function”), then  $\mathbf{Df}$  is the identity matrix. As another example, suppose that all output units are logistic sigmoidal units so that the activation level of the  $i$ th output unit is given by  $\mathcal{S}(u_i)$  where  $u_i$  is the netinput to the  $i$ th output unit and  $\mathcal{S}(u_i) = 1/(1 + \exp(-u_i))$ . Assume there are only three units in a layer. In this case,  $\mathbf{f}$  will be a three-dimensional vector whose  $i$ th on-diagonal element is defined by  $\mathcal{S}(u_i)$  and  $\mathbf{Df}$  will be a three-dimensional matrix whose off-diagonal elements are zero and whose  $i$ th on-diagonal element is  $\mathcal{S}(u_i)(1 - \mathcal{S}(u_i))$ .

If the prediction error is a sum-squared error signal then

$$\delta_k^{(m)} = -(\mathbf{t}_k - \mathbf{a}^{(m)}) \quad (6.8)$$

If the prediction error is a cross-entropy error function, then

$$\delta_k^{(m)} = -(\mathbf{t}_k - \mathbf{a}^{(m)}) ./ [\mathbf{a}^{(m)} .* (\mathbf{1} - \mathbf{a}^{(m)})] \quad (6.9)$$

where it is assumed that the  $m$  layer consists of logistic sigmoidal functions. Note the notation  $./$  refers to element by element division.

## 6.2 Parameter Estimation Formulas

### 6.2.1 Calculation of the gradient vector (and overall error).

For each data record ( $k = 1, \dots, N$ ), compute the following:

- *Step 1: Forward Activation Propagation.* Given stimulus  $\mathbf{s}_k$ , propagate activation levels forward layer by layer through the network using the forward propagation equations (equations (6.4) and (6.5)) in order to generate output activation vector  $\mathbf{a}_k^{(m)}$  for the last layer (layer  $m$ ).
- *Step 2: Compute data record error.* Compute the output error  $E_k$  using equation (6.1).

- *Step 3: Backward error Propagation.* Compute the output error signal  $\delta_k^{(m)}$  for layer  $m$  and propagate this backwards so that an error signal is obtained for layers  $m - 1$  through layer 1 using equation (6.7).
- *Step 4: Compute and store data record gradient.* Let the column vector  $\mathbf{f}_k^{(j-1)}$  be defined such that  $\mathbf{f}_k^{(j-1)} = [\mathbf{a}_k^{(j-1)}, 1]^T$ .

$$\mathbf{G}_k^{(j)} = \delta_k^{(j)} \mathbf{D} \mathbf{f}_k^{(j)} [\mathbf{f}_k^{(j-1)}]^T + (1/\sigma_w^2) [\mathbf{W}, \mathbf{b}] \quad (6.10)$$

for  $j = 1, \dots, m$ .

Let the notation  $STACK[\mathbf{G}_k^{(j)}]$  refer to a vector formed by “stacking” the rows of the matrix  $\mathbf{G}_k^{(j)}$  one on top of another. Thus, if  $\mathbf{G}_k^{(j)}$  has  $d$  rows, then the first  $d$  elements of  $STACK[\mathbf{G}_k^{(j)}]$  will be the first row of  $\mathbf{G}_k^{(j)}$ , the second  $d$  elements of  $STACK[\mathbf{G}_k^{(j)}]$  will be the second row of  $\mathbf{G}_k^{(j)}$ , and so on. It will also be convenient to have the function  $UNSTACK$  which is the inverse of the function  $STACK$ . Now define the column vectors:  $\mathbf{g}_k^{(j)} = STACK[\mathbf{G}_k^{(j)}]$  and  $\mathbf{g}_k = [\mathbf{g}_k^{(1)}, \dots, \mathbf{g}_k^{(m)}]$ . The vector  $\mathbf{g}_k$  is called the *data record gradient* for the  $k$ th data record in the training data and should be temporarily stored once it is computed.

### 6.2.2 Example learning algorithms.

**Important indexing issues and definitions.** Define  $\mathbf{y}^{(j)} = STACK[\mathbf{W}^{(j)} \mathbf{b}^{(j)}]$  and  $\mathbf{y} = [\mathbf{y}^{(1)}, \dots, \mathbf{y}^{(m)}]$ . We will use the index  $t$  to denote the *iteration number* of the learning algorithm. *The index  $t$  is not the same as the index  $k$ !! The  $k$  index ranges over the set of  $N$  data records in the training set, while the  $t$  index identifies the learning algorithm iteration!!* Using this notation,  $E_k(\mathbf{y}_t)$  will be used to refer to the error for the  $k$ th data record at iteration number  $t$  of the learning algorithm. Similarly,  $\mathbf{g}_k(\mathbf{y}_t)$  refers to the gradient of the error for the  $k$ th data record at iteration number  $t$  of the learning algorithm. Similarly, the variable  $N$  is **NEVER** equal to the number of learning trials but is **ALWAYS** defined as the number of data records in the training set.

**Initialization issues.** In all of the following learning algorithms the choice for the initial value  $\mathbf{y}_0$  is absolutely crucial. The elements of this vector should be chosen so that they are independent and identically distributed with mean zero and variance  $\sigma_0^2$ . The parameter  $\sigma_0^2$  should be “highly visible” to the user. Learning then proceeds until the parameters of the network converge. If  $\sigma_0^2$  is too small, then the learning process will be very long. If  $\sigma_0^2$  is too large, then the learning process will rapidly converge to a crummy solution. It is best to err on the side of selecting  $\sigma_0^2$  to be too small.

**Classic on-line backprop.** The *classic on-line* backpropagation learning algorithm is given by the formula:

$$\mathbf{y}_{t+1} = \mathbf{y}_t - \eta \mathbf{g}_k(\mathbf{y}_t) \quad (6.11)$$

where  $k$  (the particular training data record index) is chosen at random on iteration  $t$  of the learning algorithm. The parameter  $\eta$  is a strictly positive

number called the *learning rate* or *stepsize* for the learning algorithm (e.g.,  $\eta = 1/N$  is a good choice).

**Stochastic approximation on-line backprop.** An improved version of the *classic on-line* algorithm called *stochastic approximation on-line* is given by the formula:

$$\mathbf{y}_{t+1} = \mathbf{y}_t - \left( \frac{\eta}{t_0 + t} \right) \mathbf{g}_k(\mathbf{y}_t) \quad (6.12)$$

where  $t_0$  is a positive integer whose magnitude is roughly equal to the number of records,  $N$ , in the training data set. The stepsize  $\eta$  is a positive number (e.g.,  $\eta = 1$  is a good choice).

**Classic off-line (“batch”) backprop.** The *classic off-line* or “classic batch” backpropagation learning algorithm is given by the formula:

$$\mathbf{y}_{t+1} = \mathbf{y}_t - \eta \mathbf{g}(\mathbf{y}_t) \quad (6.13)$$

where the *average gradient*  $\mathbf{g}$  is defined by the formula:

$$\mathbf{g} = (1/N) \sum_{k=1}^N \mathbf{g}_k. \quad (6.14)$$

A good choice for  $\eta$  for this algorithm would be  $\eta = 1$  but as usual  $\eta$  must be a strictly positive real number.

**Offline backprop with automatic stepsize selection.** Define the *average error*  $E$  by the formula:

$$E = (1/N) \sum_{k=1}^N E_k. \quad (6.15)$$

An improved version of the *classic off-line* backpropagation learning algorithm is given by the formula:

$$\mathbf{y}_{t+1} = \mathbf{y}_t - \eta_t \mathbf{g}(\mathbf{y}_t) \quad (6.16)$$

where  $\mathbf{g}$  is defined as in (6.14) and  $\eta_t$  is chosen such that quantity  $E(\mathbf{y}_t - \eta \mathbf{g}(\mathbf{y}_t))$  is minimized with respect to  $\eta$  (treating  $\mathbf{y}_t$  as a known constant) subject to the constraint that  $0 \leq \eta_t \leq 1$ . A “line search” routine from *Numerical Recipes* may be used to solve this one-dimensional optimization problem.

## 6.3 Data Analysis Algorithms

### Conditions for Statistical Inference to be Valid.

The four assumptions that we make here are as follows. All four of these assumptions must be satisfied in order to guarantee reliable statistical inferences.

**Log-likelihood Objective Function Assumption.** First, it is assumed that the prediction error objective function  $e_k$  may be interpreted as the negative natural logarithm of the likelihood of a training data record given the network’s

parameters. When the target  $\mathbf{t}_k$  takes on a continuous range of values and the least-squares error function defined in (6.2) then this assumption is satisfied by assuming a multivariate conditional Gaussian distribution for the target vector given the input vector and network parameters. When the target consists of elements taking on the values of either zero or one and the output units are sigmoidal, then this assumption is satisfied by interpreting the activation level of an output unit as the probability that the target for that output unit will take on the value of one and using the cross-entropy function defined in (6.3). The best way to satisfy this assumption is to make sure that the user only gets to choose from a set of valid log-likelihood objective functions.

**Goodness-of-Fit Assumption.** Second, it is assumed that the predictions of the model with respect to the training data would be “perfect” if the model was exposed to a sufficiently large set of training data and was provided with a sufficiently large number of learning iterations. That is, we assume the model provides a good fit to the observed training data. More technically, it is assumed that the probabilistic modeling assumptions introduced in the “log-likelihood objective function assumption” are valid.

We will call the parameters which provide a good fit to the data  $\mathbf{y}^*$ . Note that in many real-world situations, this assumption will not be satisfied. For example, suppose we teach a linear ANN the “exclusive-or” problem. No matter how long we train the ANN, the ANN will never “fit the data”.

To check this assumption, provide the user with plots of training and test set performance. Warn user that statistical inferences may not be reliable if the ANN is not a good data model.

**Convergence Assumption.** Third, it is assumed that the gradient vector,  $\mathbf{g}(\mathbf{y}^*)$ , is sufficiently small in magnitude. Typically, we check this assumption by seeing if the element in  $\mathbf{g}_t$  with the largest absolute value is less than  $\epsilon$ . The constant  $\epsilon$  is typically about 0.0001. If this assumption is not satisfied, the user should be recommended to adjust the parameters of the learning procedure or alter the initial conditions (initialization of the ANN connection weights).

**Convergence to a Strict Local Minimum Assumption.** Fourth, it is required that this local uniqueness assumption be satisfied. Typically, we check this assumption by computing the matrix:

$$\mathbf{H}^* = (1/N) \sum_{k=1}^N \mathbf{g}_k(\mathbf{y}^*) [\mathbf{g}_k(\mathbf{y}^*)]^T. \quad (6.17)$$

All eigenvalues of  $\mathbf{H}^*$  can be shown to be always non-negative (if you see any negative eigenvalues then there are numerical problems with the eigenvalue calculation routine). Compute the condition number of  $\mathbf{H}^*$  which is defined as the largest eigenvalue of  $\mathbf{H}^*$  divided by the smallest eigenvalue of  $\mathbf{H}^*$ . If the condition number is less than some constant  $K$  then we say the local uniqueness assumption is satisfied. A constant  $K$  with a value of 1000 is acceptable. Most statisticians use a constant  $K$  of value of 100. Usually,  $K > 1000000$  is too large. If this assumption is not satisfied, the user should consider collecting more data, changing the network architecture, or decreasing the value of  $\sigma_w$ . Of

course the preferred course of action is collecting more data. Note that altering the value of  $\sigma_w$  is formally equivalent to changing the network architecture.

### Wilks Likelihood Ratio Test

<sup>1</sup> Wilk’s Likelihood Ratio Test works as follows. Estimate the parameters of the network. Call that set of parameters  $\mathbf{y}_{full}$ . Now constrain  $r$  free parameters of the network equal to the value of zero and re-estimate the network parameters using  $\mathbf{y}_{full}$  as the “initialization” for the learning algorithm. The resulting parameter estimates will be called  $\mathbf{y}_{reduced}$ . It is assumed that  $\mathbf{y}_{full}$  satisfies all four assumptions and  $\mathbf{y}_{reduced}$  satisfies all assumptions except for the Goodness-of-Fit assumption. It is assumed that  $\mathbf{y}_{reduced}$  and  $\mathbf{y}_{full}$  are sufficiently close to one another in the sense that under the null hypothesis that any difference between  $\mathbf{y}_{reduced}$  and  $\mathbf{y}_{full}$  is due to the fact that we only have a finite random sample.

Using equation (6.15), compute

$$G^2 = -2N[E(\mathbf{y}_{full}) - E(\mathbf{y}_{reduced})]. \quad (6.18)$$

If  $G^2$  exceeds the critical value of a chi-square random variable with  $r$  degrees of freedom at the  $\alpha$  significance level ( $r$ =number of parameters in full model minus number of parameters in reduced model), then we reject the null hypothesis at the  $\alpha$  significance level that the  $r$  parameters could be set equal to zero without hurting the fit of the model to the data.

### Another Version of Wilks Test called the Wald Test

<sup>2</sup> A generalization of the Z-test can be used to answer the same questions as Wilks test. This statistical test is called the Wald Test. It has the advantage that you can do the statistical test with only one set of parameter estimates. First, you estimate the parameter vector which we will call  $\mathbf{y}^*$ . Now suppose you want to test the null hypothesis that  $r$  specific elements of  $\mathbf{y}^*$  are equal to zero. We will denote this subvector by the  $r$ -dimensional vector  $\mathbf{q}^*$ . Obviously this is just as general as Wilks test because you can select the  $r$  elements to the connection strengths leaving a particular input unit (for example).

Now you define a *selection matrix*  $\mathbf{S}$  which has  $r$  rows and  $h$  columns where  $h$  is the dimension of  $\mathbf{y}^*$ . You want to select  $\mathbf{S}$  in just such a way so that  $\mathbf{q}^* = \mathbf{S}\mathbf{y}^*$ . This can always be done and it might make sense to have a set of “canned” selection matrix generation routines available so the user doesn’t have to figure this one out.

For example, suppose  $\mathbf{y}^* = [1, 2, 11, 4, 6]$  and we want to test the null hypothesis that the third and fourth weights are both equal to zero. Then we would

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<sup>1</sup>S. S. Wilks, “The large-sample distribution of the likelihood ratio for testing composite hypotheses,” *The Annals of Mathematical Statistics* 9(1), 60–62, 1938, <https://doi.org/10.1214/aoms/1177732360>.

<sup>2</sup>A. Wald, “Tests of statistical hypotheses concerning several parameters when the number of observations is large,” *Transactions of the American Mathematical Society* 54(3), 426–482, 1943, <https://doi.org/10.1090/S0002-9947-1943-0012401-3>.

construct a selection matrix  $\mathbf{S}$  with two rows whose first row is  $[0, 0, 1, 0, 0]$  and whose second row is  $[0, 0, 0, 1, 0]$ . The only technical restriction on the choice of the selection matrix  $\mathbf{S}$  is that the rank of the matrix  $\mathbf{S}$  must be exactly equal to the number of rows in  $\mathbf{S}$  (i.e., the rows of  $\mathbf{S}$  must be linearly independent vectors). clearly in this example the dot product of the two row vectors in  $\mathbf{S}$  is equal to zero implying that the two row vectors are orthogonal and thus linearly independent.

Once you've got the selection matrix the remaining calculations are straightforward. Compute the formulas (using equation (6.17)):

$$\mathbf{C}^* = \mathbf{S}^*[\mathbf{H}^*]^{-1}(\mathbf{S}^*)^T \quad (6.19)$$

and

$$\mathcal{W}_N = N(\mathbf{S}\mathbf{y}^*)^T[\mathbf{C}^*]^{-1}(\mathbf{S}\mathbf{y}^*) \quad (6.20)$$

where  $N$  is the number of data records.

If the Wald statistic  $\mathcal{W}_N$  exceeds the value of a chi-square random variable with  $r$  degrees of freedom (where  $r$  is the number of rows of  $\mathbf{S}$ ) at the  $\alpha$  significance level, then reject the null hypothesis that the  $r$  selected weights take on the value of zero at the  $\alpha$  significance level.

## Chapter 7

# Documentation

There is a reason that documentation is described even before the most general features of neUroN design. Students of computer science often think of documentation as ancillary to code. Nothing could be further from the truth. When one realizes that code is secondary to documentation, one becomes a mature programmer.

The reason for this transcendence of documentation is actually quite simple. Code is merely an instance of an abstract idea, the algorithm. Different computer languages yield different instances, but the algorithm is at the core. In order to access the algorithm, a method is coded with inputs and outputs. These inputs and outputs are specified by the programmer, and allow the method to become a “black box” which can be used at will, and, if all goes well, as part of a library of methods that can be used and reused often.

The job of a student programmer is to write code. The job of a master programmer is to write the specifications for the inputs and outputs of methods, which when combined, solve a specific problem. After the specifications are written, then the algorithms within the methods are coded. Documentation ultimately becomes these specifications.

Documentation thus has several purposes. It serves the programmer who wants to reuse code written in the past, but that use is for minor programming tasks. It serves the programmer who wants the code used by other programmers, a legacy for major programming tasks. Ultimately, it serves the programmer who wants the specification to be used by other programmers, and that is the stuff which drives, for example, national economies.

The neUroN programmer will document code in several ways, but in all, *documentation will be written before the code*. Not after, not even while. Documentation will be written *before* the code.

Each method (and feature corresponding to a set of methods) coded in neUroN is to be documented in 3 specific places:

1. Here, in the Design Manifest. Here a natural language description, algorithmic description if applicable, and specification is to be documented.

2. Immediately prior to the method in the code, as a specification.
3. Throughout the code.

As an example, here is the code for the dot product method in matrix.h:

## 7.1 Documentation as it appears in the Design Manifest:

Given a matrix object  $M$ , and vectors  $\vec{v}$  and  $\vec{o}$ , where  $M \cdot \vec{v} = \vec{o}$ , e.g.

$$\begin{bmatrix} M_{11} & M_{12} & \cdots & M_{1c} \\ M_{21} & M_{22} & \cdots & M_{2c} \\ \vdots & \vdots & \vdots & \vdots \\ M_{r1} & M_{r2} & \cdots & M_{rc} \end{bmatrix} \cdot \begin{bmatrix} \vec{v}_1 \\ \vec{v}_2 \\ \vdots \\ \vec{v}_c \end{bmatrix} = \begin{bmatrix} \vec{o}_1 \\ \vec{o}_2 \\ \vdots \\ \vec{o}_r \end{bmatrix}$$

the following dot product operators are available in the Matrix class:

- $M.\text{dotprod}(\text{vector\& } v)$  returns a new vector object which is the dot product of the matrix and the passed vector
- $M.\text{dotprod}(\text{vector\& } v_{in}, \text{vector\& } v_{out})$  which populates the output vector  $v_{out}$  with the dot product of the matrix and the passed vector  $v_{in}$
- $M.\text{dotprod}(\text{vector\& } v_{in}, \text{vector\& } v_{out}, \text{unsigned } begin, \text{unsigned } end)$  which populates the output vector range  $v_{out}[begin]$  to  $v_{out}[end]$  with the dot product of the matrix and the passed vector  $v_{in}$

### Examples

```
// Matrix A has 4 rows and 3 columns, vector v1 has 3 elements
vector< double > v2;
v2 = A.dotprod( v1 ); // v1 is unchanged
                        // v2 now contains the dot product

// vector v3 has 4 elements
A.dotprod( v1, v3 ); // v1 is unchanged
                    // v3 now contains the dot product

// vector v4 has 5 elements
A.dotprod( v1, v4, 0, 3 ); // v4[0] through v4[3] now
                           // contains the dot product
```

## Notes

For `M.dotprod( v )`, the number of columns in *M* *must equal* the number of elements in *v*. For `M.dotprod( vin, vout )`, the number of columns in *M* *must equal* the number of elements in *v<sub>in</sub>*, and the number of rows in *M* *must equal* the number of elements in *v<sub>out</sub>*. Because `M.dotprod( v )` must construct a new vector object, it is less efficient than `M.dotprod( vin, vout )`.

### 7.1.1 General principles of documentation in the Manifest: emphasis on use of the method

The emphasis of documentation in the Manifest is on how the method is used. The programmer calling the code should not need to know anything about the inner workings of the method, only how it is called, and what it returns.

### 7.1.2 Emphasis on readability

Documentation in the manifest is written in a conversational tone so that it is easy to read and understand.

### 7.1.3 Examples

Good, simple examples are provided to demonstrate the use of the method.

### 7.1.4 Important notes

Any specific behavior of the code is noted so that it may be used in the most effective manner.

## 7.2 Documentation as it would appear immediately before the procedure in the code:

```
// Dot product
// For vectors o and v and Matrix M, where M . v = o
//
//                               [ v1
// [ M11 M21 M31 M41      .   v2 = [ o1
//   M12 M22 M32 M42 ]      v3   o2 ]
//                               v4 ]
//
// ( Note that the number of columns in M, 4, is the number of rows in
//   v )
//   by convention, ALL vectors are assumed to be vertical
// Calling sequence: M.dotprod( v, o );
```

### 7.2.1 General principles of documentation immediately before the procedure in the code: clear argument specification

Inputs and when applicable, returned values to the method are clearly and succinctly stated so that casual inspection will allow the code to be used correctly.

### 7.2.2 Critical notes

Critical usage issues (such as the assumption that all vectors are assumed to be vertical in this example) are included in the comments immediately prior to the procedure.

## 7.3 Documentation in the code itself:

Finally, the code itself is written and commented *while it is being written*. Debugging is the *last* step, and comments are *modified* as the method is debugged.

```
template< class T >
vector< T >& Matrix< T >::dotprod( const vector< T >& iVec, vector< T
>& oVec ) const
{
    // Number of Matrix columns must equal number of input vector
    elements
    // Number of Matrix rows must equal number of output vector
    elements
    assert ( iVec.size() == ncols_ && oVec.size() == nrows_);

    // Set a pointer ( pData ) to Matrix's data array
    T* pData = data_;

    // Set a const iterator ( pi ) to the input vector
    vector< T >::const_iterator pi;

    // Set an iterator ( po ) to the output vector
    vector< T >::iterator po;

    // Clear output vector by setting all elements to zero
    std::fill( oVec.begin(), oVec.end(), 0 );

    // Calculate the dot product using vector iterators
    for ( po = oVec.begin(); po != oVec.end(); ++po )
        // Calculate an output vector element = Matrix row . input
        vector
        for ( pi = iVec.begin(); pi != iVec.end(); ++pi )
            *po += ( ( *pData++ ) * ( *pi ) );
```

```
    return oVec; // enables use in vector formulae
}
```

### 7.3.1 General principles of documentation within code:

*Note how easy the code is to read. These basic principles are to be followed:*

- *Comments are meaningful.* Nothing is more useless than a comment which simply restates the code, for example:

```
// Set the end of outputText to NULL
outputText[ 9 ] = '\0';
```

is truly useless, whereas

```
// Truncate output string
outputText[ 9 ] = '\0';
```

while less verbose, is actually more meaningful.

- *Comments are capitalized correctly.* Using all capitals is equivalent to email shouting, and equally offensive. Use

```
// Set the end of outputText to NULL
```

instead of:

```
// SET THE END OF OUTPUTTEXT TO NULL
```

- *Pretty printing makes the code easier to read.* Consistent tabbing and bracket placement go a long way towards making clear code.
- *Frequent meaningful comments make the code easier to read.* Every single code block is documented with a meaningful comment.

### 7.3.2 Variable names:

Variables are to be meaningfully named. It is difficult to sort through previously written code when variables assume generic names like 'x' or 'px'. Verbose variable names are rarely a hindrance. Often student programmers dislike verbose variable names because they do not allow easy packing of variables into 1 line of code. However, this very restriction often makes the code much easier to read by requiring the programmer to split the code into several lines. Consider the example:

```
irpp = (EXP(LN((EXP(n*LN(1+(i/n))))-1)+1)/(ap*ppy))-1)*ap*ppy;
```

While it does fit on 1 line, it is thoroughly unreadable. As such, it is difficult for the programmer to debug while programming, and nearly impossible for succeeding programmers to intelligently use in future code. Consider the alternative:

```
// Use times_per_year_compounded = 2 for Canada, 12 for U.S.
incremental_rate = interest_rate / times_per_year_compounded;

// Calculate effective annual rate
effective_rate = EXP( times_per_year_compounded
    * LN( 1 + incremental_rate ) ) - 1;

// Finally, calculate incremental_interest = interest rate per period:
amortization = amortization_period * payments_per_year;
incremental_interest = (EXP( LN( effective_rate + 1 ) / amortization )
- 1 )
    * amortization;
```

Not only is the algorithm that the programmer used to calculate the interest rate per period manifestly clear, the variables are so well named as to make the ‘//’ comments complimentary rather than critical. And when a future programmer sees the variable ‘incremental\_interest’ 2000 lines of code later, he or she will have half a clue about what that variable means, whereas the meaning of ‘irpp’ would be completely opaque.

### 7.3.3 Space

One of the simplest ways to make code readable is to insert space around operators in code. Consider the example in section 7.3.2,

```
irpp = (EXP(LN((EXP(n*LN(1+(i/n))))-1)+1)/(ap*ppy))-1)*ap*ppy;
```

One of the difficulties in reading this code is the lack of spaces between operators.

Rules for spaces in coding neURon2++ are:

- Put spaces after binary operators (like = ):

```
a = b + c;
```

rather than:

```
a=b+c;
```

- After control keywords like ‘for’ and ‘while’, put spaces around the ( )s, e.g.:

```
for ( i = 1; i <= 10; i++ ) {
```

rather than:

```
for(i = 1; i <= 10; i++){
```

- Inside of brackets, braces or parentheses, put spaces right after the first bracket, brace, or parenthesis, and right before the second, e.g.:

```
a[ i ] = 10;
```

rather than:

```
a[i] = 10;
```

## Chapter 8

# C++ coding and building

neURON was written in C, and took advantage of the high- and low-level compilation features available in that language. While neURON2++ was written in C++, coding began shortly after the first standardized C++ implementation became available. Because C++ is a superset of C, neURON++ became a hodgepodge of C and C++ coding styles, and ultimately, became too cumbersome to maintain, let alone extend. neURON2++ is designed to take advantage of the extensibility offered by the object-oriented nature of C++, but it is also designed to take advantage of other significant advances in C++ such as operator and method overloading. The following C++ style rules are expected to be followed closely, but are not intended to be an exhaustive list. For those interested in an excellent source of good C++ coding style rules, see Deitel and Deitel's *C++ How to Program*.<sup>1</sup>

In general, while coding, think of the 3 cardinal rules of good programming: code must be

- *Readable* (see Chapter 7 on rules for documentation.)
- *Efficient*
- *Bulletproof*

### 8.1 Current build and portability

neuron 3.0 is a C++17 project built with CMake. GSL and a thread library are its compiled dependencies; the browser page is embedded into the executable at configure time. A Release build is important because an unoptimized engine is approximately an order of magnitude slower.

```
cmake -B build -DCMAKE_BUILD_TYPE=Release
```

---

<sup>1</sup>Deitel and Deitel, C++ How to Program, 3rd edition, Prentice Hall, 2001, ISBN 0-13-089571-7

```
cmake --build build --parallel
./build/neuron --version
```

On macOS, install the prerequisites with `brew install cmake gsl`. On Ubuntu or Debian, use `apt install build-essential cmake libgsl-dev`. There are no hand-maintained IDE project files or compiler-specific RTTI settings.

Portability is expressed in `CMakeLists.txt` and verified by continuous integration, rather than maintained as parallel compiler projects.

## 8.2 Use friends sparingly

Friend methods are to be used *only* when absolutely necessary. Consider using a friend a failure to find a good solution. Obviously, friends are used in overloading the ostream operator `<<` and istream operator `>>`, e.g.:

```
friend ostream& operator << ( ostream&, const ... );
```

because there is simply no other good way to overload the ostream operator for a new object. However, friends should *not* be used to overload mathematical operators such as `+` and `*`. *Do not use a friend unless you discuss it with the neURon2++ project group first.*

## 8.3 Use const correctness

Const correctness provides one way to write efficient and bulletproof code right from the start. By being const-correct, you think about which objects can be mutated in a method and which cannot. By not allowing certain methods to mutate objects, you prevent particularly intractable runtime errors, and by judiciously allowing certain methods to mutate objects (e.g. passed array references), you know exactly where your code is behaving efficiently, and exactly where runtime errors may be introduced. *The label const is to be applied while code is written, not after. Constness applied as a patch to already written code is uniformly considered to be very poor programming practice.*

## 8.4 Validate invariants at the appropriate boundary

Assertions remain useful for programmer invariants in development builds, but Release builds define `NDEBUG` and therefore cannot depend on them for user input or memory safety. Public handlers validate request values before calling the engine. Matrix element access performs unconditional bounds checks and throws `BoundsViolation` in every build, and the vector operations of section 9.1.1 likewise reject bounds and dimension violations in every build. Both halves of the

numerical layer are required: until 2026, every precondition in “vector\_ops.h” was an assertion, and because Release is the default configuration none of them had ever executed in the test suite.

An assertion can still document an internal precondition such as equal matrix dimensions:

```
// Catch different size matrices
assert ( nrows_ == rhs.nrows_ && ncols_ == rhs.ncols_ );
```

Code reachable from data, GUI, or API input must additionally reject invalid dimensions through ordinary control flow or an unconditional exception.

## Chapter 9

# vector and Matrix

Establishment of numerical vector and 2-dimensional matrix objects in neURon2++ serves as the basis by which clear, concise, efficient neural computational code may be written. The header file “vector\_ops.h” extends the STL `<vector>` by adding simple unary and binary mathematical operations, and overloaded ostream `<<` and istream `>>` operators for simplified output and input. The header file “matrix.h” implements a matrix class for simplified 2-dimensional matrix operations.

### 9.1 vector\_ops.h

Inclusion of the header file “vector\_ops.h” overloads operators and adds methods to the `<vector>` STL. Please see `<vector>` STL documentation for standard `<vector>` STL methods.

#### 9.1.1 Bounds and dimension violations

Many of the operations below walk two vectors in lockstep, or index into a vector by position. Violating such a precondition is not a data condition to be handled; it is a programming error that reads or writes memory the operation does not own. These preconditions are therefore checked *unconditionally*, and the checks are not assertions: Release builds define `NDEBUG` (see section 8.4), so an assertion cannot protect memory in the shipped binary.

Three exception types are declared in **namespace nvec**:

- **nvec::SizeMismatch**: two parallel containers have incompatible lengths.
- **nvec::RangeViolation**: a position, range or extent lies outside the container it indexes.
- **nvec::EmptyVector**: the operation has no value on an empty set. This applies to `maxabs` and `minabs` only; a `sum` has an identity, so `squared`, `dotprod` and `sumSquaredDifference` all return 0 over empty operands.

Each operation below states its own precondition and which of these it throws. The checks are performed once, before the elementwise loop, never per element.

### 9.1.2 Unary mathematical operators

The following operators perform element by element addition, subtraction, multiplication and division on the right hand side (rhs) vector and left hand side (lhs) vector, returning the result in the left hand side (lhs) vector:

- `+=`: unary addition
- `-=`: unary subtraction
- `*=`: unary multiplication
- `/=`: unary division

#### Example

```
// Standard C-style arrays
double data1[ 4 ] = { 1.0, 2.0, 3.0, 4.0 },
       data2[ 4 ] = { 2, 3, 4, 5 },
       data3[ 3 ] = { 8, 9, 10 };

// vector's std array ctor
vector< double > v1( data1, data1 + 4 ), v2( data2, data2 + 4 );

// Unary addition element by element
v1 += v2;

// Different size vectors
v3 += v1; // v1[ 3 ] will be ignored
```

#### Notes

The vectors may have different numbers of elements, but the right hand side (rhs) vector size *must be* greater than the left hand side (lhs) vector. If the vectors have different sizes, the remaining elements at the end of the rhs vector that are not represented in the lhs vector will be ignored.

This inequality is deliberate and is relied upon: the result carries the lhs size, which is how the bias slot is dropped when a hidden error vector of  $n$  elements is multiplied by an output weight vector of  $n + 1$ . A rhs *shorter* than the lhs is the error, because the operation would then read past its end, and throws `nvec::SizeMismatch` (section 9.1.1).

### 9.1.3 Unary mathematical operators with scalars

The following operators perform scalar addition, subtraction, multiplication and division on a right hand side (rhs) scalar and left hand side (lhs) vector, returning the result in the left hand side (lhs) vector:

- `+=`: unary scalar addition
- `-=`: unary scalar subtraction
- `*=`: unary scalar multiplication
- `/=`: unary scalar division

#### Example

```
// Standard C-style array
double data1[ 4 ] = { 1.0, 2.0, 3.0, 4.0 };

// vector's std array ctor
vector< double > v1( data1, data1 + 4 );

// Unary scalar addition
v1 += 2.0;
```

#### Notes

Type resolution may be a problem if the right hand side argument is not specified, e.g. “`v1 += 2;`” may give an error. To fix, either cast appropriately, or include decimal points in double values, e.g. “`v1 += 2.0;`”.

### 9.1.4 Binary mathematical operators

The following operators perform element by element addition, subtraction, multiplication and division on the right hand side (rhs) vectors, returning the result for allocation in a new left hand side (lhs) vector:

- `+`: binary addition
- `-`: binary subtraction
- `*`: binary multiplication
- `/`: binary division

### Example

```
// Standard C-style arrays
double data1[ 4 ] = { 1.0, 2.0, 3.0, 4.0 },
       data2[ 4 ] = { 2, 3, 4, 5 },
       data3[ 3 ] = { 8, 9, 10 };

// vector's std array ctor
vector< double > v1( data1, data1 + 4 ),
                 v2( data2, data2 + 4 ), v3( data3, data3 + 3 ),
                 v4, v5;

// Binary addition element by element
v4 = v1 + v2;

// Binary addition with different size vectors
v5 = v3 + v2; // v5 will have 3 elements
```

### Notes

In the passed arguments to the overloaded operator, the left hand side (lhs) is *not* a reference, but a local copy of the lhs vector, and a local copy is returned. There is thus some inefficiency introduced in the use of binary vector operators. The vectors may have different numbers of elements, but like their unary counterparts, the right hand side (rhs) size *must be* greater than the size of the lhs vector. The final size of the returned vector will be the same as the lhs vector size.

### 9.1.5 Binary mathematical operators with scalars

The following operators perform scalar addition, subtraction, multiplication and division on a vector and a scalar, and return the result for allocation in a new vector:

- $+$ : binary scalar addition
- $-$ : binary scalar subtraction
- $*$ : binary scalar multiplication
- $/$ : binary scalar division

### Example

```
// Standard C-style arrays
double data1[ 4 ] = { 1.0, 2.0, 3.0, 4.0 };

// vector's std array ctor
```

```
vector< double > v1( data1, data1 + 4 ), v2;

// Binary scalar addition
v2 = v1 + 3.0;
```

### Notes

Type resolution may be a problem if the right hand side argument is not specified, e.g. “`v1 + = 2;`” may give an error. To fix, either cast appropriately, or include decimal points in double values, e.g. “`v1 + = 2.0;`”. In the passed arguments to the overloaded operator, the left hand side (lhs) is *not* a reference, but a local copy of the lhs vector, and a local copy is returned. There is thus some inefficiency introduced in the use of binary vector operators.

### 9.1.6 Overloaded `==` and `!=`

The boolean operators `==` and `!=` perform element by element comparisons between vectors, `==` returning *true* if they are equal, and `!=` returning *false* if they are not.

#### Example

```
vector< double > v1, v2;

// Code to load the vectors would follow...

if ( v1 == v2 )
    cout << "They're equal" << endl;
else
    cout << "They're not equal" << endl;
```

### 9.1.7 Overloaded `<<`

The ostream operator `<<` is overloaded for vector output.

#### Example

```
cout << "v1 = " << v1 << endl;
```

### Notes

This overloaded operator may also be used for vector of vector (etc.), and will insert double (triple, etc.) spaces between the output vectors.

### 9.1.8 Overloaded `>>`

The istream operator `>>` is overloaded for vector input.

### Example

```
vector< double > v1( 5 );
cout << "Enter a 5 element vector: ";
cin >> v1;
cout << "Your 5 element vector is " << v1 << endl;
```

### 9.1.9 Dot product

The *dotprod* method may be used to compute a scalar dotproduct from 2 vectors. Its signature is:

```
dotprod( vector& v1, vector& v2 )
```

### Example

```
x = dotprod( v1, v2 );
```

### Notes

$v_1$  and  $v_2$  *must* have the same number of elements, and `nvec::SizeMismatch` (section 9.1.1) is thrown if they do not. Note that this is *equality*, not the inequality the unary operators accept: a dot product taken over a prefix of the longer vector is a different scalar, and returning it silently is precisely what the check exists to prevent. Two *empty* vectors are accepted and give 0, the identity of a sum.

The method `sumSquaredDifference( vector& v1, vector& v2 )` returns  $\|v_1 - v_2\|^2$  under the same contract, without constructing the difference vector.

### 9.1.10 Function passing

The following methods allow functions to be applied to a vector element by element:

- `func( vector& vin, function f, vector& vout )`: applies function *f* element by element to vector  $v_{in}$ , populating vector  $v_{out}$  with the result.
- `func( vector& vin, function f, vector& vout, unsigned begin, unsigned end )`: applies function *f* element by element to vector  $v_{in}$  in the range  $v_{in}[begin]$  to  $v_{in}[end]$ , populating vector  $v_{out}$  in the range  $v_{out}[begin]$  to  $v_{out}[end]$  with the result.
- `func( vector& v, function f )`: applies function *f* element by element to vector  $v$ , and returns a new vector object.
- `func( vector& v, function f, unsigned begin, unsigned end )`: applies function *f* element by element to vector  $v$  in the range  $v[begin]$  to  $v[end]$ , and returns a new vector object of size  $v[begin] - v[end] + 1$ .

### Example

```
\index{Transfer function!Sigmoidal}

#include <vector>
#include <math.h>

using namespace std;

#include "vector_ops.h"

// Define sigmoidal function as a class which extends
// unary_function. This is necessary for algorithm::transform.
class sigmoidal : public unary_function< double, double >
{
public:
    inline double operator()( const double& x )
    {
        return 1.0 / ( 1.0 + exp( -x ) );
    }
};

int main() {
    // Standard C-style arrays
    double data1[ 4 ] = { 1.0, 2.0, 3.0, 4.0 },
           data2[ 4 ] = { 2, 3, 4, 5 };

    // vector's std array ctor
    vector< double > v1( data1, data1 + 4 ),
                     v2( data1, data1 + 4 ), v3( 4 ), v4;

    // Apply sigmoidal function to each element of v1
    // Populate v3 with the result
    func( v1, sigmoidal(), v3 );

    // Apply sigmoidal function to v1 with itself as result
    func( v1, sigmoidal(), v1 );

    // Apply sigmoidal function to each element of v2
    // After this function call, v4 will contain the result
    v4 = func( v2, sigmoidal() );

    // Copy vector v1 into a new vector v5
    vector< double > v5( v1 );

    // Apply the sigmoidal function to elements 1 through 2
```

```

// in vector v1, with the result placed in vector v5
// elements 1 through 2
func( v1, sigmoidal(), v5, 1, 2 );

// The receiving vector may be smaller than the sending
// vector
vector< double > v6( 3 );
func( v1, sigmoidal(), v6, 0, 2 ); // v1 has 4 elements
                                   // v6 has 3 elements

// Make a new vector
vector< double > v7;

// And put the sigmoidal function applied to vector v1
// elements 1 through 2 into it
v7 = func( v1, sigmoidal, 1, 2 ); // v7 has 2 elements

return 0;
}

```

## Notes

The passed function *must* be defined as a class extension of unary\_function as demonstrated in the example. Common functions useful for neural computation (e.g. sigmoidal) are packaged with neURon2++.

If a vector to be populated is passed without bounds arguments, it *must have* the same number of elements as the input vector. If a vector to be populated is passed with bounds arguments, the size of the vector *must be* greater than the upper bound. As func( vector& v, function f ) and func( vector& v, function f, unsigned begin, unsigned end ) must construct new vectors, they are less efficient than func( vector& v<sub>in</sub>, function f, vector& v<sub>out</sub> ) and func( vector& v<sub>in</sub>, function f, vector& v<sub>out</sub>, unsigned begin, unsigned end ). func( vector& v<sub>in</sub>, function f, vector& v<sub>out</sub> ) and func( vector& v<sub>in</sub>, function f, vector& v<sub>out</sub>, unsigned begin, unsigned end ) are preferred.

To use a class extension of unary\_function with a scalar, the following may be written (using sigmoidal as an example):

```

double x, y;
y = sigmoidal()( x );

```

is valid if sigmoidal is defined as described above.

## Notes

The two forms taking v<sub>out</sub> write into a vector the caller supplies; neither ever resizes it, because silently growing a destination would conceal a defect in the

shape of the model that supplied it. Their preconditions, and the exceptions of section 9.1.1:

- `func(  $v_{in}$ ,  $f$ ,  $v_{out}$  )`:  $v_{in}$  and  $v_{out}$  *must have* the same number of elements, and `nvec::SizeMismatch` is thrown if they do not. This is equality rather than the inequality the unary operators take, deliberately: the operation writes  $v_{in}.size()$  elements into  $v_{out}$ , so a longer input runs off the end of the destination. A caller that wants only part of a vector says which part, through the ranged form below.
- `func(  $v_{in}$ ,  $f$ ,  $v_{out}$ ,  $begin$ ,  $end$  )`:  $begin$  *must not exceed*  $end$ , and  $end$  *must be within both*  $v_{in}$  and  $v_{out}$ . `nvec::RangeViolation` is thrown otherwise.
- `func(  $v$ ,  $f$ ,  $begin$ ,  $end$  )`:  $begin$  *must not exceed*  $end$ , and  $end$  *must be within*  $v$ ; the returned vector is allocated at the correct size and cannot be overrun. `nvec::RangeViolation` is thrown otherwise.

#### 9.1.11 Calculating the sum of squares of a vector elements

The method `squared( vector&  $v$  )` returns the value of the sum of squares of vector  $v$  elements.

##### Example

```
vector< double > v; // v is then populated with double values...
double sum;
sum = squared( v );
```

#### 9.1.12 Calculating the maximum absolute value of vector elements

The method `maxabs( vector&  $v$  )` returns the maximum absolute value of vector  $v$  elements.

##### Example

```
vector< double > v; // v is then populated with double values...
double max;
max = maxabs( v );
```

##### Notes

$v$  *must not be empty*, and `nvec::EmptyVector` (section 9.1.1) is thrown if it is. The maximum of an empty set has no value, and returning 0 would be acted upon: this method supplies the largest absolute gradient to the training stopping rule, where a value of 0 means *converged*.

### 9.1.13 Calculating the minimum absolute value of vector elements

The method `minabs( vector& v )` returns the minimum absolute value of vector *v* elements.

#### Example

```
vector< double > v; // v is then populated with double values...
double min;
min = minabs( v );
```

#### Notes

*v must not be empty*, and `nvec::EmptyVector` (section 9.1.1) is thrown if it is.

### 9.1.14 Randomization

#### namespace nvec

The method `random( vector& v, double n )` populates a double precision vector *v* with double precision random numbers between  $-n$  and  $+n$ .

#### Example

```
vector< double > v( 6 ); // Make a 6 element vector
// Populate it with random numbers between -0.5 and +0.5,
// and print it out
cout << "The random vector is " << nvec::random( v, 0.5 );
```

#### Notes

This method is only coded for vectors of type double, e.g. `vector< double >`.

### 9.1.15 Random positions

#### namespace nvec

The method `random_positions( vector& v )` populates an unsigned vector *v* with a series of unique integers from 0 to  $n - 1$ , where *n* is the number of elements in the vector.

#### Example

The example code:

```
vector< unsigned > v( 6 );
cout << nvec::random_positions( v ) << endl;
```

May yield:

```
5 2 1 3 4 0
```

## Notes

For obvious reasons, this method is only coded for vectors of type unsigned, e.g. `vector< unsigned >`.

### 9.1.16 Converting a vector of vector into a vector

The following methods allow conversion of a vector of vector into a vector, thus "flattening" the vector of vector:

- `flatten( vector< vector >& container, vector& v )` converts the data structure vector of vector *container* to passed vector *v*, returns a reference to vector *v*.
- `flatten( vector< vector >& container )` converts the data structure vector of vector *container* to a new vector, returns a reference to the new vector.

## Example

See section [10.9](#) for method *variable\_parse*.

```
// Create a vector of vector using util::variable_parse
string parse_test = "0-3,5; 4,8; 6,7";
vector< vector< unsigned > > parse_numbers;
parse_numbers = util::variable_parse( parse_test );
cout << "Here it is: " << parse_numbers << endl;

// Now flatten it
cout << endl << "Flattening:" << endl;
vector< unsigned > v_flat;
flatten( parse_numbers, v_flat );
cout << v_flat << endl;
cout << flatten( parse_numbers ) << endl;
```

### 9.1.17 Binning a vector into a vector of vector

- `bin( const vector& v, unsigned b, bool binFlag, vector< vector >& vv )`: partitions vector *v* into the data structure vector of vector *vv* according to specified bin size, with flag *binFlag* indicating number in the last bin:
  - **false** if the last bin contains  $\leq$  bin size (last bin = remaining elements,
  - **true** if the last bin contains  $\geq$  bin size (remaining elements are added to the last bin.
- `bin( const vector& v, unsigned b, bool binFlag )`: partitions vector *v* into a new data structure vector of vector according to specified bin size, with flag *binFlag* indicating number in the last bin:

- **false** if the last bin contains  $\leq$  bin size (last bin = remaining elements,
- **true** if the last bin contains  $\geq$  bin size (remaining elements are added to the last bin.

and returns a reference to the new data structure vector of vector.

### Example

```
double vpart_[ 10 ] = { 0.0, 1.0, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0, 8.0,
9.0 };
vector< double > vpart( vpart_, vpart_ + 10 );
vector< vector< double > > vbin;

bin( vpart, 3, false, vbin );
cout << vpart << endl << "partitioned with small last bin =" << endl <<
< vbin;
cout << endl << "partitioned with large last bin =" << endl <<
    bin( vpart, 3, true ) << endl;
```

Produces the following output:

```
0 1 2 3 4 5 6 7 8 9
partitioned with small last bin =
0 1 2 3 4 5 6 7 8 9
partitioned with large last bin =
0 1 2 3 4 5 6 7 8 9
```

## 9.2 matrix.h and the Matrix class

Including the header file matrix.h allows implementation of the Matrix class, and allows simplified numerical operations on 2 dimensional arrays.

### 9.2.1 Constructors

The following constructors are implemented in the Matrix class:

- Matrix(): Constructs a Matrix object of zero size, with zero rows and zero columns.
- Matrix( unsigned  $r$ , unsigned  $c$  ): Constructs a Matrix object with  $r$  rows and  $c$  columns.
- Matrix( unsigned  $r$ , unsigned  $c$ , *value* ): Constructs a Matrix object with  $r$  rows,  $c$  columns, and fills it with *value*.
- Matrix( vector&  $q$ , vector&  $p^t$  ): Constructs a Matrix object which is the outer product of vectors  $q$  and the transpose of  $p$  (see 9.2.17).

- `Matrix( string& filename, unsigned c )`: Constructs a Matrix object, and loads it from a file named “*filename*” with *c* columns.
- `Matrix( bool report, string& filename )`: Constructs a Matrix object, and loads it from a file named “*filename*”. Delimiters are any whitespace character, or any character that is non-numeric (+, -, e and E are considered numeric, as is a decimal point.) Rows are delimited by carriage returns in the file, and the number of columns is the maximum number of elements in any one row. Elements in unpopulated rows will be filled with 0, beginning from the *right* hand side of the row. If the bool flag *report* is true, a report will be printed indicating which rows were found to be unpopulated.

### Examples

```
// Make some Matrices
Matrix< double > A( 4, 3 ), B, C( 5, 4, 0.01 );

// Get a file name
string filename;
cout << "The file name is ";
cin >> filename;

// Make a Matrix from the file
Matrix< double > C( filename, 3 ), // It is known to have 3 columns
                  D( true, filename ); // Number of columns unknown,
                                      // report will be printed

// Make a matrix from the outer product of vectors v1 and v2
Matrix< double > E( v1, v2 );
```

### Notes

A Matrix constructed by *Matrix( r, c )* is **not** initially populated. Consider it to be filled with garbage.

### 9.2.2 Accessors

Given a Matrix object *M*, the following accessors are implemented in the Matrix class:

- `M( unsigned r, unsigned c )`: overloaded () to access the element at row *r* and column *c* in Matrix *matrix\_object*
- `M.rows()`: returns number of rows in Matrix
- `M.cols()`: returns number of columns in Matrix

## Examples

```
// Make a Matrix with 4 rows and 3 columns
Matrix< double > A( 4, 3 );

// Fill Matrix A with numbers
for ( i = 0; i < 4; i++ )
    for ( j = 0; j < 3; j++ )
        A( i, j ) = 3 * i + j + 1;

// Print out number of rows and number of columns in Matrix
cout << "Matrix has " << A.rows() << " rows and ";
cout << A.cols() << " columns." << endl;
```

### 9.2.3 Submatrix accessors

The following accessors return a submatrix from a Matrix:

- `M.submatrix( unsigned row1, unsigned rowN, unsigned col1, unsigned colN, Matrix& Mout )`: populates passed matrix *M<sub>out</sub>* with Matrix *M* rows *row<sub>1</sub>* to *row<sub>N</sub>*, and columns *col<sub>1</sub>* to *col<sub>N</sub>*.
- `M.submatrix( unsigned row1, unsigned rowN, unsigned col1, unsigned colN )`: returns a new Matrix object with Matrix *M* rows *row<sub>1</sub>* to *row<sub>N</sub>*, and columns *col<sub>1</sub>* to *col<sub>N</sub>*.

## Examples

```
// Make some matrices
Matrix< double > P( 5, 6, 2.5 ), R( 2, 4 ), S;

// Populate matrix R with the resulting submatrix
P.submatrix( 3, 4, 2, 5, R );

// Construct a new matrix for result
S = P.submatrix( 3, 4, 2, 5 );
```

## Notes

Row and column dimensions must be within the bounds of the originating matrix. If a matrix is passed as an argument, its number of columns *must equal* *col<sub>N</sub> - col<sub>1</sub> + 1*, and its number of rows *must equal* *row<sub>N</sub> - row<sub>1</sub> + 1*. A passed matrix to be populated is more efficient than constructing a new matrix for the result.

### 9.2.4 Row and column accessors

The following accessors return a vector from a row or column of a Matrix:

- `M.row( unsigned  $r$ , vector&  $v$  )`: populates vector  $v$  with row  $r$  from Matrix  $M$ .
- `M.row( unsigned  $r$  )`: returns a new vector with row  $r$  from Matrix  $M$ .
- `M.col( unsigned  $c$ , vector&  $v$  )`: populates vector  $v$  with column  $c$  from Matrix  $M$ .
- `M.col( unsigned  $c$  )`: returns a new vector with column  $c$  from Matrix  $M$ .

#### Examples

```
// Make a matrix with 5 rows and 6 columns, filled with Pi
Matrix< double > P( 5, 6, 3.14159 );

// Make some vectors
vector< double > v5( 5 ), v6( 6 ), v7, v8;

// Put row 2 of Matrix P in vector v6
P.row( 2, v6 );

// Construct a new vector v7 for row 2 of Matrix P
v7 = P.row( 2 );

// Put column 2 of Matrix P in vector v5
P.col( 2, v5 );

// Construct a new vector v8 for column 2 of Matrix P
v8 = P.col( 2 );
```

#### Notes

For the methods which take passed vectors as arguments, the size of the vector  $v$  *must equal* the number of Matrix columns for `M.row(  $r$ ,  $v$  )`, and the size of the vector  $v$  *must equal* the number of Matrix rows for `M.col(  $c$ ,  $v$  )`. The methods which take passed vectors as arguments are more efficient than the methods which construct vectors.

### 9.2.5 Methods which append vectors to Matrices

The following methods append a vector to a Matrix:

- `M.addrow( vector&  $v$ , Matrix  $N$  )`: populates Matrix  $N$  by appending vector  $v$  to the *end* of Matrix  $M$  as a *row*.

- `M.addrow( vector& v )`: returns a new Matrix with vector  $v$  appended to the *end* of Matrix  $M$ .
- `M.addcol( vector& v, Matrix N )`: populates Matrix  $N$  by appending vector  $v$  to the *right* of Matrix  $M$  as a *column*.
- `M.addrow( vector& v )`: returns a new Matrix with vector  $v$  appended to the *right* of Matrix  $M$ .

### Examples

```
// Matrix A has 3 rows and 4 columns, vector v1 has 4 elements
// Matrix B has 4 rows and 4 columns
A.addrow( v1, B );
C = A.addrow( v1 );

// Matrix D has 3 rows and 4 columns, vector v2 has 3 elements
// Matrix E has 3 rows and 5 columns
D.addrow( v2, E );
F = D.addrow( v2 );
```

### Notes

The number of columns in  $M$  and  $N$  *must* be the same as the number of elements of vector  $v$  for `M.addrow`, and the number of rows in  $M$  and  $N$  *must* be the same as the number of elements of vector  $v$  for `M.addcol`.

Because they construct new Matrices, `M.addrow( vector& v )` and `M.addcol( vector& v )` are less efficient than `M.addrow( vector& v, Matrix N )` and `M.addcol( vector& v, Matrix N )`. `M.addrow( vector& v, Matrix N )` and `M.addcol( vector& v, Matrix N )` are to be preferred.

`M.addcol` is computationally expensive, and its use should be minimized.

## 9.2.6 Methods which replace a row or column of a Matrix with a vector

The following methods replace a row or column of a Matrix with a vector:

- `M.replacrow( unsigned n, vector& v )`: replaces row  $n$  in Matrix  $M$  with vector  $v$ , returning a reference to Matrix  $M$  with the result.
- `M.replacecol( unsigned n, vector& v )`: replaces column  $n$  in Matrix  $M$  with vector  $v$ , returning a reference to Matrix  $M$  with the result.

### Examples

```
// Replace row 3 of Matrix A with vector v1, and print the result
cout << A.replacrow( 3, v1 );
```

```
// Replace column 2 of Matrix A with vector v2, and print the result
cout << A.replacecol( 2, v2 );
```

## Notes

The number of columns in  $M$  *must* be the same as the number of elements of vector  $v$  for  $M.replacecol$ , and the number of rows in  $M$  *must* be the same as the number of elements of vector  $v$  for  $M.replacecol$ .

### 9.2.7 Column inclusion and exclusion

The following operators include or exclude those columns indicated by the passed vector, which includes the (unsigned) column values:

- $M.includecols( Matrix\& M_{out}, vector< unsigned >\& v )$ : includes in the returned Matrix  $M_{out}$  only those columns from Matrix  $M$  indicated by  $v$ , returning a reference to Matrix  $M_{out}$  with the result.
- $M.includecols( vector< unsigned >\& v )$ : returns a new Matrix *with* only those columns from Matrix  $M$  indicated by  $v$
- $M.excludecols( Matrix\& M_{out}, vector< unsigned >\& v )$ : excludes in the returned Matrix  $M_{out}$  those columns from Matrix  $M$  indicated by  $v$ , returning a reference to Matrix  $M_{out}$  with the result.
- $M.excludecols( vector< unsigned >\& v )$ : returns a new Matrix *without* those columns from Matrix  $M$  indicated by  $v$

## Examples

```
// Make a 5x4 Matrix, fill with some numbers
Matrix< double > M( 5, 4 );
for ( unsigned row = 0; row < 5; row++ )
    for ( unsigned column = 0; column < 4; column++ )
        M( row, column ) = 4 * column + row;
cout << "Here is the original Matrix:" << M;
```

```
// Make some smaller Matrices
Matrix< double > M1( 5, 2 ), M2, M3( 5, 2 ), M4;
```

```
// Make a vector to indicate columns 1 & 3
vector< unsigned > v;
v.push_back( 1 );
v.push_back( 3 );
```

```
// Passing a Matrix to include columns
M.includecols( M1, v );
cout << "From the matrix above, including columns "
```

```

    << v << ": " << M1;

// A new Matrix to include columns
M2 = M.includecols( v );
cout << "Returning a new Matrix: " << M2;

// Passing a Matrix to exclude columns
M.excludecols( M3, v );
cout << "From the matrix above, excluding columns "
    << v << ": " << M3;

// A new Matrix to exclude columns
M4 = M.excludecols( v );
cout << "Returning a new Matrix: " << M4;

```

## Notes

If the Matrix  $M_{out}$  to be populated is passed as an argument, its rows *must* be the same as the number of rows in  $M$ , and its number of columns *must* be the same as the difference between the number of columns in  $M$  and the number of elements in the passed vector  $v$ . The elements in the passed vector  $v$  *must* be *unique* (no duplicated values), and no element can be  $\geq$  to the number of columns in Matrix  $M$ .

These methods are useful to include or exclude variables or nodes represented by columns in a Matrix.

## 9.2.8 Boolean equality operators

The following operators perform boolean equality and inequality comparisons, element by element:

- `==`: equality
- `!=`: inequality

### Example

```

// Make 2 Matrices with 4 rows and 3 columns
Matrix< double > A( 4, 3 ), B( 4, 3 );

// Fill Matrices A and B with numbers

for ( i = 0; i < 4; i++ )
{
    for ( j = 0; j < 3; j++ )
    {
        A( i, j ) = 3 * i + j;
    }
}

```

```

        B( i, j ) = 3 * i + j + 1;
    }
}

// Compare Matrix A to B:
if ( A == B )
    cout << "They're equal" << endl;
if ( A != B )
    cout << "They're not equal" << endl;

```

### 9.2.9 Unary mathematical operators

The following operators perform element by element addition, subtraction, multiplication and division on the right hand side (rhs) Matrix and left hand side (lhs) Matrix, returning the result in the left hand side (lhs) Matrix:

- +=: unary addition
- -=: unary subtraction
- \*=: unary multiplication
- /=: unary division

#### Example

```

// Make 2 Matrices with 4 rows and 3 columns
Matrix< double > A( 4, 3 ), B( 4, 3 );

// Fill Matrices A and B with numbers

for ( i = 0; i < 4; i++ )
{
    for ( j = 0; j < 3; j++ )
    {
        A( i, j ) = 3 * i + j;
        B( i, j ) = 3 * i + j + 1;
    }
}

// Add matrix B to A element by element
A += B;

```

### 9.2.10 Unary mathematical operators with scalars

The following operators perform scalar addition, subtraction, multiplication and division on a right hand side (rhs) scalar and left hand side (lhs) Matrix, returning the result in the left hand side (lhs) Matrix:

- `+=`: unary scalar addition
- `-=`: unary scalar subtraction
- `*=`: unary scalar multiplication
- `/=`: unary scalar division

### Example

```
// Make a Matrix with 4 rows and 3 columns, filled with 1.0
Matrix< double > A( 4, 3, 1.0 );

// Add 3.0 to each element of A
A += 3.0;
```

## 9.2.11 Binary mathematical operators

The following operators perform element by element addition, subtraction, multiplication and division on the right hand side (rhs) Matrices, returning a new Matrix object:

- `+`: binary addition
- `-`: binary subtraction
- `*`: binary multiplication
- `/`: binary division

### Example

```
// Make 2 Matrices with 4 rows and 3 columns, and a 3rd empty Matrix
Matrix< double > A( 4, 3 ), B( 4, 3 ), C;

// Fill Matrices A and B with numbers
for ( i = 0; i < 4; i++ )
{
    for ( j = 0; j < 3; j++ )
    {
        A( i, j ) = 3 * i + j;
        B( i, j ) = 3 * i + j + 1;
    }
}

// Add matrix A and B element by element, the result is matrix C
C = A + B;
```

## Notes

The Matrices *must have* the same number of rows and columns. As a new matrix object must be constructed, unary operations are preferred for efficiency.

### 9.2.12 Binary mathematical operators with scalars

The following operators perform scalar addition, subtraction, multiplication and division on a Matrix, returning a new Matrix object:

- `+`: binary scalar addition
- `-`: binary scalar subtraction
- `*`: binary scalar multiplication
- `/`: binary scalar division

#### Example

```
// Make some Matrices
Matrix< double > A( 4, 3, 1.0 ), B;

// Add 3.0 to each element of A, B is the result
B = A + 3.0;
```

## Notes

As a new matrix object must be constructed, unary operations are preferred for efficiency. **Known bug: the scalar must be on the right side of the Matrix for all operators.**

### 9.2.13 Transpose

The following operators transpose a Matrix:

- `M.t()` transposes matrix  $M$ , and returns a matrix object
- `M.t( O )` populates passed matrix  $O$  with the transpose of  $M$

#### Example

```
// A is a 3 x 4 matrix
Matrix< double > B( 4, 3 ), C;

A.t( B ); // B now contains the transpose of A
C = A.t(); // C now contains the transpose of A
```

## Notes

For  $M.t(O)$ , the number of columns in  $M$  *must equal* the number of rows in  $O$  and the number of rows in  $M$  *must equal* the number of columns in  $O$ .

Because  $M.t()$  must construct a new matrix object, it is less efficient than  $M.t(O)$ .

### 9.2.14 Overloaded <<

The ostream operator << is overloaded for Matrix output.

#### Example

```
cout << "Matrix A = " << A << endl;
```

#### setHeader

The setHeader method:

- $M.setHeader(\text{bool } flag)$

is provided for ostream output. If  $flag$  is true, a header line containing the format:

$r$  rows and  $c$  columns:

is output to the stream prior to the Matrix, whereas if  $flag$  is false, the header line is not output, only the Matrix. *By default, this flag is set to **true**.*

#### Examples

```
cout << "Matrix A alone = " << A.setHeader( false ) << endl;
A.setHeader( true );
cout << "Matrix A with header = " << A << endl;
```

### 9.2.15 Overloaded >>

The istream operator >> is overloaded for Matrix input.

#### Example

```
Matrix< double > A( 3, 2 );
cout << "Enter Matrix A: ";
cin >> A;
```

### 9.2.16 Dot product

Given a matrix object  $M$ , and vectors  $\vec{v}$  and  $\vec{o}$ , where  $M \cdot \vec{v} = \vec{o}$ , e.g.

$$\begin{bmatrix} M_{11} & M_{12} & \cdots & M_{1c} \\ M_{21} & M_{22} & \cdots & M_{2c} \\ \vdots & \vdots & \ddots & \vdots \\ M_{r1} & M_{r2} & \cdots & M_{rc} \end{bmatrix} \cdot \begin{bmatrix} \vec{v}_1 \\ \vec{v}_2 \\ \vdots \\ \vec{v}_c \end{bmatrix} = \begin{bmatrix} \vec{o}_1 \\ \vec{o}_2 \\ \vdots \\ \vec{o}_r \end{bmatrix}$$

or other matrix objects  $N$  and  $O$ , where  $M \cdot N = O$ , the following dot product operators are available in the Matrix class:

- $M.\text{dotprod}(\text{vector\& } v)$  returns a new vector object which is the dot product of the matrix and the passed vector
- $M.\text{dotprod}(\text{vector\& } v_{in}, \text{vector\& } v_{out})$  which populates the output vector  $v_{out}$  with the dot product of the matrix and the passed vector  $v_{in}$
- $M.\text{dotprod}(\text{vector\& } v_{in}, \text{vector\& } v_{out}, \text{unsigned } begin, \text{unsigned } end)$  which populates the output vector range  $v_{out}[begin]$  to  $v_{out}[end]$  with the dot product of the matrix and the passed vector  $v_{in}$
- $M.\text{dotprodt}(\text{vector\& } v)$  returns a new vector object which is the dot product of the *transpose* of the matrix and the passed vector
- $M.\text{dotprodt}(\text{vector\& } v_{in}, \text{vector\& } v_{out})$  which populates the output vector  $v_{out}$  with the dot product of the *transpose* of the matrix and the passed vector  $v_{in}$
- $M.\text{dotprodt}(\text{vector\& } v_{in}, \text{vector\& } v_{out}, \text{unsigned } begin, \text{unsigned } end)$  which populates the output vector range  $v_{out}[begin]$  to  $v_{out}[end]$  with the dot product of the *transpose* of the matrix and the passed vector  $v_{in}$
- $M.\text{dotprod}(N)$  returns a new Matrix object which is the dot product of the matrix  $M$  and the passed matrix  $N$
- $M.\text{dotprod}(N, O)$  which populates the passed matrix  $O$  with the dot product of the matrix  $M$  and the passed matrix  $N$

#### Examples

(`dotprodt(...)` follows the same form as `dotprod(...)`.)

```
// Matrix A has 4 rows and 3 columns, vector v1 has 3 elements
vector< double > v2;
v2 = A.dotprod( v1 ); // v1 is unchanged
                      // v2 now contains the dot product

// vector v3 has 4 elements
```

```

A.dotprod( v1, v3 ); // v1 is unchanged
                      // v3 now contains the dot product

// vector v4 has 5 elements
A.dotprod( v1, v4, 0, 3 ); // v4[0] through v4[3] now
                          // contains the dot product

// Matrix B has 3 rows and 5 columns
Matrix< double >C ( 4, 5 ), D;
A.dotprod( B, C ); // C now contains the dot product
D = A.dotprod( B ); // D now contains the dot product

```

## Notes

For  $M.\text{dotprod}( v )$ , the number of columns in  $M$  *must equal* the number of elements in  $v$ . For  $M.\text{dotprod}( v_{in}, v_{out} )$ , the number of columns in  $M$  *must equal* the number of elements in  $v_{in}$ , and the number of rows in  $M$  *must be equal to or greater than* the number of elements in  $v_{out}$ . If the number of rows in  $M$  is greater than the number of elements in  $v_{out}$ , the resulting vector will be truncated to the size of  $v_{out}$ , and the remainder ignored.

For  $M.\text{dotprodt}( v )$ , the number of rows in  $M$  *must equal* the number of elements in  $v$ . For  $M.\text{dotprodt}( v_{in}, v_{out} )$ , the number of rows in  $M$  *must equal* the number of elements in  $v_{in}$ , and the number of columns in  $M$  *must be equal to or greater than* the number of elements in  $v_{out}$ . If the number of columns in  $M$  is greater than the number of elements in  $v_{out}$ , the resulting vector will be truncated to the size of  $v_{out}$ , and the remainder ignored.

For  $M.\text{dotprod}( N )$ , the number of columns in  $M$  *must equal* the number of rows in  $N$ . For  $M.\text{dotprod}( N, O )$ , the number of rows in  $M$  *must equal* the number of rows in  $O$ , the number of columns in  $N$  *must equal* the number of columns in  $O$ , and the number of columns in  $M$  *must equal* the number of rows in  $N$ .

Because  $M.\text{dotprod}( v )$  and  $M.\text{dotprodt}( v )$  must construct new vector objects, and  $M.\text{dotprod}( N )$  must construct a new matrix object, these methods are less efficient than  $M.\text{dotprod}( v_{in}, v_{out} )$ ,  $M.\text{dotprodt}( v_{in}, v_{out} )$  and  $M.\text{dotprod}( N, O )$ .

## dotprod\_row

The *dotprod\_row* method takes a single row from a Matrix, and transposes it to be used as the right hand side argument for a dot product. It is useful for extracting a single row from a dataset and passing it to the first layer of a neural network. It is functionally equivalent to  $M \cdot \vec{d}^t = \vec{o}$ , where  $M$  is the matrix,  $\vec{d}^t$  is the transpose of the vector  $\vec{d}$  derived from a single row of the dataset matrix, and  $\vec{o}$  is the output vector.

Suppose that  $d$  is a single row of the dataset matrix  $D$ , e.g.:

$$\begin{bmatrix} \cdots & \cdots & \cdots & \cdots \\ d_{r1} & d_{r2} & \cdots & d_{rc} \\ \vdots & \vdots & \vdots & \vdots \\ \cdots & \cdots & \cdots & \cdots \end{bmatrix}$$

Then  $M.\text{dotprod\_row}( D, r, o )$  is equivalent to:

$$\begin{bmatrix} M_{11} & M_{12} & \cdots & M_{1c} \\ M_{21} & M_{22} & \cdots & M_{2c} \\ \vdots & \vdots & \vdots & \vdots \\ M_{r1} & M_{r2} & \cdots & M_{rc} \end{bmatrix} \cdot \begin{bmatrix} \vec{d_{r1}} \\ \vec{d_{r2}} \\ \vdots \\ \vec{d_{rc}} \end{bmatrix} = \begin{bmatrix} \vec{o_1} \\ \vec{o_2} \\ \vdots \\ \vec{o_r} \end{bmatrix}$$

where  $r$  is the row number. The following `dotprod_row` operators are available in the `Matrix` class:

- $M.\text{dotprod\_row}( \text{Matrix\& } D, \text{ unsigned } r, \text{ vector\& } v )$  which populates the passed vector  $v$  with the dot product of Matrix  $M$  and the transposed row  $r$  in matrix  $D$
- $M.\text{dotprod\_row}( \text{Matrix\& } D, \text{ unsigned } r )$  which returns a new `Matrix` object with the dot product of Matrix  $M$  and the transposed row  $r$  in matrix  $D$
- $M.\text{dotprod\_row}( \text{Matrix\& } D, \text{ unsigned } r, \text{ vector\& } v, \text{ unsigned } begin, \text{ unsigned } end )$  which populates the range in the passed vector  $v[begin]$  to  $v[end]$  with the dot product of Matrix  $M$  and the transposed row  $r$  in matrix  $D$

### Examples

```
// Matrix A and D both have 3 columns, vector v1 has 3 elements
A.dotprod_row( D, 3, v1 ); // v1 now contains the dot product
                             // of A and the transpose of D's 3rd row

vector< double > v2;
v2 = A.dotprod_row( D, 3 ); // v2 now contains the dot product
                             // of A and the transpose of D's 3rd row

// Matrix A has 4 rows, vector v3 has 5 elements
A.dotprod_row( D, 3, v3, 0, 3 ); // v3[0] through v3[3] now
                                   // contains the dot product of A
                                   // and the transpose of D's 3rd row
```

## Notes

Both matrices *must have* the same number of columns. If a vector is passed, it *must have* the same number of elements as the matrices have columns. Because `M.dotprod_row( D, r )` must construct a new vector object, it is less efficient than `M.dotprod_row( D, r, v )`.

### 9.2.17 Outer product

The `outprod` method is available in the Matrix class to populate a Matrix with the outer product of 2 vectors. For example, given 2 vectors  $\vec{p}$  and  $\vec{q}$ , e.g.:

$$\vec{q} = \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_m \end{bmatrix}, \vec{p} = \begin{bmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{bmatrix}$$

The outer product  $\vec{q}\vec{p}^t$  is:

$$\vec{q}\vec{p}^t = \begin{bmatrix} q_1p_1 & q_1p_2 & \cdots & q_1p_n \\ q_2p_1 & q_2p_2 & \cdots & q_2p_n \\ \vdots & \vdots & \vdots & \vdots \\ q_mp_1 & q_mp_2 & \cdots & q_mp_n \end{bmatrix}$$

The outer product method is called as in:

`M.outprod( vector& q, vector& p )`

## Example

```
// Make some vectors
double array1[ 4 ] = { 0, 1, 2, 3 },
      array2[ 5 ] = { 0, 1, 2, 3, 4 };
vector< double > vec1( array1, array1 + 4 ),
      vec2( array2, array2 + 5 );

// Make the right size Matrix
Matrix< double > U( 4, 5 );

// Populate the Matrix with the outer product of the vectors
U.outprod( vec1, vec2 );
```

## Notes

The number of elements in the first vector argument *must equal* the number of Matrix rows, and the number of elements in the second vector argument *must equal* the number of Matrix columns.

### 9.2.18 Function passing

The following methods allow functions to be applied to a Matrix element by element:

- `func( Matrix&  $M_{in}$ , function  $f$ , Matrix&  $M_{out}$  )`: applies function  $f$  element by element to Matrix  $M_{in}$ , populating Matrix  $M_{out}$  with the result.
- `func( Matrix&  $M$ , function  $f$  )`: applies function  $f$  element by element to Matrix  $M$ , and returns a new Matrix object.

#### Example

```
\index{Transfer function!Sigmoidal}

#include <vector>
#include <math.h>

using namespace std;

#include "vector_ops.h"

// Define sigmoidal function as a class which extends
// unary_function. This is necessary for algorithm::transform.
class sigmoidal : public unary_function< double, double >
{
public:
    inline double operator()( const double& x )
    {
        return 1.0 / ( 1.0 + exp( -x ) );
    }
};

int main() {
    // Make some Matrices
    Matrix < double > M1( 4, 3, 0.5 ), M2( 4, 3, 0.0 ),
        M3( 4, 3, 2.0 ), M4;

    // Apply sigmoidal function to each element of M1
    // Populate M2 with the result
    func( M1, sigmoidal(), M2 );

    // Apply sigmoidal function to a Matrix, with itself as result
    func( M3, sigmoidal(), M3 );

    // Apply sigmoidal function to each element of M1
    // After this function call, M4 will contain the result
```

```

    M4 = func( M1, sigmoidal() );

    return 0;
}

```

## Notes

The passed function *must* be defined as a class extension of unary\_function as demonstrated in the example. Common functions useful for neural computation (e.g. sigmoidal) are packaged with neUOn2++.

If a Matrix to be populated is passed, it *must have* the same number of rows and columns as the input Matrix. As func( Matrix& M, function f ) must construct a new vector, it is less efficient than func( Matrix& M<sub>in</sub>, function f, Matrix& M<sub>out</sub> ).

### 9.2.19 Randomization

The method M.random( double n ) populates a double precision Matrix M with double precision random numbers between -n and +n.

## Example

```

Matrix< double > Z( 6, 5 ); // Make a 6 x 5 Matrix
// Populate it with random numbers between -0.5 and +0.5,
// and print it out
cout << "The random matrix is " << Z.random( 0.5 );

```

## Notes

This method is only coded for Matrices of type double, e.g. Matrix< double >.

### 9.2.20 Loading a Matrix from a file

The following methods will load a matrix from a file:

- M.loadfile( string& filename, unsigned c ): loads Matrix M from a file named “filename” with c columns. The Matrix will be resized if necessary, and if the file does not exist, the user will be queried until a good filename is entered.
- M.loadfile( bool report, string& filename ): loads Matrix M from a file named “filename”. Delimiters are any whitespace character, or any character that is non-numeric (+, -, e and E are considered numeric, as is a decimal point.) Rows are delimited by carriage returns in the file, and the number of columns is the maximum number of elements in any one row. Elements in unpopulated rows will be filled with 0, beginning from the *right* hand side of the row. If the bool flag *report* is true, a report will be printed indicating which rows were found to be unpopulated. The

Matrix will be resized if necessary, and if the file does not exist, the user will be queried until a good filename is entered.

### Examples

```
Matrix< double > A, B;
string filename;
unsigned cols;

cout << "What is the name of the data file? ";
cin >> filename;
cout << "How many columns does it have? ";
cin >> cols;

A.loadfile( filename, cols );

cout << "What is the name of another data file? ";
cin >> filename;

B.loadfile( true, filename ); // a report will be printed
```

### 9.2.21 Saving a Matrix to a file

The following method will save a matrix to a file:

- `bool M.savefile( string& filename )`: saves Matrix  $M$  to a file named "*filename*", and returns a Boolean flag indicating whether the save operation was successful.

### Example

```
Matrix< double > A( 5, 4, 3.14159 ); // fill a 5 x 4 Matrix with pi
A.savefile( "pi.txt" ); // and save it to a file named pi.txt
```

### 9.2.22 Resizing a Matrix

The method `M.resize( unsigned  $r$ , unsigned  $c$  )` resizes Matrix  $M$  to  $r$  rows and  $c$  columns.

### Example

```
A.resize( 5, 4 ); // resize Matrix A to 5 rows and 4 columns
```

### Notes

The following Matrix so resized is filled with **garbage**.

### 9.2.23 Clearing a Matrix

The method `M.clear()` clears Matrix `M`, setting its dimensions and data to zero.

#### Example

```
A.clear(); // clear Matrix A
```

### 9.2.24 Populating a Matrix with a value

The method `M.fill( v )` populates Matrix `M` with value `v` of its type.

#### Example

```
Matrix< double > A( 3, 2 ); // Make a Matrix of type double with
                           // 3 rows and 2 columns

A.fill( 3.14159 ); // and fill it with pi
```

### 9.2.25 Calculating the sums of columns of a Matrix

The method `M.colsums()` returns a new vector with each element the sum of the respective column in `M`. The method `M.colsums( vector& v )` returns the result in the passed vector `v`.

#### Example

```
#include <iostream>
#include <string>
#include "matrix.h"
#include "vector_ops.h"

string filename;
cout << "The file name is ";
cin >> filename;
Matrix< double > M( true, filename ); // load Matrix from file
vector< double > v;
v = M.colsums(); // v will contain the sum of the columns in M
cout << "Sum of columns = " << v << endl;
```

#### Note

If the vector to be returned is passed as an argument, its size *must* be the same as the number of columns in the Matrix.

### 9.2.26 Calculating the sum of squares of a Matrix elements

The method `M.squared()` returns a the value of the sum of squares of Matrix elements.

#### Example

```
Matrix< double > M; // M is then populated with double values...
double sum;
sum = M.squared();
```

### 9.2.27 Calculating the maximum absolute value of Matrix elements

The method `M.maxabs()` returns the maximum absolute value of Matrix elements.

#### Example

```
Matrix< double > M; // M is then populated with double values...
double max;
max = M.maxabs();
```

### 9.2.28 Finding the row elements with value 1 in a Matrix

The method (bool) `M.rowindex( vector< unsigned >& v )` populates a passed (unsigned) vector `v` with each vector element representing in which Matrix column each of the Matrix row elements was 1, takes passed vector as argument, returns true if successful.

#### Example

```
vector< unsigned > v( M.rows() );
if ( M.rowindex( v ) )
    cout << v << endl;
else
    cout << "can't do it: goofy Matrix." << endl;
```

#### Note

The passed Matrix must consist of rows that have a single row element with a value of 1, and the remainder with values of 0. For example, the Matrix

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

would return true and populate the passed vector with

$$\begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$$

whereas Matrices

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

would all return false.

The size of the vector *must* be the same as the number of rows in the Matrix.

### 9.2.29 Converting a Matrix to a vector, and a vector to a Matrix

A Matrix may be converted row by row to a vector, e.g. Matrix

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{bmatrix}$$

would be converted to vector

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 \end{bmatrix}$$

A vector may also be converted to a Matrix. Methods to perform the conversion include:

- `M.toVector( vector& v )`: converts Matrix *M* to passed vector *v* which is populated with the result, and returns a reference to vector *v*
- `M.toVector()`: converts Matrix *M* to a new vector, and returns a reference to the new vector
- `toMatrix( Matrix& M, vector& v )`: converts vector *v* to passed Matrix *M* which is populated with the result, and returns a reference to Matrix *M*
- `toMatrix( vector& v, unsigned r, unsigned c )`: converts vector *v* to a new Matrix *M* with *r* rows and *c* columns, and returns a reference to the new Matrix

### Example

```
// Make a Matrix as an example, and print it out
Matrix< double > A( 4, 3 );
for ( unsigned i = 0; i < 4; i++ )
    for ( unsigned j = 0; j < 3; j++ )
        A( i, j ) = 3 * i + j + 1;
cout << "Matrix A: " << A << endl;

// Convert the Matrix to vectors
vector< double > v1( 12 ), v2;
A.toVector( v1 );
cout << "toVector passed existing vector: " << v1 << endl;
v2 = A.toVector();
cout << "toVector passed new vector: " << v2 << endl;

// Convert the vectors back to Matrices
Matrix< double > B( 4, 3 ), C;
toMatrix( B, v2 );
cout << "back to an existing Matrix: " << B << endl;
C = toMatrix( v2, 4, 3 );
cout << "back to a new Matrix: " << C << endl;
```

### Notes

If a vector is passed as an argument to `toVector`, its size *must be* the same as the product of the rows and the columns of the Matrix. If a Matrix is passed as an argument to `toMatrix`, the product of its rows and columns *must be* the same as the size of the vector which is also passed. If a Matrix is not passed as an argument to `toMatrix`, the product of the 2<sup>nd</sup> and 3<sup>rd</sup> arguments *must be* the same as the size of the passed vector.

### 9.2.30 First order covariance of a Matrix

The method `M.covariance()` returns a new Matrix with the first-order covariance Matrix of `M`. The method `M.covariance( Matrix< double >& N )` populates the passed Matrix `N` with the result, and returns a reference to Matrix `N`.

### Example

```
#include <iostream>
#include <string>
#include "matrix.h"

string filename;
cout << "The file name is ";
cin >> filename;
```

```

Matrix< double > Data( true, filename ), // load data Matrix from file
    C; // for the covariance Matrix

C = Data.covariance();
cout << "Covariance Matrix = " << C << endl;

```

## Notes

The covariance method is only coded for Matrices of type `< double >`. If the Matrix to be populated with the result is passed as an argument, its dimensions must be  $Col \times Col$ , where  $Col$  is the number of columns in the data Matrix.

### 9.2.31 Matrix inverse and determinant

The following methods calculate the inverse and determinant of a Matrix:<sup>1</sup>

- `M.inverse( Matrix< double >& I )`: calculates the inverse of Matrix  $M$  by LU decomposition, populating Matrix  $I$  with the result, and returns a reference to Matrix  $I$
- `M.inverse()`: calculates the inverse of Matrix  $M$  by LU decomposition, and returns a new Matrix with the result
- `M.inverse( Matrix< double >& I, double& d )`: calculates the inverse of Matrix  $M$  by LU decomposition, populating Matrix  $I$  with the result, returns the determinant in the passed argument  $d$ , and returns a reference to Matrix  $I$
- `M.inverse( double& d )`: calculates the inverse of Matrix  $M$  by LU decomposition, returns the determinant in the passed argument  $d$ , and returns a new Matrix with the result
- `M.determinant()`: returns the determinant of Matrix  $M$  by LU decomposition
- `M.inverse( Matrix< double >& I, unsigned choice )`: allows the user to force type of inverse calculation by setting *choice* to “GaussJordan,” (default is LU decomposition) populating Matrix  $I$  with the result, and returns a reference to Matrix  $I$
- `M.inverse( unsigned choice )`: allows the user to force type of inverse calculation by setting *choice* to “GaussJordan” (default is LU decomposition), and returns a reference to the new Matrix

In the case of a singular Matrix, these methods throw exception `Matrix< double >::Singular& e`, where `e.what()` is “singular Matrix”.

---

<sup>1</sup>Adapted from W. H. Press et. al, Numerical Recipes in C, 2nd edition, Cambridge University Press, Cambridge, 1992, pages 36-48

### Example

```
#include <iostream>
#include <string>
#include "matrix.h"

string inverse_filename;
cout << "The file name for Matrix inverse is ";
cin >> inverse_filename;
Matrix< double > I( true, inverse_filename ), J, K;
double determinant;

try
{
    // Test populating a passed Matrix
    J.resize( I.rows(), I.cols() ); // resize receiving Matrix to input
    Matrix
    cout << "The inverse of " << I << "is " << I.inverse( J ) << endl;

    // Test returning a new Matrix
    K = I.inverse();
    cout << "The inverse of " << I << "is " << K << endl;

    // Test populating a passed Matrix, and returning a determinant
    cout << "The inverse of " << I << "is " << I.inverse( J,
    determinant )
    << endl;
    // Store the inverse and determinant separately for clarity:
    cout << "The determinant is " << determinant << endl;

    // Test returning a new Matrix with a determinant
    K.clear();
    K = I.inverse( determinant );
    cout << "The inverse of " << I << "is " << K
    << endl << "The determinant is " << determinant << endl;

    // Test returning just the determinant
    cout << "That determinant again is " << I.determinant() << endl;

    // Test populating a passed Matrix, and forcing type of inverse
    procedure
    cout << "The Gauss Jordan inverse of " << I << "is "
    << I.inverse( J, GaussJordan ) << endl;

    // Test returning a new Matrix and forcing type of inverse
    procedure
```

```

K.clear();
K = I.inverse( GaussJordan );
cout << "The Gauss Jordan inverse of " << I << "is " << K << endl;

// Examine the difference between LU decomposition and Gauss Jordan
cout << "The difference between LU decomposition and Gauss Jordan
inverse of "
    << I << "is " << I.inverse() - I.inverse( GaussJordan ) << endl;
}
catch ( Matrix< double >::Singular& e )
{
    cout << "Can't do it: " << e.what() << endl;
}

```

## Notes

Use a **try block with these methods to catch the Singular exception**. If you don't, your program may bomb out if a singular Matrix is encountered.

These methods are only coded for Matrices of type `< double >`. Matrices to invert must be square, e.g. dimensions must be  $Col \times Col$ , where *Col* is the number of columns in the Matrix. If the Matrix to be populated with the result is passed as an argument, it's dimensions must be the same as the Matrix object.

### 9.2.32 Eigenvalues

The calculation of the eigenvalues of a symmetric matrix is the only current instance of using pre-existing code and libraries, as these calculations can be quite intensive, and are currently beyond the scope of the current program. Gaurav Bansal ported the eigenvalue routines for a symmetric matrix from the Gnu Scientific Library (GSL) in 2003, recoding those portions of GSL to integrate with the matrix specification in neUROn2++. **As a result, do not expect GSL functionality beyond eigenvalue routines in the neUROn2++ environment.**

The best way to understand calling eigenvalue routines in neUROn2++ is to study the given example. A GSL matrix is first created from a neUROn2++ matrix using *gsl\_matrix\_view*, as is a GSL vector using *gsl\_vector*. A GSL workspace is then created using *gsl\_eigen\_symm\_workspace* for symmetric matrix eigenvalue calculations, and the eigenvalues are calculated using *gsl\_eigen\_symm*. Finally, the GSL vector is returned to an STL vector containing the eigenvalues using *gsl\_vector\_ptr*, and memory is freed using *gsl\_eigen\_symm\_free*.

For those interested in GSL, it may be found at <http://sources.redhat.com/gsl/> as of the time of this writing (August 11, 2003).

### Example

```
#include "gsl_eigen.h"

// Create a symmetric matrix using our matrix class, and fill with
random values
Matrix< double > m1( 4, 4 );
m1.random( 5 );

// Check to see that we perform eigenvalue routines to symmetric
routines only
assert ( m1.rows() == m1.cols() );

// Create a gsl matrix from our matrix--begin function is cautioned to
use but
//   here we need the value as double*
gsl_matrix_view m = gsl_matrix_view_array( m1.begin(), 4, 4 );

// Create a gsl vector to store all the eigenvalues
gsl_vector *eval = gsl_vector_alloc( 4 );

// Create gsl workspace for eigenvalue calculations
gsl_eigen_symm_workspace *w = gsl_eigen_symm_alloc( 4 );

// Calculate eigenvalues
gsl_eigen_symm( &m.matrix, eval, w );

// Create vector< double > from gsl vector
vector< double > eigenv( gsl_vector_ptr( eval, 0 ), gsl_vector_ptr(
eval, 4 ) );

// Free workspace
gsl_eigen_symm_free( w );

// Output results
cout << "Matrix: " << m1 << endl << "Eigenvalues are: " << eigenv <<
endl;
cout << "Maximum value = " << maxabs( eigenv ) << endl;
cout << "Minimum value = " << minabs( eigenv ) << endl;
```

### Notes

GSL is discovered and linked by CMake through `find_package(GSL REQUIRED)` and the imported targets `GSL::gsl` and `GSL::gslcblas`. Do not add local GSL copies, compiler search paths, or IDE-specific library settings to the source tree.

### 9.3 An example of the Matrix and vector classes

The following example constructs a simple neural network with 3 hidden nodes on one layer to train the exclusive-or problem. The file “xor.set” contains the exclusive-or dataset:

```
-0.9 -0.9 0.1
-0.9 0.9 0.9
0.9 -0.9 0.9
0.9 0.9 0.1
```

The following is the program:<sup>2</sup>

```
#include <iostream>
#include <iomanip>
#include <string>
#include <math.h>

#include "matrix.h"
#include "vector_ops.h"
#include "function_defs.h"

using namespace std;

int main()
{
    // Assumes a file named xor.set
    string filename = "xor.set";

    // Load the XOR dataset from the file xor.set
    Matrix< double > dataset( filename, 3 ); // it has 3 columns

    // Get the input Matrix In from the dataset as a submatrix
    Matrix< double > In = dataset.submatrix( 0, 3, 0, 1 );

    // Get the output vector y from the dataset's last column
    vector< double > y = dataset.col( 2 );

    // Make the 3 x 2 hidden weight Matrix hW and the hW update Matrix
    hWup
    Matrix< double > hW( 3, 2 ), hWup( 3, 2 );
    hW.random( 0.5 ); // randomize hW

    // Make the 2 element vector for the inputs I
```

---

<sup>2</sup>References are to James A. Freeman and David M. Skapura, Neural Networks, Algorithms, Applications and Programming Techniques, Addison-Wesley, 1991, ISBN 0-201-51376-5

```

// and the 3 element vector hidden output h0
// and the 3 element vector output weights oW
vector< double > I( 2 ), h0( 3 ), oW( 3 );
nvec::random( oW, 0.5 ); // randomize oW

// o is the neural net's output, o_err the output error term,
// eta is the learning rate
double o, o_err, eta = 0.05;

// h_err is the vector with hidden error terms
vector< double > h_err( 3 );

for ( unsigned iteration = 0; iteration < 1000000; iteration++ )
{
    for ( unsigned example = 0; example < 4; example++ )
    {
        // Get a single row from the input Matrix
        // (Freeman & Skapura p. 102, #1)
        In.row( example, I );

        // Take the dot product of the hidden weights Matrix hW and
        // the
        // transpose of a row of the input Matrix I, and apply the
        // sigmoidal function to the resulting vector
        // (Freeman & Skapura p. 102, #2 & #3)
        func( hW.dotprod( I, h0 ), sigmoidal(), h0 );

        // Take the dotproduct of the hidden output vector and the
        // output
        // weights vector, and apply the sigmoidal function to the
        // result
        // (Freeman & Skapura p. 102, #4 & #5)
        o = sigmoidal()( dotprod( h0, oW ) );

        // Calculate the error term for the output unit
        // (Freeman & Skapura p. 102, #6)
        o_err = ( y[ example ] - o ) * d_sigmoidal()( o );

        // Calculate the error terms for the hidden units
        // (Freeman & Skapura p. 102, #7)
        // Note that using func which takes the output vector as the
        // last argument, and *= instead of *, is the most efficient
        // way
        ( func( h0, d_sigmoidal(), h_err ) *= o_err ) *= oW;

        // Update the output weights, note again the use of *= for

```

```

    efficiency
    // (Freeman & Skapura p. 102, #8)
    oW += ( h0 *= ( eta * o_err ) );

    // Update the hidden weights
    // (Freeman & Skapura p. 102, #9)
    hW += ( hWup.outprod( h_err, I ) *= eta );
}

// Every 1000 iterations, print out the error
if ( iteration % 1000 == 0 )
    cout << setw( 8 ) << iteration << ": LMS error = " <<
        0.5 * ( y[ 3 ] - o ) * ( y[ 3 ] - o ) << endl;
}

return 0;
}

```

## Chapter 10

# Utility methods

The following are contained in the header file “utility.h”, and contain global utility methods. These methods throw exception `util::utilErr& e`, where `e.what()` specifies the module from which the error originated, and the nature of the error.

### 10.1 Random number generator

#### **namespace util**

The following utilities seed or generate random numbers of double precision:

- `d_random()`: seeds the random number generator with an internal timestamp
- `d_random( double n )`: returns a double precision random number between  $-n$  and  $+n$
- `d_random( double lower, double upper )`: returns a double precision random number between *lower* and *upper*

#### 10.1.1 Examples

```
util::d_random(); // seed the generator
```

```
double x, y;
```

```
x = util::d_random( 0.5 ); // Generates a random number between -0.5  
and +0.5
```

```
y = util::d_random( 1.0, 10.0 ); // Between 1.0 and 10.0
```

## 10.2 Timestamp

The `timestamp( unsigned  $x$  )` utility inserts a timestamp of the form `HHHH:MM:SS` (where `H` is hour, `M` is minute, `S` is second) into the output stream, given  $x$  seconds.

### 10.2.1 Example

```
unsigned start = time( 0 ); // gets the current time in seconds

// Do something for a while
for ( i = 0; i < 100000; i++ )
    cout << ".";

// Calculate the elapsed time
unsigned increment = time( 0 ) - start;

// Print out how long it took
cout << endl << "That took " << timestamp( increment ) << endl;
```

## 10.3 Round

### `namespace util`

The following method will round a number to a specified number of significant digits:

- `round( double  $x$ , unsigned  $d$  )`: rounds the passed value  $x$  to  $d$  significant digits, returns the rounded value

### 10.3.1 Example

```
double x;
unsigned d;

cout << "x = ";
cin >> x;
cout << "d = ";
cin >> d;
cout << "round( x, d ) = " << util::round( x, d ) << endl;
```

### 10.3.2 Notes

Zero significant digits is not allowed. Rounds positive numbers up, and negative numbers down, e.g. `util::round( 30.9999, 2 ) = 31`, and `util::round( -30.9999, 2 ) = -31`.

## 10.4 Queries with bounds checking

### namespace util

The *askI* and *askD* methods allow user queries with bounds checking. Specifications include:

- *askI*( string& *query*, int *lower* ) which queries the user with the string *query* repeatedly if necessary until the user enters an integer value  $\geq$  *lower*, and returns the value
- *askI*( string& *query*, int *lower*, int *upper* ) which queries the user with the string *query* repeatedly if necessary until the user enters an integer value  $\geq$  *lower* and  $\leq$  *upper*, and returns the value
- *askD*( string& *query*, double *lower* ) which queries the user with the string *query* repeatedly if necessary until the user enters a value  $\geq$  *lower*, and returns the value
- *askD*( string& *query*, double *lower*, double *upper* ) which queries the user with the string *query* repeatedly if necessary until the user enters a value  $\geq$  *lower* and  $\leq$  *upper*, and returns the value

### 10.4.1 Example

```
int i, h;
double d;

i = util::askI( "How many input nodes? ", 1 );
h = util::askI( "How many hidden nodes? ", 1, 10 );
d = util::askD( "What is the learning rate? ", 0, 1 );
```

## 10.5 Query for yes/no answer

### namespace util

The *askYesNo* method allows user queries with yes/no answers. Specifications include:

- *askYesNo*( string& *query* ) queries the user with the string *query* repeatedly if necessary until the user enters “yes”, “y”, “no” or “n”, and returns boolean true for yes, false for no

### 10.5.1 Example

```
bool yesNoAnswer;
yesNoAnswer = util::askYesNo( "Shall we clean the closet? " );
```

## 10.6 Query for existing file name

### namespace util

The *getGoodFile* method tests for a file name to see if it actually exists, then queries for an existing file name repeatedly until a file that actually exists is entered, and returns that file name as a *string*:

- `getGoodFile( string& filename )` tests *filename* to see if it actually exists, then if it doesn't, queries the user for a file name repeatedly if necessary until the user enters an existing file name, and returns it as a string.

### 10.6.1 Example

```
#include <iostream>
#include <fstream>
#include <string>
#include "utility.h"

string filename; // for the name of a loaded file

cout << "What is the name of your data file (it must exist)? ";
cin >> filename;

// Open the file as read-only, and associate it with the 'label'
ifstream inputFile( ( util::getGoodFile( filename ) ).c_str(), ios::in
);
```

## 10.7 Remove a carriage return

### namespace util

The *chopEndl* method removes ASCII 13, generally a carriage return, from the end of a string if it exists, and returns a reference to the string.

- `chopEndl( string& stringObj )` removes the last character from *stringObj* if it is ASCII 13, and returns a reference to *stringObj*.

### 10.7.1 Example

```
#include <iostream>
#include <fstream>
#include <string>
#include "utility.h"

string lineString; // string to hold the 1st line of file
```

```
// Open the file "test.txt" for input
ifstream loadfile( "test.txt", ios::in );

getline( loadfile, lineString ); // get the 1st line of file
util::chopEndl( lineString ); // remove <cr> from end of string if
exists

loadfile.close(); // close the input file
```

## 10.8 Parse a variable definition string

### namespace util

- `variable_parse( string& stringObj )` parses *stringObj* for variable representations from nodes (see below), and returns a vector of vector of unsigned representing the nodes.
- `variable_parse( ifstream& fileObj )` parses the file *fileObj* for variable representations from nodes (see below), and returns a vector of vector of unsigned representing the nodes.

The *variable\_parse* method parses a string passed as the argument to determine variable representation from nodes, and returns a *new* vector of vector of unsigned. In the string,

- `;` delimits variables
- `,` delimits nodes
- `-` specifies a range of nodes
- spaces are ignored

For example, passed string “0-3,5; 4,8; 6,7” returns vector of vector of unsigned (each line is a vector):

```
0 1 2 3 5
4 8
6 7
```

util::utilErr exceptions thrown by this method include e.what()

- “util variable\_parse error: bad character found at string position *position* ( *character* )”
- “util variable\_parse error: too many hyphens”
- “util variable\_parse error: hyphen without a number on both ends ”
- “util variable\_parse error: 1st number in range greater than 2nd ”

- “util variable\_parse error: numbers in range are the same”
- “util variable\_parse error: missing zero”
- “util variable\_parse error: duplicated positions”
- “util variable\_parse error: missing positions”

### 10.8.1 Example

```
#include <iostream>
#include <fstream>
#include <vector>
#include <string>

#include "vector_ops.h"
#include "utility.h"

using namespace std;

int main()
{
    // The returned structures
    vector< vector< unsigned > > parse_numbers, parse2;

    string parse_test = "0-3,5; 4,8; 6,7"; // a test string

    // Parse the string
    try
    {
        parse_numbers = util::variable_parse( parse_test );

        // Then print out the returned structure, a vector at a time
        cout << "Here it is: " << endl;
        unsigned pCount;
        for ( pCount = 0; pCount < parse_numbers.size(); pCount++ )
            cout << parse_numbers[ pCount ] << endl;
    }
    catch ( util::utilErr& e )
    {
        cout << e.what() << endl;
    }

    // Do the same from a file
    string reply;
    cout << "What is the name of the file? ";
    cin >> reply;
```

```

    ifstream inputFile( ( util::getGoodFile( reply ) ).c_str(), ios::in
    );

    try
    {
        parse2 = util::variable_parse( inputFile );
        cout << "Here it is: " << endl;
        for ( pCount = 0; pCount < parse2.size(); pCount++ )
            cout << parse2[ pCount ] << endl;
    }
    catch ( util::utilErr& e )
    {
        cout << e.what() << endl;
    }

    return 0;
}

```

**variable\_parse( ifstream& *fileObj* )** will close and clear the file for you. Parsing a file will support multiple line definitions of variables by concatenating the lines prior to parsing the resulting string.

## 10.9 Remove a variable from a vector of vector of unsigned data structure

namespace util

- **remove\_variable( vector< vector< unsigned > >& *variables*, unsigned *n*, vector< vector< unsigned > >& *result* )** removes the subvector representing variable *n* from the vector of vector of unsigned data structure *variables*, populates *result* as a vector of vector of unsigned data structure representing the result, returns a reference to the resulting data structure.
- **remove\_variable( vector< vector< unsigned > >& *variables*, unsigned *n* )** as above, except returns a new vector of vector of unsigned data structure with the result.

The *remove\_variable* method removes a single subvector, representing a single variable, from a vector of vector of unsigned data structure which represents a set of variables. As neURon2++ variables may contain more than 1 input node (e.g. a variable may contain 2 nodes, the value of the variable itself, and whether or not that variable was present in the dataset), this method is used to remove a single variable from a set of variables.

### 10.9.1 Example

```
// Create a vector of vector of unsigned with some variables
string parse_test = "0-3,5; 4,8; 6,7";
vector< vector< unsigned > > parse_numbers, parse_new, parse_new2;
parse_numbers = util::variable_parse( parse_test );

// Test remove_variable
parse_new = parse_numbers;
unsigned position;
position = util::askI( "What is the position to remove? ", 0 );
util::remove_variable( parse_numbers, position, parse_new );
cout << "After I've removed variable " << position << ": " << endl;
cout << "Here is the original: " << parse_numbers << endl;
cout << "Here is the result: " << parse_new << endl;

parse_new2 = util::remove_variable( parse_numbers, position );
cout << "Making a new data structure : " << endl;
cout << "Here is the original: " << parse_numbers << endl;
cout << "Here is the result: " << parse_new2 << endl;
```

## Chapter 11

# Statistical methods

The following are contained in the header file “stats.h”, and implemented in “stats.cpp”, and contain global statistical methods. Those that are not classes are contained within namespace “stats”, and throw exception stats::statsErr& e, where e.what() specifies the module from which the error originated, and the nature of the error.

### 11.1 Log gamma

#### namespace stats

The following method returns the natural logarithm of the gamma function  $\log \Gamma(a)$ <sup>1</sup> where

$$\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t} dt \quad (11.1)$$

- `gammln( double a )`: returns the double value  $\log \Gamma(a)$

#### 11.1.1 Example

```
double x;  
cout << "For log gamma, x = ";  
cin >> x;  
cout << "Log gamma " << x << " = " << stats::gammln( x ) << endl;
```

### 11.2 Incomplete gamma by fraction

#### namespace stats

---

<sup>1</sup>Adapted from W. H. Press et. al, Numerical Recipes in C, 2nd edition, Cambridge University Press, Cambridge, 1992, p. 214

The following method returns the incomplete gamma function

$$P(a, x) \equiv \frac{\gamma(a, x)}{\Gamma(a)} \equiv \frac{1}{\Gamma(a)} \int_0^x e^{-t} t^{a-1} dt \quad (a > 0) \quad (11.2)$$

by continued fraction development:<sup>2</sup>

$$\Gamma(a, x) = e^{-x} x^a \left( \frac{1}{x + \frac{1-a}{1 + \frac{1-a}{x + \frac{2-a}{1 + \frac{2-a}{x + \dots}}}}} \right) \quad (x > 0) \quad (11.3)$$

- `gcf( double  $a$ , double  $x$  )`: returns the double value  $\Gamma(a, x)$

### 11.2.1 Example

```
double a, x;
cout << "For gcf, a = ";
cin >> a;
cout << "x = ";
cin >> x;
try {
    cout << "gcf( " << a << " , " << x << " ) = " << stats::gcf( a, x
    );
} catch ( stats::statsErr& e ) {
    cout << e.what();
}
cout << endl;
```

## 11.3 Incomplete gamma by series

### namespace stats

The following method returns the incomplete gamma function (equation 11.2) by series: <sup>3</sup>

$$\gamma(a, x) = e^{-x} x^a \sum_{n=0}^{\infty} \frac{\Gamma(a)}{\Gamma(a+1+n)} x^n \quad (11.4)$$

- `gser( double  $a$ , double  $x$  )`: returns the double value  $\gamma(a, x)$

<sup>2</sup>Adapted from W. H. Press et. al, Numerical Recipes in C, 2nd edition, Cambridge University Press, Cambridge, 1992, p. 219

<sup>3</sup>Adapted from W. H. Press et. al, Numerical Recipes in C, 2nd edition, Cambridge University Press, Cambridge, 1992, pp. 218–219

### 11.3.1 Example

```
double a, x;
cout << "For gser, a = ";
cin >> a;
cout << "x = ";
cin >> x;
try {
    cout << "gser( " << a << " , " << x << " ) = " << stats::gser( a,
        x );
} catch ( stats::statsErr& e ) {
    cout << e.what();
}
cout << endl;
```

## 11.4 Incomplete gamma composite

### namespace stats

The following method returns the incomplete gamma function ([equation 11.2](#)) by either continued fraction or series, based on speed of convergence:<sup>4</sup>

- `gammp( double  $a$ , double  $x$  )`: returns the double value  $P(a, x)$

### 11.4.1 Example

```
double a, x;
cout << "For gammp, a = ";
cin >> a;
cout << "x = ";
cin >> x;
try {
    cout << "gammp( " << a << " , " << x << " ) = " << stats::gammp(
        a, x );
} catch ( stats::statsErr& e ) {
    cout << e.what();
}
cout << endl;
```

## 11.5 Incomplete gamma complement

### namespace stats

---

<sup>4</sup>Adapted from W. H. Press et. al, Numerical Recipes in C, 2nd edition, Cambridge University Press, Cambridge, 1992, p. 218

The following method returns the complement of the incomplete gamma function

$$Q(a, x) \equiv 1 - P(a, x) \quad (11.5)$$

(see [equation 11.2](#) for the definition of  $P(a, x)$ ): <sup>5</sup>

- `gammq( double a, double x )`: returns the double value  $Q(a, x)$

### 11.5.1 Example

```
double a, x;
cout << "For gammq, a = ";
cin >> a;
cout << "x = ";
cin >> x;
try {
    cout << "gammq( " << a << " , " << x << " ) = " << stats::gammq(
        a, x );
} catch ( stats::statsErr& e ) {
    cout << e.what();
}
cout << endl;
```

## 11.6 Beta function

**namespace stats**

The following method returns the beta function  $B(z, w)$  <sup>6</sup> where

$$B(z, w) = B(w, z) = \int_0^1 t^{z-1} (1-t)^{w-1} dt = \frac{\Gamma(z)\Gamma(w)}{\Gamma(z+w)} \quad (11.6)$$

- `beta( double z, double w )`: returns the double value  $B(z, w)$

### 11.6.1 Example

```
double z, w;
cout << "For beta function, z = ";
cin >> z;
cout << "w = ";
cin >> w;
cout << "Beta = " << stats::beta( z, w ) << endl;
```

---

<sup>5</sup>Adapted from W. H. Press et. al, Numerical Recipes in C, 2nd edition, Cambridge University Press, Cambridge, 1992, p. 218

<sup>6</sup>Adapted from W. H. Press et. al, Numerical Recipes in C, 2nd edition, Cambridge University Press, Cambridge, 1992, pp. 215–216

## 11.7 Incomplete Beta function

### namespace stats

The following method returns the incomplete beta function  $I_x(a, b)$ <sup>7</sup> where

$$I_x(a, b) \equiv \frac{B_x(a, b)}{B(a, b)} \equiv \frac{1}{B(a, b)} \int_0^x t^{a-1} (1-t)^{b-1} dt \quad (11.7)$$

Note that the continued fraction representation is also available, but as a utility function should not be necessary to call directly:

$$I_x(a, b) = \frac{x^a(1-x)^b}{aB(a, b)} = \left( \frac{1}{1 + \frac{d_1}{1 + \frac{d_2}{1 + \dots}}} \right) \quad (11.8)$$

where

$$d_{2m+1} = -\frac{(a+m)(a+b+m)x}{(a+2m)(a+2m+1)} \quad (11.9)$$

and

$$d_{2m} = \frac{m(b-m)x}{(a+2m-1)(a+2m)} \quad (11.10)$$

- `betai( double a, double b, double x )`: returns the double value  $I_x(a, b)$
- `betacf( double a, double b, double x )`: returns the double value  $I_x(a, b)$  by continued fraction evaluation, used by `betai`

### 11.7.1 Example

```
double a, b, x;
cout << "For incomplete beta function, a = ";
cin >> a;
cout << "b = ";
cin >> b;
cout << "x = ";
cin >> x;
cout << "Incomplete beta = " << stats::betai( a, b, x ) << endl;
```

---

<sup>7</sup>Adapted from W. H. Press et. al, Numerical Recipes in C, 2nd edition, Cambridge University Press, Cambridge, 1992, pp. 226–228

## 11.8 Error functions

### namespace stats

The following methods return the error function (commonly referred to as *erf*)<sup>8</sup> and its complement

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt = P\left(\frac{1}{2}, x^2\right) \quad (11.11)$$

$$\operatorname{erfc}(x) \equiv 1 - \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2} dt = Q\left(\frac{1}{2}, x^2\right) \quad (11.12)$$

- `erff( double x )`: returns the double value  $\operatorname{erf}(x)$
- `erfc( double x )`: returns the double value  $\operatorname{erfc}(x)$
- `erffc( double x )`: returns the double value  $\operatorname{erfc}(x)$  by Chebyshev fitting

### 11.8.1 Example

```
double x;
cout << "x = ";
cin >> x;
try {
    cout << "erf( " << x << " ) = " << stats::erff( x );
} catch ( stats::statsErr& e ) {
    cout << e.what();
}
cout << endl;
try {
    cout << "erfc( " << x << " ) = " << stats::erfc( x );
} catch ( stats::statsErr& e ) {
    cout << e.what();
}
cout << endl;
```

### 11.8.2 Notes

Note the 2 *f* s in `erff`, so that this is not confused with the innate compiler function `erf`.

## 11.9 Error functions inverses

### namespace stats

---

<sup>8</sup>Adapted from W. H. Press et. al, Numerical Recipes in C, 2nd edition, Cambridge University Press, Cambridge, 1992, pp. 220–221

The following methods return the inverse of the error function and the inverse of the complement of the error function:<sup>9</sup>

- `invErff( double y )`: returns the double value  $erf^{-1}(y)$
- `invErfc( double y )`: returns the double value  $erfc^{-1}(y)$

### 11.9.1 Example

```
double y;
cout << "y = ";
cin >> y;
try {
    cout << "inverse erf( " << y << " ) = " << stats::invErff( y );
} catch ( stats::statsErr& e ) {
    cout << e.what();
}
cout << endl;
try {
    cout << "inverse erfc( " << y << " ) = " << stats::invErfc( y );
} catch ( stats::statsErr& e ) {
    cout << e.what();
}
cout << endl;
```

## 11.10 Gaussian distribution

### namespace stats

The following method returns the unit Gaussian function

$$\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2} \quad (11.13)$$

where given mean  $\mu$  and standard deviation  $\sigma$  the following transformation  $x \rightarrow z$  is made:

$$z \equiv \frac{x - \mu}{\sigma}$$

$x$ ,  $\mu$  and  $\sigma$  may also be passed, where the Gaussian function is then

$$\varphi(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)} \quad (11.14)$$

- `gaussD( double z )`: returns the double value  $\varphi(z)$
- `gaussD( double x, double  $\mu$ , double  $\sigma$  )`: returns the double value  $\varphi(x, \mu, \sigma)$

---

<sup>9</sup>Algorithm for  $erfc^{-1}(y)$  by Takuya Ooura © 1996,  $erf^{-1}(y) = erfc^{-1}(1 - y)$

### 11.10.1 Example

```
double z, x, u, s;
cout << "for gaussD, z = ";
cin >> z;
cout << "gaussD = " << stats::gaussD( z ) << endl;
cout << "for gaussD, x = ";
cin >> x;
cout << "u = ";
cin >> u;
cout << "s = ";
cin >> s;
cout << "gaussD = " << stats::gaussD( x, u, s ) << endl;
```

## 11.11 z Area

### namespace stats

The following method returns the area under a unit Gaussian distribution

$$\Phi(z) \equiv \frac{1}{\sqrt{2\pi}} \int_0^z e^{-z^2/2} dz = \frac{1}{2} \operatorname{erf}\left(\frac{z}{\sqrt{2}}\right) \quad (11.15)$$

- Zarea( double  $z$  ): returns the double value  $\Phi(z)$

### 11.11.1 Example

```
double z;
cout << "for z Area, z = ";
cin >> z;
try {
    cout << "z Area = " << stats::Zarea( z );
} catch ( stats::statsErr& e ) {
    cout << e.what();
}
cout << endl;
```

## 11.12 z Area inverse

### namespace stats

The following method returns the inverse of the area under a unit Gaussian distribution

$$z = Z(A) = \Phi^{-1}(A) = \sqrt{2} \operatorname{erf}^{-1}(2A - 1) \quad (11.16)$$

- invZarea( double  $A$  ): returns the double value  $Z(A) = \Phi^{-1}(A)$

### 11.12.1 Example

```
double A;
cout << "for invZarea, A = ";
cin >> A;
try {
    cout << "invZarea ( " << A << " ) = " << stats::invZarea( A ) <<
    endl;
} catch ( stats::statsErr& e ) {
    cout << e.what();
}
```

## 11.13 Population metrics

The *Population* class provides statistical metrics such as mean, variance, standard deviation, skew and kurtosis. It's constructor takes `vector< double >` as its argument, representing the population from which to generate metrics. Errors are caught by *PopulationErr* (see below for examples).

- `Population( vector< double > x )` defines a population from `vector< double > x` from which to generate statistical metrics

Accessors include:

- `P.mean()`: returns the mean
- `P.var()`: returns the variance
- `P.std()`: returns the standard deviation
- `P.skew()`: returns the skew
- `P.kurtosis()`: returns the kurtosis

### 11.13.1 Example

```
double numbers[ 4 ] = { 1.0, 2.0, 3.0, 4.0 }; // define the dataset
vector< double > dset( numbers, numbers + 4 );
Population P( dset ); // define the population from the dataset
cout << "average = " << P.mean() << endl;
cout << "variance = " << P.var() << endl;
cout << "standard deviation = " << P.std() << endl;
try {
    cout << "skew = " << P.skew() << endl;
    cout << "kurtosis = " << P.kurtosis() << endl;
} catch ( Population::PopulationErr& e ) {
    cout << e.what() << endl;
}
```

### 11.13.2 Notes

The only allowable constructor takes `vector< double >` as an argument, and there is no copy constructor or overloaded `=` operator. The `vector< double >` argument *must* contain 2 or more elements.

## 11.14 F and Student's t

### namespace stats

The following methods return probabilities for the F-test that 2 sample variances are different ( $p=1.0$  if they are equal), and Student's t- tests that 2 samples are different. All take 2 vectors of double as the samples  $S_1$  and  $S_2$ .

- `ftest( vector< double > S1, vector< double > S2 )`: returns a double value with the probability for an F-test.
- `ttest( vector< double > S1, vector< double > S2 )`: returns a double value with the probability for a Student's t- test with equal sample variance.
- `tutest( vector< double > S1, vector< double > S2 )`: returns a double value with the probability for a Student's t- test with unequal sample variance.
- `tptest( vector< double > S1, vector< double > S2 )`: returns a double value with the probability for a paired Student's t-test.

Note that<sup>10</sup>

- For the F-test,

$$\alpha = I_{\frac{\nu_2}{\nu_2 + \nu_1 F}}\left(\frac{\nu_2}{2}, \frac{\nu_1}{2}\right) \quad (11.17)$$

(see [equation 11.7](#)), where  $F = \frac{\sigma_1^2}{\sigma_2^2}$  ( $\sigma_1^2 > \sigma_2^2$ ), and  $\nu_1 = N_1 - 1$  and  $\nu_2 = N_2 - 1$

- For Student's distribution,  $\alpha = 1 - A(t|\nu)$ , where

$$A(t|\nu) = \frac{1}{\sqrt{\nu} B(\frac{1}{2}, \frac{\nu}{2})} \int_{-t}^t \left(1 + \frac{x^2}{\nu}\right)^{\frac{\nu+1}{2}} dx \quad (11.18)$$

(see [equation 11.6](#)) and is related to the incomplete beta function by

$$A(t|\nu) = 1 - I_{\frac{\nu}{\nu + t^2}}\left(\frac{\nu}{2}, \frac{1}{2}\right) \quad (11.19)$$

(see [equation 11.7](#))

---

<sup>10</sup>Adapted from W. H. Press et. al, Numerical Recipes in C, 2nd edition, Cambridge University Press, Cambridge, 1992, pp. 228–229, 615–619

- For Student's t with equal sample variance,

$$t = \frac{\mu_1 - \mu_2}{s_D} \quad (11.20)$$

where

$$s_D = \sqrt{\frac{\sum_i^{N_1} (x_i - \mu_1)^2 + \sum_i^{N_2} (x_i - \mu_2)^2}{N_1 + N_2 - 2} \left( \frac{1}{N_1} + \frac{1}{N_2} \right)} \quad (11.21)$$

and  $\nu = N_1 + N_2 - 2$

- For Student's t with unequal sample variance,

$$t = \frac{\mu_1 - \mu_2}{\sqrt{\frac{\sigma_1^2}{N_1} + \frac{\sigma_2^2}{N_2}}} \quad (11.22)$$

and

$$\nu = \frac{\left( \frac{\sigma_1^2}{N_1} + \frac{\sigma_2^2}{N_2} \right)^2}{\frac{\left( \frac{\sigma_1^2}{N_1} \right)^2}{N_1 - 1} + \frac{\left( \frac{\sigma_2^2}{N_2} \right)^2}{N_2 - 1}} \quad (11.23)$$

- For paired Student's t,

$$t = \frac{\mu_1 - \mu_2}{s_D} \quad (11.24)$$

where

$$s_D = \sqrt{\frac{\sigma_1^2 + \sigma_2^2 - 2Cov(x_1, x_2)}{N}} \quad (11.25)$$

where

$$Cov(x_1, x_2) \equiv \frac{1}{N-1} \sum_{i=1}^N (x_{i1} - \mu_1)(x_{i2} - \mu_2) \quad (11.26)$$

and  $\nu = N - 1$ , where  $N$  is the *number of pairs*.

### 11.14.1 Example

```
double d1[ 9 ] = { 3, 4, 5, 8, 9, 1, 2, 4, 5 };
double d2[ 9 ] = { 6, 19, 3, 2, 14, 4, 5, 17, 1 };
vector< double > v1( d1, d1 + 9 ), v2( d2, d2 + 9 );

cout << "F-test = " << stats::ftest( v1, v2 ) << endl;
cout << "Student's t for same variance = " << stats::ttest( v1, v2 ) <
< endl;
cout << "Student's t for unequal variance = " << stats::tutest( v1, v2
) << endl;
cout << "Paired Student's t = " << stats::tptest( v1, v2 ) << endl;
```

### 11.14.2 Notes

Note that in all cases, sample (passed vector) sizes *must be*  $> 1$ , and for the paired Student's t-test, they *must be* equal.

## 11.15 Chi-square

### namespace stats

The following method returns the probability  $p$  for  $df$  degrees of freedom and Chi-square value  $\chi^2$ :

- `pX2( unsigned  $df$ , double  $\chi^2$  )`: returns a double value with the probability.

Note that the probability  $p$  for the Chi-square value  $\chi^2$  with  $df$  degrees of freedom is  $Q(\frac{df}{2}, \frac{\chi^2}{2})$  (see [equation 11.5](#)).

### 11.15.1 Example

```
double x; // chi-square value
unsigned df; // degrees of freedom

// Query the user for the value and degrees of freedom
cout << "For chi-square, df = ";
cin >> df;
cout << "X2 = ";
cin >> x;

// Then calculate the probability p
try {
    cout << "p = " << stats::pX2( df, x );
} catch ( stats::statsErr& e ) { // catch a stats error
    cout << e.what();
}
cout << endl;
```

## 11.16 Kolmogorov-Smirnov statistic

### namespace stats

The following method returns the Kolmogorov-Smirnov probability  $Q_{KS}$ <sup>11</sup>

$$Q_{KS}(\lambda) = 2 \sum_{j=1}^{\infty} (-)^{j-1} e^{-2j^2\lambda^2} \quad (11.27)$$

- `double probks( double  $\lambda$  )`: returns a double value with  $Q_{KS}(\lambda)$ .

---

<sup>11</sup>Adapted from W. H. Press et. al, Numerical Recipes in C, 2nd edition, Cambridge University Press, Cambridge, 1992, pages 623-626

### 11.16.1 Example

```
double alam;
cout << "For Kolmogorov-Smirnov, lambda = ";
cin >> alam;
cout << "Q_KS " << alam << " = " << stats::probks( alam ) << endl;
```

## 11.17 Functions

The *Funct* class provides functional manipulation such as root finding, etc. The *Funct* constructor takes as its arguments an instance of the function (generally *\*this*) followed by a reference to the *member function of a class*, for example:

```
class TestIt {
public:
    TestIt(){}
    ~TestIt(){}
    Funct< TestIt > squarer(){ return Funct< TestIt >( *this, &
        TestIt::xsqrd ); }

private:
    double xsqrd( double x ){ return x * x - 4; }
};
```

Note that *Funct* is a template class, and takes as its template argument the name of the class containing the member function. Errors are caught by *functErr* (see below for examples).

### 11.17.1 Bracketing

The following methods bracket a function:<sup>12</sup>

- `F.mnbrak( double x1, double x2 )` takes boundary points  $x1$  and  $x2$  to bracket function  $F$ . Accessors below return the bracketed result, where  $ax$ ,  $bx$  and  $cx$  are 3 sequential  $x$ -axis points, with  $fb = f(bx)$  less than  $fa = f(ax)$  and  $fc = f(cx)$ . All returned values are of type `double`.
- `F.mnbrak_ax()`: accessor returns  $ax$
- `F.mnbrak_bx()`: accessor returns  $bx$
- `F.mnbrak_cx()`: accessor returns  $cx$
- `F.mnbrak_fa()`: accessor returns  $fa$
- `F.mnbrak_fb()`: accessor returns  $fb$
- `F.mnbrak_fc()`: accessor returns  $fc$

---

<sup>12</sup>Adapted from W. H. Press et. al, Numerical Recipes in C, 2nd edition, Cambridge University Press, Cambridge, 1992, pp. 400–401

### Example

```
class TestIt {
public:
    TestIt(){}
    ~TestIt(){}
    void mnbrak( double x1, double x2 )
    {
        try
        {
            Funct< TestIt > f( *this, &TestIt::xsqrd );
            f.mnbrak( x1, x2 );
            cout << "ax = " << f.mnbrak_ax() << endl;
            cout << "bx = " << f.mnbrak_bx() << endl;
            cout << "cx = " << f.mnbrak_cx() << endl;
            cout << "fa = " << f.mnbrak_fa() << endl;
            cout << "fb = " << f.mnbrak_fb() << endl;
            cout << "fc = " << f.mnbrak_fc() << endl;
        }
        catch ( Funct< TestIt >::functErr& e ) { cout << e.what() <<
            endl; }
    }

private:
    double xsqrd( double x ){ return x * x - 4; }
};

int main()
{
    TestIt tester;
    double x1, x2;
    cout << "For testing mnbrak, x1 = ";
    cin >> x1;
    cout << "x2 = ";
    cin >> x2;
    tester.mnbrak( x1, x2 );

    return 0;
}
```

### 11.17.2 Find Minimum

The following methods isolate the minimum of function  $F$  using Brent's method:<sup>13</sup>

---

<sup>13</sup>Adapted from W. H. Press et. al, Numerical Recipes in C, 2nd edition, Cambridge University Press, Cambridge, 1992, pp. 402–405

- `F.brent( double ax, double bx, double cx, double tol )`: takes initial bracketing triplet of abscissas *ax*, *bx* and *cx* and tolerance *tol* to isolate minimum of function *F*.
- `F.brent_xmin()`: returns the isolated minimum (type double) of function *F*.

### Example

```
class TestIt {
public:
    TestIt(){}
    ~TestIt(){}
    void brent( double a1, double b1, double c1, double tol )
    {
        try
        {
            Funct< TestIt > f( *this, &TestIt::xsqrd );
            cout << "brent = " << f.brent( a1, b1, c1, tol ) << endl;
            cout << "xmin = " << f.brent_xmin() << endl;
        }
        catch ( Funct< TestIt >::functErr& e ) { cout << e.what() <<
            endl; }
    }

private:
    double xsqrd( double x ){ return x * x - 4; }
};

int main()
{
    TestIt tester;
    double a, b, c, tol;
    cout << "For testing brent, a = ";
    cin >> a;
    cout << "b = ";
    cin >> b;
    cout << "c = ";
    cin >> c;
    cout << "tolerance = ";
    cin >> tol;
    tester.brent( a, b, c, tol );

    return 0;
}
```

### 11.17.3 Find Roots

The following method finds roots of function  $F$  using Brent's method:<sup>14</sup>

- `F.zbrent( double  $x1$ , double  $x2$ , double  $tol$  )`: takes 2  $x$ -axis bounds  $x1$  and  $x2$ , and tolerance  $tol$ , and returns the root of function  $F$  between those bounds.

#### Example

```
class TestIt {
public:
    TestIt(){}
    ~TestIt(){}
    void zbrent( double x1, double x2, double tol )
    {
        try
        {
            Funct< TestIt > f( *this, &TestIt::xsqrd );
            cout << "zbrent = " << f.zbrent( x1, x2, tol ) << endl;
        }
        catch ( Funct< TestIt >::functErr& e ) { cout << e.what() <<
            endl; }
    }

private:
    double xsqrd( double x ){ return x * x - 4; }
};

int main()
{
    TestIt tester;
    double x1, x2, tol;
    cout << "For testing zbrent, x1 = ";
    cin >> x1;
    cout << "x2 = ";
    cin >> x2;
    cout << "tolerance = ";
    cin >> tol;
    tester.zbrent( x1, x2, tol );

    return 0;
}
```

---

<sup>14</sup>Adapted from W. H. Press et. al, Numerical Recipes in C, 2nd edition, Cambridge University Press, Cambridge, 1992, pp. 359–362

## 11.18 Linear Regression

The *XY* class performs least squares linear regression. The *XY* constructors takes as their arguments vector< double >s containing the x- and y-coordinates, and if present, the x- and y-standard deviations:<sup>15</sup>

- *XY*( vector< double >& *x*, vector< double >& *y* ): loads an XY dataset with x-coordinates *x* and y-coordinates *y* without standard deviations
- *XY*( vector< double >& *x*, vector< double >& *y*, vector< double >&  $\sigma_x$  ): loads an XY dataset with x-coordinates *x*, y-coordinates *y*, and *x* standard deviations  $\sigma_x$
- *XY*( vector< double >& *x*, vector< double >& *y*, vector< double >&  $\sigma_x$ , vector< double >&  $\sigma_y$  ): loads an XY dataset with x-coordinates *x*, y-coordinates *y*, *x* standard deviations  $\sigma_x$ , and *y* standard deviations  $\sigma_y$

The following accessors return results of least squares linear regression on XY dataset *D* (all of type double):

- *D.a*() : returns coefficient *a* (y-intercept) in least squares linear fit  $y = a + bx$
- *D.b*() : returns coefficient *b* (slope) in least squares linear fit  $y = a + bx$
- *D.x0*() : returns x-intercept  $x_0$  in least squares linear fit  $y = a + bx$
- *D.siga*() : returns standard errors on *a*
- *D.sigb*() : returns standard errors on *b*
- *D.chi2*() : returns minimized chi-squared value of least squares fit
- *D.q*() : returns chi-squared probability of least squares fit (small is a poor fit)
- *D.r*() : returns correlation coefficient *r*

### 11.18.1 Notes

Errors are caught by *XYErr* (see below for examples).

---

<sup>15</sup>Adapted from W. H. Press et. al, Numerical Recipes in C, 2nd edition, Cambridge University Press, Cambridge, 1992, pp. 661–670

### 11.18.2 Examples

```
double xs[ 5 ] = { 1.0, 2.0, 3.0, 4.0, 5.0 };
double ys[ 5 ] = { 1.0, 2.2, 3.0, 4.0, 6.0 };
double stdxs[ 5 ] = { 0.2, 0.4, 0.2, 0.2, 1 };
double stdys[ 5 ] = { 0.2, 0.3, 0.2, 0.2, 0.8 };

vector< double > xv( xs, xs + 5 ), yv( ys, ys + 5 ), stdx( stdxs,
stdxs + 5 ),
    stdy( stdys, stdys + 5 );

try {
    XY xyset( xv, yv );
    cout << "For dataset without errors, a = " << xyset.a() << endl;
    cout << "b = " << xyset.b() << endl;
    cout << "x0 = " << xyset.x0() << endl;
    cout << "siga = " << xyset.siga() << endl;
    cout << "sigb = " << xyset.sigb() << endl;
    cout << "chi2 = " << xyset.chi2() << endl;
    cout << "q = " << xyset.q() << endl;
    cout << "r = " << xyset.r() << endl << endl;
}
catch ( XY::XYErr& e ) { cout << e.what() << endl;

try {
    XY xyset2( xv, yv, stdx );
    cout << "For dataset with x errors only, a = " << xyset2.a() <<
endl;
    cout << "b = " << xyset2.b() << endl;
    cout << "x0 = " << xyset2.x0() << endl;
    cout << "siga = " << xyset2.siga() << endl;
    cout << "sigb = " << xyset2.sigb() << endl;
    cout << "chi2 = " << xyset2.chi2() << endl;
    cout << "q = " << xyset2.q() << endl;
    cout << "r = " << xyset2.r() << endl << endl;
}
catch ( XY::XYErr& e ) { cout << e.what() << endl;

try {
    XY xyset3( xv, yv, stdx, stdy );
    cout << "For dataset with x & y errors, a = " << xyset3.a() <<
endl;
    cout << "b = " << xyset3.b() << endl;
    cout << "x0 = " << xyset3.x0() << endl;
    cout << "siga = " << xyset3.siga() << endl;
    cout << "sigb = " << xyset3.sigb() << endl;
```

```
    cout << "chi2 = " << xyset3.chi2() << endl;
    cout << "q = " << xyset3.q() << endl;
    cout << "r = " << xyset3.r() << endl << endl;
}
catch ( XY::XYErr& e ) { cout << e.what() << endl; }
```

## Chapter 12

# Object Design

The design of the network object pursues 2 objectives. The first is a well organized, object oriented representation of a generic network object that is flexible and easily extended to specific network types. In fact, the current design allows for all models, including statistical ones.

The second objective of the network object design is that the code be readable. This was the logic of constructing efficient yet readable matrix and vector operations, as described in Chapter 9.

Figure 12.1 demonstrates the current inheritance hierarchy and the services which compose experiments around it. Rounded boxes are objects in the inheritance or ownership hierarchy. Component-shaped boxes are namespace services or helper objects; their arrows show use and data flow rather than inheritance.

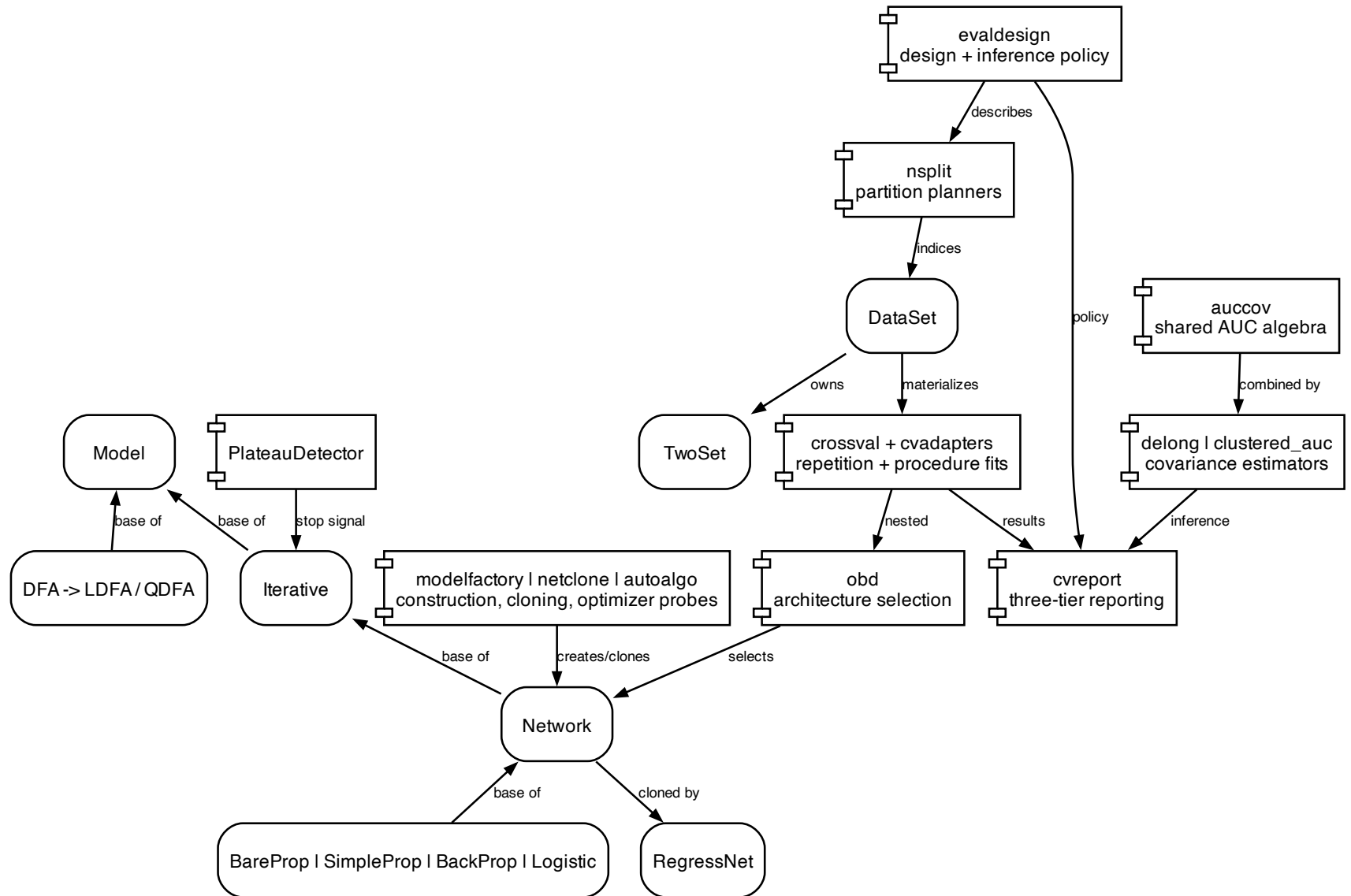


Figure 12.1: Object design

- **DataSet** contains and manipulates the data to be modeled.
- **TwoSet** is a specialized object designed to calculate metrics for *one* output *Models*, including sensitivity, specificity, predictive value positive, predictive value negative, and ROC area.
- **Model** is the base object. *Model* provides access to the dataset, and contains the virtual methods common to all modeling techniques.
- **Iterative** extends *Model*. Neural computational algorithms are iterative, and thus use *Iterative* as a base. *Iterative* manages methods and parameters related to iterating through a dataset.
- **Network** extends *Iterative*. Algorithms with hidden layers, weights, learning rates, transfer functions and other general properties of neural networks are encapsulated in *Network*.
- **RegressNet** is a compound object including a *Network* object which implements stepwise regression. See Chapter 13.
- **BackProp** extends *Network*. *BackProp* is a multilayer multiple output back-propagation neural network.
- **SimpleProp** also extends *Network*. *SimpleProp* is a backpropagation neural network which has 1 output node, 1 hidden node layer, and biases.
- **BareProp** also extends *Network*. *BareProp* is an example *Network* object for illustrative purposes which has 1 output node, 1 hidden node layer, and no biases.
- **Logistic** also extends *Network*. *Logistic* implements binary logistic regression, which is mathematically equivalent to a single output node neural network with a bias, the sigmoidal transfer function, no hidden nodes, and the cross-entropy error function.
- **DFA** extends *Model*, and encodes discriminant function analysis. **QDFA** extends *DFA*, and encodes quadratic discriminant function analysis. **LDFA** extends *DFA*, and encodes linear discriminant function analysis.

## 12.1 Current object and service map

Figure 12.1 includes both the inheritance hierarchy and the modern services that compose experiments around it. A component-shaped box is not a new base class: it is a namespace service or helper object with one explicit responsibility. The arrows describe use or ownership, not inheritance unless the label says “base of.”

The important division is between mechanisms and coordination. **DataSet** owns data and materializes row partitions; **nsplit** chooses indices; **Network** objects fit models; **obd** selects an architecture; **crossval** repeats a supplied

procedure; `evaldesign` records what design was achieved and which inference it permits; the AUC covariance modules estimate uncertainty; and `cvreport` renders and writes results. No outer coordinator reimplements a lower layer's rule.

### 12.1.1 Current additions to DataSet

The original `DataSet` contract earlier in this chapter remains valid. The following public methods extend it for validation and planned partitions:

- `valLoaded()`, `getValMatrix()`, and `getNumVal()` expose the optional validation set.
- `monitorSet()` returns `MONITOR_NONE`, `MONITOR_VALIDATION`, or `MONITOR_TEST`; `monitorSetName()` returns the same choice for reports.
- `randomize3(nTest,nVal)` and `randomize3D(testRatio,valRatio)` create an outcome-stratified train/validation/test split.
- `makeFold(trainRows,testRows,valRows)` materializes explicit raw-row indices and computes every scale from the training rows only.
- `setStrataColumns`, `getStrataColumns`, `setStrataBins`, and `getStrataBins` configure outcome-by-covariate splitting.
- `setGroupColumns` and `getGroupColumns` configure exact joint-value groups which may not be divided.
- `strataKey(columns,bins)` returns dense outcome-by-covariate cell identifiers; `groupKey(columns)` returns dense exact-match group identifiers. Both read the raw matrix without changing configuration.

#### Example

```
vector<unsigned> group = data.groupKey({2, 3, 4});
nsplit::GroupFoldPlan p =
    nsplit::stratifiedGroupKFold(label, group, 5);
data.makeFold(trainRows, testRows, validationRows);
cout << data.monitorSetName();          // "validation"
```

#### Notes

Column positions passed by C++ callers are zero-based input positions. Public REST fields are one-based and are validated and converted by the handler. Grouping takes precedence over covariate stratification in the ordinary raw split. A three-way split is outcome-stratified and does not currently compose with either policy.

## 12.2 Current contracts added to established objects

The detailed specifications later in this chapter describe the established objects. The following additions are part of their current public contracts and must be read with those specifications.

### 12.2.1 Model

The following method exposes the objective needed by likelihood-based callers:

- `bool getXError() const`: returns `true` when cross-entropy is active and `false` for least-mean-square error.

Callers whose mathematics depends on likelihood, notably stepwise Wilks comparisons, query this state and refuse an incompatible source fit; they do not mutate the model's objective to make it appear compatible.

### 12.2.2 TwoSet

`TwoSet` now exposes explicit cache invalidation and structured ROC results. Its empirical AUC is a ranking probability: the probability that a random outcome-1 row receives a higher score than a random outcome-0 row, with half credit for a tie. This is the normalized Mann–Whitney statistic, not an area obtained by choosing an arbitrary number of thresholds.<sup>1</sup>

The bootstrap interval repeatedly resamples outcome-0 and outcome-1 rows within their own classes, recomputes the fitted binormal area, and reports its 2.5th and 97.5th empirical percentiles. Resampling is stratified so a replicate cannot lose an outcome class merely by chance. This is the ordinary nonparametric bootstrap applied to the statistic, with a repository-specific stratification rule.<sup>2</sup> The following methods are available:

- `void invalidate()`: clears every cached statistic after a caller changes predictions through a mutable accessor.
- `unsigned getTP(), getTN(), getFP(), and getFN()`: return the four confusion-matrix counts at the current threshold.
- `double getTrapROCarea()`: returns the exact empirical Mann–Whitney AUC over every distinct operating point.
- `const vector<double>& getROCx() and getROCy()`: return the empirical ROC coordinates, 1–specificity and sensitivity.

---

<sup>1</sup>J. A. Hanley and B. J. McNeil, “The meaning and use of the area under a receiver operating characteristic (ROC) curve,” *Radiology* 143(1), 29–36, 1982, <https://doi.org/10.1148/radiology.143.1.7063747>.

<sup>2</sup>B. Efron, “Bootstrap methods: another look at the jackknife,” *The Annals of Statistics* 7(1), 1–26, 1979, <https://doi.org/10.1214/aos/1176344552>.

- `ROCfit getROCfit()`: returns the current binormal ROC fit as a structured value; an unsupported fit has `valid=false`.
- `void bootstrapROC()`: performs the configured number of outcome-stratified bootstrap resamples and caches a percentile interval for the binormal area.
- `CI getStatCi()`: returns the cached bootstrap interval.
- `void setBootstrapResamples(unsigned n)`: sets the requested resample count; zero disables the bootstrap.
- `unsigned getBootstrapResamples()`: returns that count.
- `unsigned getStatPoints()`: returns the number of distinct interior operating points fitted by the zROC line.
- `double getStatChi2()`: returns the zROC fit chi-square.
- `double getPearsonX2()`: returns the individual-level Pearson statistic; it is not assigned a chi-square p-value on continuous data.

`ROCfit` contains the following public variables:

- `double az`: fitted binormal ROC area.
- `double chi2`: zROC fit chi-square, or not-a-number when it cannot be computed.
- `double p`: fit p-value, or not-a-number when unavailable.
- `unsigned points`: operating points included in the fit.
- `bool valid`: whether a fit was computed.

`CI` contains:

- `double lo, hi`: 2.5th and 97.5th percentiles.
- `double se`: standard deviation of successful resampled areas.
- `unsigned resamples`: successful resamples.
- `unsigned failures`: resamples for which no area was computable.
- `bool valid`: whether enough resamples succeeded.

### 12.2.3 Iterative

StopReason identifies why the last run ended:

- `STOP_NONE`: training has not run.
- `STOP_MIN_ERROR`, `STOP_CHANGE`, `STOP_WINDOW`, `STOP_GRADMAX`, `STOP_PLATEAU`, and `STOP_EARLY_STOP`: a configured stopping rule fired, so the run converged under its declared rule.
- `STOP_MAX_ITERATIONS`: the safety ceiling was exhausted; this is not convergence.
- `STOP_CANCELLED`: an external request stopped the fit.
- `STOP_PROBE_BUDGET`: an optimizer probe used its wall-clock allowance; the probe is an experiment, not a completed fit.

The following methods expose and control this state:

- `static const char* stopReasonToken(StopReason r)`: returns the single machine-readable name used in reports and JSON.
- `static bool converged(StopReason r)`: returns `true` only for the six stopping-rule outcomes listed above.
- `void setObserver(Observer* obs)` and `Observer* getObserver()`: install or return the non-owning iteration observer.
- `StopReason getStopReason()`: returns the last run's reason.
- `unsigned getIterations()`: returns iterations completed by the last run.
- `void setAutoStop(bool flag, double tol, unsigned win)`: enables or disables plateau stopping and sets its tolerance and window.
- `bool getAutoStop()`, `double getAutoStopTol()`, and `unsigned getAutoStopWindow()`: return that configuration.
- `void setQuiet(bool flag)` and `bool getQuiet()`: suppress or query narration and the reporting epilogue. Quiet mode never suppresses a calculation used by fitting or stopping.

An `Iterative::Observer` supplies:

- `bool onIteration(unsigned iteration, double setError)`: receives every completed iteration and returns `false` to stop.
- `StopReason whyStopped() const`: identifies why the observer returned `false`; the conservative default is cancellation.

### 12.2.4 Network and concrete models

The following current methods supplement the detailed **Network** specification later in this chapter:

- **double sampleTestError(unsigned stride)**: returns mean error over every **stride**-th exemplar in the authoritative validation/test monitoring set without changing cached statistics; returns  $-1$  when no one-output monitoring set exists.
- **double getCondNum() const**: returns the last computed condition number, or  $-1$  before calculation.
- **double getCondMaxEig() const** and **getCondMinEig() const**: return the extreme eigenvalues used in that condition number.

**OneHiddenNet** sits between **Network** and the two one-hidden-layer models.

**SimpleProp** and **BareProp** describe the same architecture — one hidden layer and one output node — differing only in whether bias nodes are present. They therefore carry identical state, and several of their methods differed by nothing but the class name. **OneHiddenNet** owns that state (the hidden weight, update, and gradient matrices; the input, hidden-output, output-weight, hidden-error, and output-gradient vectors; the output error term; and the hidden-unit count) and implements what does not depend on the bias architecture:

- **virtual void setHidden(unsigned n)**: pure virtual. Sizing is the bias architecture, so each concrete model sizes its own structures; **load** calls this method rather than deciding widths itself.
- **virtual void randomize()**: assigns uniform random weights within the configured limit and marks the weights set.
- **virtual bool save(string& filename)**: writes the model file — the concrete class's **outputHeader**, the hidden layer, then the output layer.
- **virtual bool load(string& filename)**: reads that file back, refusing a file whose input count disagrees with the loaded dataset.
- **virtual void pack()**: flattens the hidden and output gradients into the packed gradient vector the conjugate-direction optimizers use.
- **virtual double innerTrainSet()**: performs one training pass over the training set and returns the set error. This is the training loop itself — on-line and batch, canonical descent and the separate-gradient path, weight decay, and the optimizer call — for both models.
- **void propagate()**: applies the forward equations to the exemplar already in the input vector, with the hidden bias slot already established if the model has one. Non-virtual, and called directly by each concrete **forward()**: this is a per-exemplar path and may not acquire indirect calls in order to remove source duplication.

The training loop and the forward equations are shared with no parameterization at all, rather than with a bias flag. Their one differing statement, the hidden error term, is the same formula once its domain is written out: elements 0 through  $n_{\text{hidden}} - 1$  are the hidden units, which is every element of an unbiased hidden-output vector and all but the pinned bias slot of a biased one. The ranged operations of **Matrix** and the vector operators express that domain directly.

Three things deliberately remain in the concrete classes. **df()** is two published parameter counts,  $((n_{\text{input}} + 2)n_{\text{hidden}}) + 1$  with biases and  $(n_{\text{input}} + 1)n_{\text{hidden}}$  without. **outputHeader()** writes a model-file format line that **load** reads back. Every width that follows from the bias column and the pinned bias slot — **setDataSet**, **setHidden**, **unpack**, **removeInputs**, and the statements of **forward()** that read the exemplar and, in **SimpleProp** alone, pin the bias slot — stays where the biased and unbiased architectures can be read side by side.

Two notes matter to callers. **SimpleProp** and **BareProp** remain distinct concrete types: type identity drives cloning, the model factory, and the first line of every saved model file, and neither is assignable to the other. Nothing in the shared implementation reads **Network::biasFlag**; that flag is publicly settable, and a shared method that consulted it could reshape a live model or change what it writes to disk. The bias architecture is a property of the concrete type.

**SimpleProp** adds the following OBD mechanisms:

- **void growHidden(unsigned extra):** adds hidden units as a warm start, preserving existing weights and assigning each new unit zero outgoing contribution.
- **void removeHidden(const vector<unsigned>& indices):** removes the named non-bias hidden units while preserving the order and weights of survivors; at least one unit must remain.
- **vector<double> hiddenSaliency():** returns one saliency per non-bias hidden unit,  $|w_j|$  times the training-set standard deviation of that unit's output.

**Logistic** exposes its last coefficient audit:

- **const vector<double>& getBetas() const:** returns fitted coefficients; the last element is the intercept.
- **const vector<double>& getBetaSE() const:** returns coefficient standard errors;  $-1$  marks an unavailable value.
- **const vector<double>& getWaldP() const:** returns coefficient Wald-test p-values.

### 12.2.5 DFA, LDFA and QDFA

DFA is the discriminant-analysis base. It owns the class matrices, mean vectors, a priori probabilities and constants that both concrete analyses build, and since 2026 it also owns the run itself:

- **double train() override final**: performs one analysis. It writes the banner and header, calls the fit and then the report, catches a singular covariance and reports the refusal, writes the history and last-operation files, and returns `-1` because a discriminant analysis has no set error. It is **final** because no subclass may replace the scaffold, and it remains a virtual override of `Model::train()` — cross-validation reaches it through a `unique_ptr<Model>`.
- **virtual void fitDiscriminant()**: pure virtual, protected. The fit, and only the fit. Runs once per analysis, inside the scaffold's **try**.
- **void reportAccuracy(ostream&) override = 0**: pure virtual. Each model writes this out in full.

LDFA fits one *pooled* covariance across all classes, inverts it, and forms constants  $K_c = \frac{1}{2} \mathbf{u}_c^T S \mathbf{u}_c - \log P_c$ . Its discriminant is  $d_c = \mathbf{u}_c^T S \mathbf{x} - K_c$  and the *larger* discriminant wins; the binary graded score is  $\sigma(d_1 - d_0)$ .

QDFA fits a *separate* covariance per class, inverts each, and forms constants  $K_c = \log |C_c| - \log P_c$ . Its discriminant is  $d_c = (\mathbf{x} - \mathbf{u}_c)^T S_c (\mathbf{x} - \mathbf{u}_c) + K_c$  and the *smaller* discriminant wins, because it is a distance where the linear one is a similarity; the binary graded score is  $\sigma(d_0 - d_1)$ .

Those opposite senses are why only the scaffold is shared. Sharing the per-exemplar scoring loops would require either a virtual call per exemplar or a comparator parameter, and the base class deliberately contains no sign, winner flag, formula descriptor or type switch: the published mathematics stays in the class that owns it. Both models write a *graded* score through the logistic function rather than a hard decision, so the ROC sweep sees a curve; the classification boundary is unchanged.

### 12.2.6 RegressNet

The following methods support non-interactive, cancellable stepwise analysis:

- **void setThreshold(double t)**: supplies the p-value threshold without an interactive prompt.
- **void setProgress(ProgressFn f)**: installs the progress callback.
- **void setObserver(Iterative::Observer\* obs)**: installs the non-owning cancellation observer on every candidate clone.
- **const vector<Candidate>& getCandidates() const**: returns every candidate in evaluation order.

- `const vector<pair<double,unsigned> >& getSelectionPath()`  
`const`: returns selected p-value/variable pairs in selection order.
- `const vector<unsigned>& getFinalVariables()` `const`: returns the variables settled by completed passes.
- `unsigned getFitsCompleted()` `const`: returns candidate fits made.
- `bool getComplete()` `const`: distinguishes a completed selection from a partial audit after failure or cancellation.

**Progress** contains direction, pass number, candidate number and pass total, completed-fit count, conceptual variable, its full input-node group, phase, completion flag, convergence flag, and stop-reason token.

**Candidate** contains pass/candidate/variable identity, the full input group, prior and candidate errors, degrees of freedom,  $G^2$ , p-value, convergence, stop reason, iterations, and whether it won its pass.

### Example

```
network.setAutoStop(true, 1e-4, 100);
double error = network.train();
if (!Iterative::converged(network.getStopReason())) rejectFit();

TwoSet::ROCfit fit = data.getTestTwoSet().getROCfit();
const vector<double>& p = logistic.getWaldP();
```

### Notes

Reaching `STOP_MAX_ITERATIONS` is operation completion but not fit convergence. Quiet mode changes output only. The ROC bootstrap uses an independent resampling stream and cannot perturb training. Stepwise candidate fits must converge under cross-entropy before entering a Wilks comparison, and the original model is never mutated.

## 12.3 Partition planning: nsplit

The `nsplit` namespace is an index-level service. It never scales data, fits a model, or interprets an input column. Its allocation and validation rules are project-defined planning mechanisms, not a statistical test or a claim that one universally optimal split exists. They make three policies explicit: preserve requested strata as closely as indivisible rows permit, keep a declared group wholly on one side, and audit the completed assignment rather than trusting the construction procedure. “Largest remainder” below means that each stratum first receives its integer quota and the remaining test positions go to the largest fractional quotas until the requested total is reached.

**Holdout** contains:

- `vector<unsigned> test, train`: row positions assigned to each side.
- `unsigned n0, n1`: whole-sample outcome counts.
- `unsigned n0Test, n1Test`: achieved test-set outcome counts.

`StratHoldout` contains:

- `vector<unsigned> test, train`: row positions assigned to each side.
- `vector<unsigned> cellTotal`: whole-sample count in each stratum.
- `vector<unsigned> cellTest`: achieved test count in each stratum.

`GroupHoldout` contains:

- `vector<unsigned> test, train`: row positions assigned to each side, with every group wholly on one side.
- `unsigned nGroups`: number of distinct groups.
- `vector<unsigned> groupsInTest`: group identifiers assigned to the test side.

`FoldPlan` contains:

- `bool ok` and `string reason`: planning status and refusal.
- `vector<unsigned> foldId`: fold number for every input row.
- `unsigned k, nStrata`: requested folds and observed strata.
- `vector<unsigned> foldSize`: achieved rows per fold.
- `vector<vector<unsigned> > stratumByFold`: achieved stratum counts by fold.
- `vector<string> warnings`: non-fatal planning facts.

`GroupFoldPlan` extends `FoldPlan` with:

- `unsigned nGroups`: distinct groups in the planned rows.
- `vector<unsigned> groupsPerFold`: whole groups assigned per fold.
- `vector<array<unsigned,2> > classByFold`: negative and positive rows achieved per fold.
- `unsigned leakageCount`: groups found in more than one fold; a valid plan has zero.
- `double imbalanceScore`: worst relative row or class imbalance.

- `unsigned largestGroup`: size of the largest indivisible group.

The following functions are available:

- `string partitionError(unsigned nRows, train, test, bool coverage)`: returns an empty string for a valid two-way row partition, otherwise a reason describing an out-of-range, duplicate, overlapping, or missing row.
- `Holdout stratifiedHoldout(label, nTest)`: returns an outcome-stratified train/test partition of binary labels.
- `StratHoldout holdoutByStrata(stratum, nTest)`: apportions test positions across arbitrary stratum identifiers by largest remainder.
- `GroupHoldout groupHoldout(label, group, nTest)`: assigns each complete group to training or test while approaching the requested size and outcome balance.
- `FoldPlan stratifiedKFold(stratum, k)`: returns a balanced k-fold assignment over arbitrary stratum identifiers.
- `GroupFoldPlan stratifiedGroupKFold(label, group, k)`: returns an outcome-balanced whole-group fold assignment with measured leakage and imbalance.
- `vector<unsigned> kFold(label, k)`: returns the outcome-only case of `stratifiedKFold`.

### Example

```
util::set_seed(42);
nsplit::GroupFoldPlan plan =
    nsplit::stratifiedGroupKFold(label, group, 5);
if (!plan.ok || plan.leakageCount != 0)
    throw runtime_error(plan.reason);
```

### Notes

Every plan is reproducible under the engine seed. Group identifiers are merely equivalence labels; renumbering them must not change the partition. Achieved fold counts and leakage are recomputed from the final assignment. Indivisible groups may make exact sizes or exact class balance impossible, so imbalance is reported rather than hidden. Fewer than `k` groups is a refusal.

## 12.4 Typed evaluation design: evaldesign

`evaldesign` is a project-defined vocabulary and policy layer; it computes no statistic. It records what was actually partitioned and what the independent sampling unit is, then permits only the covariance method whose assumptions match those declarations. The purpose is to prevent a GUI, report writer or caller from inferring statistical independence from a loose string such as “grouped,” or silently choosing an estimator merely because one is available. Its enumerations are:

- `SamplingUnit`: `Unspecified`, `Row`, or `Cluster`, declaring the independent sampling unit.
- `AucInference`: `None`, `DeLongIndependent`, or `ObuchowskiClustered`, naming the permitted estimator.
- `PartitionMethod`: `OutcomeStratified`, `CovariateStratified`, or `StratifiedGroup`, naming how rows were assigned.

`Partition` contains the following public variables:

- `PartitionMethod` method: assignment method.
- `unsigned seed`: reproducibility seed.
- `unsigned k`: number of folds; zero for a holdout.
- `vector<unsigned> strataColumns`: caller’s one-based covariate columns; `strataBins` and `nStrata` give bin count and achieved cell count.
- `vector<unsigned> groupColumns`: caller’s one-based group columns; `nGroups` gives distinct groups in the partitioned rows.
- `bool developmentOnly`: whether the folds exclude a locked test.
- `unsigned nRequested`, `nAchieved`: requested and achieved holdout sizes; indivisible groups may make them differ.
- `unsigned leakage`: groups or rows found on both sides; a valid group plan requires zero.
- `vector<string> warnings`: non-fatal design facts.
- `string refusal`: reason no partition could be built.
- `foldRows`, `foldEvents`, `foldGroups`: parallel achieved counts for each fold.
- `double imbalanceScore`: worst relative deviation from a fold’s fair row or class share.

- `unsigned largestGroup`: largest indivisible cluster size.

`InferenceChoice` contains `AucInference` method and `string reason`; the reason is required when the method is `None`.

The following functions are available:

- `bool parseSamplingUnit(string text, SamplingUnit& out)`: parses the REST value `rows`, `cluster`, or an empty declaration; returns `false` for an unrecognized token.
- `samplingUnitName`, `samplingUnitPhrase`, and `independenceStatus`: return the authoritative display wording for a sampling-unit declaration.
- `inferenceName`, `inferenceText`, and `inferenceShortName`: return long, reason-bearing, and compact names for an inference choice.
- `partitionMethodName`: returns the display name of a partition method.
- `string describeFolds(const Partition& p)`: derives a fold-plan sentence from structured fields.
- `string describeHoldout(const Partition& p)`: derives the corresponding holdout sentence, including requested and achieved sizes.
- `InferenceChoice chooseInference(unit, split, estimable, reason)`: returns the only covariance estimator permitted by the declared unit and achieved split, or `None` with a reason.

Display strings must come from these functions rather than being composed by a GUI or report.

### Example

```
evaldesign::InferenceChoice c = evaldesign::chooseInference(
    evaldesign::SamplingUnit::Cluster,
    evaldesign::PartitionMethod::StratifiedGroup,
    clustered.ok, clustered.reason);
```

### Notes

A group-disjoint split prevents leakage; it does not establish row independence. An unspecified unit intentionally withholds inferential claims. Independent-row inference over a grouped design is refused, and neither covariance estimator may silently substitute for the other.

## 12.5 Shared AUC covariance algebra: `auccov`

`auccov` owns the operations that are invariant to the sampling unit. For a positive score  $X$  and negative score  $Y$ , it computes

$$A = P(X > Y) + \frac{1}{2}P(X = Y),$$

the probability that the model ranks a randomly chosen positive row above a randomly chosen negative row, with half credit for ties. In the sample this is the Mann–Whitney pair count divided by  $n_1n_0$ .<sup>3</sup> The per-positive and per-negative pair averages are the structural components called `v10` and `v01`. They are retained because both covariance estimators below use them; only the rule for combining their variation changes.

`Placements` contains:

- `bool ok` and `string reason`: computation status.
- `unsigned n0, n1`: negative and positive row counts.
- `vector<unsigned> pos, neg`: original row indices by outcome.
- `vector<vector<double>> v10, v01`: procedure-by-observation structural components.
- `vector<double> auc`: exact point AUC for each procedure.

`Interval` contains point `auc`, standard error `se`, limits `lo/hi`, and `valid`.

`Contrast` contains the two areas and their standard errors, difference `delta` and `seDelta`, each area’s interval, the difference interval, `z`, two-sided `p`, zero-variance flags `degenerate/separated`, explanatory `note`, and `valid`.

The following functions are available:

- `vector<double> midranks(const vector<double>& x)`: returns one-based ranks; tied observations receive the mean rank they span.
- `Placements compute(label, pred)`: returns positive/negative row indices, structural components, and exact point AUCs for paired procedures.
- `Interval interval(auc, cov, i, bool clamp)`: returns the normal interval for area `i`; `clamp` controls truncation to  $[0, 1]$ .
- `Contrast contrast(auc, cov, i, j, bool clamp)`: returns  $AUC_i - AUC_j$ , its standard error, interval, `z` statistic, and two-sided `p`-value.

The estimator above this layer supplies the covariance matrix; it does not reimplement these operations.

---

<sup>3</sup>H. B. Mann and D. R. Whitney, “On a test of whether one of two random variables is stochastically larger than the other,” *The Annals of Mathematical Statistics* 18(1), 50–60, 1947, <https://doi.org/10.1214/aoms/1177730491>.

### Example

```
auccov::Placements p = auccov::compute(label, predictions);
auccov::Contrast c =
  auccov::contrast(p.auc, covariance, 0, 1, false);
```

### Notes

Ties receive half credit through common mid-ranks, so the ordinary and clustered paths have identical point AUCs. A zero-variance, zero-difference contrast has  $p=1$ ; a zero-variance nonzero difference is deterministic separation with  $p=0$ . A materially negative difference variance is refused.

## 12.6 AUC covariance estimators

### 12.6.1 delong

DeLong, DeLong and Clarke-Pearson answer a specific question: when two or more procedures score the *same independent subjects*, how uncertain is each empirical AUC, and how uncertain is the difference between two correlated AUCs? Treating the areas as independent would discard the pairing and generally give the wrong standard error. Their nonparametric method views each AUC as a two-sample U-statistic and estimates the full procedure-by-procedure covariance from the positive and negative structural components.<sup>4</sup>

If  $S_{10}$  is the sample covariance of the per-positive components and  $S_{01}$  the sample covariance of the per-negative components, the implemented estimator is

$$\widehat{\text{Cov}}(A) = \frac{S_{10}}{n_1} + \frac{S_{01}}{n_0}.$$

The diagonal entries are variances of individual AUCs; off-diagonal entries preserve the correlation between procedures evaluated on the same rows. A contrast uses  $\text{Var}(A_i - A_j) = C_{ii} + C_{jj} - 2C_{ij}$ , then reports the normal z statistic and two-sided p-value. “DeLong” therefore names the covariance estimator, not a different point-AUC calculation.

`delong::Result` contains:

- `unsigned n0, n1`: negative and positive row counts.
- `vector<double> auc`: one exact point area per procedure.
- `Matrix<double> cov`: paired DeLong covariance matrix.
- `bool ok` and `string reason`: estimator status and refusal.

The following functions are available:

---

<sup>4</sup>E. R. DeLong, D. M. DeLong and D. L. Clarke-Pearson, “Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach,” *Biometrics* 44(3), 837–845, 1988, <https://doi.org/10.2307/2531595>.

- `Result analyze(label, pred)`: estimates paired DeLong covariance for procedures scored on the same independent rows.
- `Interval interval(const Result& r, unsigned i)`: returns the normal DeLong interval for procedure  $i$ , clamped to  $[0, 1]$ .
- `Contrast contrast(const Result& r, unsigned i, unsigned j)`: returns the paired area contrast  $AUC_i - AUC_j$ .

### 12.6.2 clustered\_auc

Ordinary DeLong assumes rows are independent. That is false when several rows belong to one patient, site, family or other sampling unit: those rows may move together, so counting them as independent can make an interval or comparison too confident. Obuchowski extends DeLong’s structural-component construction to clustered ROC data by centering and summing the positive and negative components within each cluster and adding the within-cluster cross-covariance term that the independent-row formula lacks.<sup>5</sup>

The point AUC remains the same row-level ranking probability. What changes is its uncertainty: the independent replication count is the number of informative clusters, not the number of rows. A cluster containing many rows supplies more pairwise information but does not become many independent experimental units. This is why the implementation requires at least two clusters carrying each outcome class and why a group-disjoint split and clustered inference solve different problems.

`clustered_auc::Result` contains:

- `bool ok` and `string reason`: estimator status and refusal.
- `unsigned nRows, nClusters`: row and independent-cluster counts.
- `unsigned clusters10, clusters01`: clusters carrying at least one positive or negative row; these are the covariance divisors’ populations.
- `unsigned n0, n1`: negative and positive row counts.
- `vector<double> auc`: exact patient-row point areas.
- `Matrix<double> cov`: Obuchowski clustered covariance matrix.
- `string method`: authoritative estimator name for reports.

The following functions are available:

- `Result analyze(label, cluster, pred)`: estimates Obuchowski’s non-parametric clustered covariance for paired procedures.

---

<sup>5</sup>N. A. Obuchowski, “Nonparametric analysis of clustered ROC curve data,” *Biometrics* 53(2), 567–578, 1997, <https://doi.org/10.2307/2533958>.

- `Interval interval(const Result& r,unsigned i):` returns the unclamped published normal interval for procedure `i`.
- `Contrast contrast(const Result& r,unsigned i,unsigned j):` returns the clustered paired area contrast.

### Example

```
delong::Result iid = delong::analyze(label, predictions);
clustered_auc::Result grouped =
    clustered_auc::analyze(label, cluster, predictions);
// The design policy decides which result may be reported.
```

### Notes

DeLong requires at least two rows of each outcome class and assumes independent rows. Obuchowski requires at least two clusters carrying each outcome class. Clustering can increase or decrease the estimated standard error; its defining property is that the cluster is the independent unit. Out-of-fold predictions are not an independent test sample for either locked-test contrast.

## 12.7 Training-control services

### 12.7.1 PlateauDetector

`PlateauDetector` is neuron's project-defined early-stopping heuristic, not a statistical convergence test. It maintains two adjacent moving-average windows of the training error and compares their relative change. A flat or rising newer window records a strike; the configured number of consecutive strikes requests a stop. Windowing prevents one noisy iteration from deciding the run, while patience requires the lack of improvement to persist. The following methods are available:

- `PlateauDetector(unsigned window,double tol,unsigned patience):` constructs a detector; `window` is at least two and `patience` at least one.
- `void reset():` clears both windows and the strike count.
- `bool update(double error):` adds one iteration error and returns `true` after the configured number of consecutive comparisons fail to achieve the required relative improvement.
- `unsigned strikes() const:` returns the current consecutive-strike count.

### 12.7.2 autoalgo

`autoalgo` is a project-defined bounded tournament among the three existing optimizers. Each optimizer starts from a clone of the same weights, runs for the same wall-clock allowance, and is ranked by the finite training error it reached; the winning clone, including the progress it already made, continues the real fit. It is an engineering choice under a time budget, not a claim that the selected optimizer is globally best.

`autoalgo::Probe` contains:

- `int algorithm` and `string name`: training-menu/API number (1 through 3) and display name.
- `double error`: final probe error.
- `unsigned iterations`: iterations completed within the probe.
- `bool usable`: whether the result may compete for selection.
- `Iterative::StopReason stop`: reason the probe ended.

`autoalgo::Result` contains:

- `int selected` and `string selectedName`: winning algorithm and display name.
- `vector<Probe> probes`: complete optimizer audit.
- `unique_ptr<Network> winner`: selected clone, including probe progress.
- `bool cancelled`: whether external cancellation ended selection.
- `Result pick(const Network& start, unsigned budgetMs, const atomic<bool>* cancel)`: probes canonical, CGD, and Shanno on clones of identical starting weights and returns the lowest-error usable clone.

### 12.7.3 modelfactory and netclone

`modelfactory` and `netclone` are project-defined construction mechanisms. They keep knowledge of concrete model types in one cold-path boundary: callers describe or clone a model without duplicating a type switch, while fitting remains virtual behavior owned by the constructed model.

`modelfactory::Spec` contains `logistic`, `bias`, hidden layer sizes, `xentropy`, `logLastop`, and `logHistory`.

- `unique_ptr<Model> build(const Spec& spec, DataSet& data, string& err)`: constructs and binds a fresh untrained model in the required order; returns null and fills `err` for an invalid request.

- `unique_ptr<Model> createByTypeName(string typeName, bool bias)`: creates the appropriate empty concrete type for a saved-network load; returns null for an unknown type name.
- `unique_ptr<Network> cloneNetwork(const Network& source)`: returns a deep copy of a known concrete network, including its dataset and training state; returns null for an unknown concrete type.

### Example

```
PlateauDetector stop(100, 1e-4, 3);
if (stop.update(trainingError)) requestStop();

modelfactory::Spec spec;
spec.logistic = true;
unique_ptr<Model> model = modelfactory::build(spec, data, error);
```

### Notes

A plateau is an eligible stopping rule; an iteration ceiling is not. Automatic optimizer probes are bounded experiments, not completed fits. Factory construction and cloning centralize type knowledge; training remains virtual behavior of the concrete model.

## 12.8 OBD architecture selection

Optimal Brain Damage originally uses a second-order approximation to predict the increase in training error caused by deleting a parameter, then removes parameters with low saliency to reduce an over-large network.<sup>6</sup> neuron applies that pruning idea at hidden-unit granularity and surrounds it with a project-defined validation search: it grows candidate hidden layers while held-out error improves, restores the best eligible size, then prunes by the engine’s saliency calculation. Thus `obd::run` is not a verbatim implementation of the paper’s entire procedure; the cited method supplies the saliency-based pruning principle, while the grow/validation/eligibility policy is neuron-specific and is specified below.

`obd::Eligibility` distinguishes `ELIGIBLE`, `INCOMPLETE_CEILING`, `INCOMPLETE_CANCELLED`, and `NUMERICAL_FAILURE`.

`SizeTrial` contains hidden-unit count, training and held-out errors and classification accuracies at the validation minimum, stop reason, growth/prune phase, eligibility, and iterations.

`Config` contains:

---

<sup>6</sup>Y. LeCun, J. S. Denker and S. A. Solla, “Optimal Brain Damage,” *Advances in Neural Information Processing Systems 2*, 598–605, 1990, <https://proceedings.neurips.cc/paper/1989/hash/6c9882bbac1c7093bd25041881277658-Abstract.html>.

- `hStart`, `hMax`: first and largest hidden-unit counts.
- `iterBudget`, `sampleEvery`: per-size ceiling and validation sampling cadence.
- `earlyStopTol`, `earlyStopPatience`: tolerated held-out reversal and consecutive samples required to stop a size.
- `growPatience`, `pruneTol`: growth and pruning decisions.
- `algorithm`: 0/1/2 for a fixed optimizer or `-1` for automatic selection.
- `plateauTol`, `plateauWindow`: training-error plateau backstop.

`Result` contains success/refusal message, selected hidden size, complete trial history, owning winner, cancellation and ceiling-exhaustion flags, resolved optimizer, and whether it was selected automatically.

The following functions are available:

- `Eligibility classify(StopReason stop, double score, double error)`: returns the authoritative trial-admission classification.
- `const char* stopToken(StopReason stop, Eligibility e)`: returns the machine-readable trial outcome.
- `Result run(DataSet& data, const Config& cfg, ProgressFn progress, const atomic<bool>* cancel)`: performs the grow-then-prune search and returns the owning selected network only after a valid search.

### Example

```
obd::Config cfg;
cfg.hMax = 8;
cfg.algorithm = -1; // automatic optimizer probe
obd::Result r = obd::run(data, cfg, nullptr, nullptr);
if (r.ok) current = std::move(r.winner);
```

### Notes

OBD requires one discrete output and a held-out monitoring set. Validation is preferred; the test set is only the two-way fallback. A needed trial that hits its ceiling is refused and never compared. In nested cross-validation the whole search, including optimizer choice, is repeated inside each fold.

## 12.9 Cross-validation objects

Cross-validation estimates out-of-sample behavior by partitioning the available rows into folds, fitting on all but one fold, predicting the held-out fold, and repeating until every row has exactly one out-of-fold prediction. Those pooled predictions describe the performance of the *procedure* that produced them, not a single model fitted once to all rows. When the procedure includes model selection, that selection must be repeated using only each outer fold’s training rows; otherwise the reported error is optimistically biased. Nested cross-validation supplies the inner selection loop and leaves the outer fold for assessment.<sup>7</sup>

The objects below separate those roles: a caller supplies one complete fitting procedure, `crossval` owns repetition and out-of-fold assembly, and `cvadapters` defines procedures whose tuning steps are contained within the fold. This ownership rule is the executable form of the statistical requirement above.

`crossval` owns repetition over an immutable fold assignment. `Metrics` contains:

- `unsigned n`: rows contributing to the calculation.
- `double az`: fitted binormal AUC.
- `double trap`: exact empirical AUC.
- `double sens, spec, ca`: sensitivity, specificity, and classification accuracy at the selected threshold.

`FoldResult` contains:

- `unsigned fold, n`: fold number and held-out row count.
- `bool ok` and `string reason`: fit status and refusal.
- `Metrics metrics`: the fold’s held-out statistics.

`RunResult` contains:

- `bool ok, cancelled` and `string message`: run status.
- `vector<FoldResult> folds`: one audit record per fold.
- `vector<double> prediction, outcome`: row-aligned out-of-fold predictions and outcomes.
- `unsigned validFolds, pooledN`: successful folds and pooled rows.
- `double pooledAz, pooledTrap`: pooled OOF binormal and exact AUCs.

The remaining coordination structures are:

---

<sup>7</sup>S. Varma and R. Simon, “Bias in error estimation when using cross-validation for model selection,” *BMC Bioinformatics* 7, 91, 2006, <https://doi.org/10.1186/1471-2105-7-91>.

- **ProcResult**: success, cancellation, refusal reason, and held-out predictions returned by one procedure invocation.
- **Progress**: stage, current procedure and total procedures, current fold and total folds.
- **FoldSelection**: selected hidden size and optimizer, plus whether automatic optimizer selection was used.
- **ProcedureSpec**: display name, procedure callback, and optional selection-audit sink.
- **Comparison**: shared outcome and fold arrays, fold count, and one timed **RunResult** entry per procedure.
- **LockedEntry**: procedure name, fit status and reason, paired test predictions, elapsed seconds, and selection audit.
- **LockedResult**: shared original test-row identities and outcomes, cancellation/message, and all **LockedEntry** values.

**Procedure** is the model-family callback which fits one materialized training fold and returns held-out predictions. **ProgressFn** reports the current procedure, fold, and stage.

The following functions are available:

- **Metrics** `metricsFor(outcome,pred,rows)`: calculates held-out classification and ROC metrics for the named rows; unavailable metrics are `-1`.
- **RunResult** `run(DataSet& data,foldId,Procedure proc,...)`: runs one procedure over every fold and scatters successful predictions back to their original row positions.
- **Comparison** `compare(DataSet& data,foldId,procs,...)`: runs several named procedures over one immutable fold plan and returns paired OOF results.
- **LockedResult** `evaluateOnce(DataSet& data,trainRows,testRows,procs,...)`: refits each procedure once on the development rows and scores the common untouched locked test.

### 12.9.1 cvadapters

**InnerSplit** contains `ok`, refusal `reason`, and raw-row `train` and `validation` vectors. The following functions adapt model families to the generic procedure contract:

- **Procedure** `trainProcedure(const Network& templateNet, unsigned maxIter)`: clones and trains a configured network from fresh weights in each fold.

- Procedure `dfaProcedure(bool quadratic)`: returns an LDFA adapter when false and QDFA adapter when true.
- Procedure `nestedObdProcedure(const obd::Config& cfg, double innerValFraction,...)`: performs the complete OBD architecture search within each outer training fold and returns only outer held-out predictions.
- `InnerSplit innerValidationSplit(trainRows,rowLabel,rowGroup, double fraction)`: returns a stratified inner split, group-disjoint when `rowGroup` is supplied; returns `ok=false` rather than falling back when the grouped split is infeasible.

### Example

```
crossval::ProcedureSpec logistic{
    "Logistic",
    cvadapters::trainProcedure(templateNet, 20000), nullptr};
crossval::Comparison c = crossval::compare(
    data, foldPlan.foldId, {logistic}, nullptr, true, 42);
```

### Notes

A failed fold contributes no fabricated prediction and remains visibly missing. All procedures share the same fold plan. Deterministic substreams key fits by seed, procedure name, and fold, so comparison membership and order cannot change a procedure's fit. Nested OBD's inner validation is group-disjoint under a grouped design; infeasibility is a failure, never a row-wise fallback.

## 12.10 Cross-validation reporting: `cvreport`

`cvreport` is neuron's project-defined presentation and artifact layer. It performs no fitting, fold assignment or covariance estimation. It receives already computed results and renders three levels of detail: a compact summary, an audit of every repetition and fold, and machine-readable files retaining raw-row identity. Keeping those jobs here ensures that the screen report, CSV/JSON artifacts and REST response describe the same run rather than recomputing or reinterpreting it independently.

`PlanInfo` contains the dataset label and whole-dataset totals, typed fold design, primary/reference names, optional per-row cluster identity, original raw-row identity, and original row count. Its methods are:

- `string foldPlanText() const`: derives the fold-plan heading from the typed partition.
- `string rawRowError(unsigned n) const`: returns an empty string for a valid absent/identity map or for exactly `n` distinct in-range raw-row identifiers; otherwise returns the defect.

LockedColumn contains:

- `string name`: procedure display name.
- `bool hasAuc` and `double auc`: point-area availability and value.
- `bool hasInterval` and `double lo, hi`: interval availability and limits.
- `string architecture`: selected architecture description.
- `string note`: refusal or qualification shown with the result.
- `vector<double> prediction`: predictions paired to locked rows.

LockedContrast contains:

- `bool hasPoint, hasInference`: availability of the AUC difference and of its inferential comparison.
- `string primary, reference`: contrasted procedure names.
- `double delta, p`: AUC difference and two-sided p-value.
- `bool significant`: whether the declared comparison crosses its significance rule.
- `bool degenerate, separated`: zero-variance case indicators.
- `string note`: refusal or qualification.

LockedInfo contains:

- `evaldesign::Partition split`: achieved locked-holdout design.
- `SamplingUnit samplingUnit`: declared independent unit.
- `AucInference inference`: covariance estimator permitted by the design.
- `string inferenceReason`: explanation when inference is withheld.
- `vector<unsigned> rawRow`: original locked-row identities.
- `vector<double> outcome`: locked outcomes.
- `vector<unsigned> cluster`: cluster identity for each locked row.
- `unsigned nClusters`: distinct clusters present in the locked sample.
- `vector<LockedColumn> columns`: procedure results.
- `LockedContrast contrast`: prespecified paired comparison.

`LockedInfo` derives every displayed design string from typed state. Its `clusterError()` method validates cluster identity before clustered inference or artifact writing.

`ArtifactResult` contains attempted artifact name and path, success, and error text. The following functions are available:

- `string tier1(cmp,info,locked)`: returns the one-screen headline table and verdict.
- `string tier2(cmp,info,locked)`: returns per-fold, selection, timing, design, and locked-test detail.
- `vector<ArtifactResult> writeArtifacts(cmp,info,dir,locked)`: writes `cv_predictions.csv`, `cv_metrics.csv`, `cv_run.json`, and, when applicable, `cv_locked_predictions.csv`; returns one status per attempted file.
- `void render(ostream& out,cmp,info,dir,locked)`: writes Tier 2, then Tier 1, then the machine-readable artifacts.

### Example

```
cvreport::PlanInfo info;  
info.folds = achievedFoldDesign;  
cvreport::LockedInfo locked;  
locked.split = achievedLockedDesign;  
cvreport::render(cout, comparison, info, runDirectory, locked);
```

### Notes

The reporter chooses neither partitions nor covariance estimators. Every design sentence is derived from typed state. Original raw-row and cluster identifiers are validated before artifacts are written so development and locked prediction files remain joinable. Tier 1 is the conclusion, Tier 2 is descriptive detail, and Tier 3 is the machine-readable audit substrate.

## 12.10.1 AsyncJob

`AsyncJob` owns the lifecycle every long GUI operation shares: training, hidden-layer sizing, cross-validation and stepwise regression. It answers one question — how a request hands work to a background thread and how the result comes back — and it answers it once. Before it existed the same launch sequence appeared at four call sites, character for character, differing only in the payload each thread carried.

- `bool start(function<string()> body)`: reaps the previous worker, clears every per-run progress field, clears the stop latch, marks the job running, and starts `body` on a thread. Returns false only when no worker

could be started, having first published a failure and released the job, so a failed launch cannot leave the engine owned forever.

- `bool requestStop()`: sets the cancellation latch, or returns false when nothing is running. Cancellation is cooperative: the engine polls the latch and finishes the current iteration.
- `void clearStopRequest()`: clears the latch without starting anything, for a blocking operation that shares its implementation with the asynchronous form and would otherwise inherit an earlier run's `Stop`.
- `void joinForShutdown()`: cancels and joins a worker that outlived the server loop.

### The ordering contract

The worker stores its result under the progress mutex and clears the running flag only after releasing it. A reader must therefore hold that mutex across *both* reads. Its critical section then totally orders against the worker's publication: either entirely before it, and the job still reads as running, or entirely after it, and the result is visible. Reading one of the two outside the lock allows a finished run to be reported as an idle one with nothing to show. The constraint binds the caller, which is why it is stated on the class rather than in the status handler that happens to satisfy it.

The lock order is the other half. The mutex is never held across a join, a thread construction, or an engine call. The join in particular precedes taking it: a departing worker's last act is to take that same mutex to publish, so joining while holding it would deadlock.

### The worker boundary

An exception escaping a thread's function calls `std::terminate`, which would end the server while the page was still polling for status. Every escape becomes a published failure instead — a `std::exception` carrying its own message, anything else a generic one — and the publish-then-clear order holds on the throwing paths too. How a failure message becomes a response body is supplied by the caller, so the class speaks no wire format.

#### 12.10.2 procguard

`procguard::run(body, err, program)` is the command-line program's top-level exception boundary: it runs the body and returns its exit status, and converts anything that reaches the top into a named diagnostic and status 1. Before it existed a contract failure ended the program with no message at all.

It is a separate unit from `AsyncJob` deliberately. Both are boundaries around an uncaught exception, but the command-line entry point has no worker, no cancellation and no result to publish — it has an exit status and a diagnostic,

and filing that under asynchronous-job behaviour would misdescribe it. Taking the body and the stream as arguments is what makes it testable: a test supplies a throwing body and an in-memory stream, and the shipped program gains no way to inject a fault.

## 12.11 Function definitions

For clarity and ease of use, function definitions are placed in the file “function\_defs.h”. The implementation of function definitions in this file may take a number of forms, depending on how the function is to be used in the code. The reader is strongly encouraged to study “function\_defs.h” to see the various forms functions may take.

### 12.11.1 In-line functions

The simplest form is the in-line function. An example is that which calculates the least mean squared error for a single output node:

```
// LMS error for a single output, takes known value y and model's
// guess o as arguments, returns LMS error
inline double singleLMSError( double y, double o )
{
    return 0.5 * ( y - o ) * ( y - o );
}
```

### 12.11.2 Function classes

For functions that are to be passed to *vector* or *Matrix* objects using the **func** method, classes must be constructed. An example is that which implements the sigmoidal transfer function:

```
// The sigmoidal function
class sigmoidal : public unary_function< double, double >
{
public:
    inline double operator()( const double& x )
    {
        return 1.0 / ( 1.0 + exp( -x ) );
    }
};
```

### 12.11.3 The errorFunction object

A very important object defined in **function\_defs.h** is *errorFunction*, which handles the calculation of model error. As neural network sigmoidal output is defined as

$$\vec{o} = \frac{1}{1 + e^{-\vec{x}}}$$

calculation of the error function may depend on  $\vec{x}$  as well as  $\vec{o}$  (see the **special note about numerical precision and the cross-entropy error** below), and thus *errorFunction* methods may be passed  $\vec{x}$  as well as  $\vec{o}$ . *errorFunction* consists of

- *Public Methods*

- *errorFunction(double y, double o, double x, bool errorType)*: Constructor for a *single* output, which takes known value  $y$ , model output  $o$ , argument  $x$  in sigmoidal transfer function  $1/(1 + e^{-x})$  and bool flag *errorType*, where

- *errorType* = 0 denotes the least mean squared error

$$E = \frac{1}{2}(y_i - o_i)^2$$

- *errorType* = 1 denotes the cross-entropy error

$$E = -y \log(o) - (1 - y) \log(1 - o)$$

- *errorFunction(vector< double >  $\vec{y}$ , vector< double >  $\vec{o}$ , vector< double >  $\vec{x}$ , bool errorType)*: Constructor for *multiple* outputs, which takes known values  $\vec{y}$ , model outputs  $\vec{o}$ , arguments  $\vec{x}$  in sigmoidal transfer function  $1/(1 + e^{-\vec{x}})$  and bool flag *errorType*, where

- *errorType* = 0 denotes the least mean squared error for  $m$  output nodes

$$E = \frac{1}{2} \sum_{i=1}^m (y_i - o_i)^2$$

- *errorType* = 1 denotes the cross-entropy error for  $m$  output nodes

$$E = \sum_{i=1}^m -y \log(o) - (1 - y) \log(1 - o)$$

- *double value()*: Accessor returns numerical error result
- *bool boundsErr()*: Accessor returns true if numerical bounds error (e.g.  $\log(0)$ ) (see the **special note about numerical precision and the cross-entropy error** below)

### Example

```
// Calculate x-entropy error for single output and add to set error
//   which is previously defined as double setError (double y, double
o
//   and double x have also previously been calculated)
errorFunction E( y, o, x, 1 );
setError += E.value();

if ( E.boundsErr() ) // check for out of bounds error in log(o)
    cout << "ERROR: Numerical bounds error" << endl;
```

### A special note about numerical precision and the cross-entropy error

As the [cross-entropy error function](#) is defined for network output  $o$  and target value  $y$ :

$$E = -y \log(o) - (1 - y) \log(1 - o)$$

And for the sigmoidal transfer function,

$$o = \frac{1}{1 + e^{-x}}$$

Then for large positive ( $\gg 0$ ) and large negative ( $\ll 0$ )  $x$ ,  $o \rightarrow 1$ , and  $o \rightarrow 0$  respectively, and thus  $\log(1 - o) \rightarrow \infty$  and  $\log(o) \rightarrow \infty$ . In these cases, the following transformations are made (which require those methods to be passed the sigmoidal transfer function argument  $x$  for the cross-entropy error.)

### Large negative $x$

If  $x \ll 0$ ,

$$o = \frac{1}{1 + e^{-x}} \approx \frac{1}{e^{-x}} = e^x$$

and thus

$$\log(o) = \log(e^x) = x$$

since  $o \rightarrow 0$ ,

$$\log(1 - o) \rightarrow \log(1) = 0$$

thus

$$E = -y \log(o) - (1 - y) \log(1 - o) \approx -yx - 0 = -yx$$

**Large positive  $x$**

$$1 - o = 1 - \frac{1}{1 + e^{-x}} = \frac{1 + e^{-x}}{1 + e^{-x}} - \frac{1}{1 + e^{-x}} = \frac{e^{-x}}{1 + e^{-x}}$$

If  $x \gg 0$ , since  $1 + e^{-x} \rightarrow 1$

$$1 - o = \frac{e^{-x}}{1 + e^{-x}} \approx e^{-x}$$

thus

$$\log(1 - o) \approx \log(e^{-x}) = -x$$

if  $x \gg 0$ ,  $o \rightarrow 1$  and  $-y \log(o) \rightarrow 0$ , thus

$$E = -y \log(o) - (1 - y) \log(1 - o) \approx 0 - (1 - y)(-x) = (1 - y)x$$

## 12.12 What you really need to know about DataSet and TwoSet

All models use training and test sets. The model builds itself on the training set, and is tested on the test set. *DataSet* is designed for generating and manipulating training and test sets, and *TwoSet* is a special object that is designed to calculate accuracies for one-output models *only*.

All training and test sets are expressed by Matrices in the following format, where  $i$  is an input to the model, and  $o$  is an output:

$$\begin{array}{cccccccc} i_{11} & i_{12} & \dots & i_{1I} & o_{11} & o_{12} & \dots & o_{1O} \\ i_{21} & i_{22} & \dots & i_{2I} & o_{21} & o_{22} & \dots & o_{2O} \\ i_{31} & i_{32} & \dots & i_{3I} & o_{31} & o_{32} & \dots & o_{3O} \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ i_{N1} & i_{N2} & \dots & i_{NI} & o_{N1} & o_{N2} & \dots & o_{NO} \end{array}$$

for  $N$  examples (“exemplars”) in the set,  $I$  inputs and  $O$  outputs. Although there are many highly useful features in *DataSet*, a programmer building a *Model* need only know how to access the training and test Matrices, whether they’ve been loaded into the *DataSet* object, how many inputs and outputs there are, whether the output is discrete (held to 0 or 1), and the number of examples in each set. These methods are shown in **red** in the *DataSet* specification.

In addition, the programmer of a *Model* needs only to understand that the methods `setTrainTwoSet` and `setTestTwoSet` must be called in order for the method `metricsReport`, which reports accuracy metrics of a one-output *Model*, to be called. These methods are also shown in **red** in the *DataSet* specification.

## 12.13 DataSet

*DataSet* is a standalone object that contains and manipulates the dataset that *Model* will access. All *DataSets* have raw, training and test datasets, and know the number of inputs and outputs. *DataSet* consists of:

- *Public* methods
    - Raw dataset methods:
      - \* void loadRaw( string& *filename* ): Takes *filename* as argument (string), and loads a file into the *Raw* dataset Matrix, returns nothing
      - \* bool rawLoaded(): Returns a flag to indicate if *Raw* dataset is loaded
      - \* Matrix< double >& getRawMatrix(): returns the *Raw* dataset
      - \* void setRawMatrix( Matrix< double >& *inMatrix* ): loads Matrix *inMatrix* into the *Raw* dataset Matrix
    - Training set methods:
      - \* void loadTrain( string& *filename* ): Takes *filename* as argument (string), and loads a file into the *TrainSetData* Matrix, returns nothing
      - \* bool trainLoaded(): Returns a flag to indicate if *TrainSetData* Matrix is loaded
      - \* bool saveTrain( string& *filename* ): Takes *filename* as argument (string), and saves the *TrainSetData* Matrix into a file, returns flag indicating if save operation was successful
      - \* bool saveScales( string& *filename* ): Takes *filename* as argument (string), and saves the scaling factors for the training set into the file, returns flag indicating if save operation was successful
      - Note:** see [equation 12.2](#) below for an explanation of scaling. Saved to the file will be *lbound* (1 value), and for each *x*, *S* and *min*.
- $$x_{norm} = lbound + S(x - min) \quad (12.1)$$
- where  $x_{norm}$  is the normalized result, *lbound* is the lower limit of normalized values, *S* is the scaling factor, and *min* is the minimum value of each *x* in the dataset.
- \* Matrix< double >& getTrainMatrix(): returns the *TrainSetData* Matrix
  - \* void setTrainMatrix( Matrix< double >& *inMatrix* ): loads Matrix *inMatrix* into the *TrainSetData* Matrix
  - *TwoSet* accessors for the training set:

- \* `bool setTrainTwoSet()`: Constructs an internal *TwoSet* object from the training set, populating the ‘real’ column of the *TwoSet* object with the output column from the training set *Matrix*, returns true if successful
- \* `TwoSet& getTrainTwoSet()`: Returns the constructed *TwoSet* object from the training set
- \* `bool TrainTwoSetLoaded()`: Returns true if the internal training *TwoSet* object was successfully constructed
- \* `bool saveTrainTwoSet( string& filename )`: Takes *filename* as argument (string), and saves the *TwoSet* object from the training set into a file, returns flag indicating if save operation was successful
- Test set methods:
  - \* `void loadTest( string& filename )`: Takes *filename* as argument (string), and loads a file into the *TestSetData* Matrix, returns nothing
  - \* `bool testLoaded()`: Returns a flag to indicate if *TestSetData* Matrix is loaded
  - \* `bool saveTest( string& filename )`: Takes *filename* as argument (string), and saves the *TestSetData* Matrix into a file, returns flag indicating if save operation was successful
  - \* `Matrix< double >& getTestMatrix()`: returns the *TestSetData* Matrix
  - \* `void setTestMatrix( Matrix< double >& inMatrix )`: loads Matrix *inMatrix* into the *TestSetData* Matrix
- *TwoSet* accessors for the test set:
  - \* `bool setTestTwoSet()`: Constructs an internal *TwoSet* object from the test set, populating the ‘real’ column of the *TwoSet* object with the output column from the test set *Matrix*, returns true if successful
  - \* `TwoSet& getTestTwoSet()`: Returns the constructed *TwoSet* object from the test set
  - \* `bool TestTwoSetLoaded()`: Returns true if the internal test *TwoSet* object was successfully constructed
  - \* `bool saveTestTwoSet( string& filename )`: Takes *filename* as argument (string), and saves the *TwoSet* object from the test set into a file, returns flag indicating if save operation was successful
- General methods:
  - \* `void setInput( unsigned I )`: Set the number of input nodes to *I*
  - \* `unsigned getInput()`: Returns number of input nodes
  - \* `void setOutput( unsigned O )`: Set the number of output nodes to *O*

- \* `unsigned getOutput()`: Returns number of output nodes
- \* `unsigned getNumTrain()`: Returns the number of examples in *TrainSetData*, the training set
- \* `unsigned getNumTest()`: Returns the number of examples in *TestSetData*, the test set
- \* `void removeInputs( vector< unsigned >& v )`: Given a vector representing input positions within a dataset, removes the corresponding columns in *Raw*, *TrainSetData*, and *TestSetData*, thus effectively removing those inputs from a dataset

**Note:** In the following methods, a variate  $x$  is normalized according to:

$$x_{norm} = lbound + \left[ \left( \frac{x - min}{max - min} \right) \times (hbound - lbound) \right] \quad (12.2)$$

where  $x_{norm}$  is the normalized value,  $lbound$  is the lower limit of the normalized result,  $hbound$  is the upper limit of the normalized result,  $min$  is the minimum value of  $x$  in the dataset, and  $max$  is the maximum value of  $x$  in the dataset.

- Methods related to normalizing dataset variates
  - \* `void setInUpper( double x )`: Sets the upper limit of normalized input variates to  $x$
  - \* `double getInUpper()`: Returns the upper limit of normalized input variates
  - \* `void setInLower( double x )`: Sets the lower limit of normalized input variates to  $x$
  - \* `double getInLower()`: Returns the lower limit of normalized input variates
  - \* `void setOutUpper( double x )`: Sets the upper limit of normalized output variates to  $x$
  - \* `double getOutUpper()`: Returns the upper limit of normalized output variates
  - \* `void setOutLower( double x )`: Sets the lower limit of normalized output variates to  $x$
  - \* `double getOutLower()`: Returns the lower limit of normalized output variates
- Methods related to identifying output as discrete (held to 0 or 1)
  - \* `void setDiscrete( bool flag )`: Sets output to be discrete if *flag* is true, continuous if false
  - \* `bool getDiscrete()`: Returns the state of discrete output
  - \* `void setThreshold( double x )`: Sets the threshold value for discrete output to be  $x$ , any output  $\geq x$  will be 1, and any output  $< x$  will be 0

- \* `double getThreshold()`: Returns the threshold value
- Methods related to transforming datasets:
  - \* `bool raw2train()`: Converts the raw dataset to a training set by normalizing its input variates, and if nondiscrete, its output variates, returning true if the operation was successful
  - \* `bool randomize( unsigned n )`: Randomizes the raw dataset into training and test sets which are then normalized, preserving the frequency of outcomes in the training and test sets, takes the number of exemplars  $n$  to be placed in the test set as an argument ( *number of outputs must be only 1, and it must be discrete* ), returns true if successful
  - \* `bool randomize( unsigned n, unsigned d )`: Randomizes the raw dataset into training and test sets which are then normalized, preserving the frequency of outcomes in the training and test sets, takes the numerator  $n$  and denominator  $d$  of a fraction representing the portion of exemplars to be placed in the test set as arguments, ( *number of outputs must be only 1, and it must be discrete* ), returns true if successful
  - \* `bool randomizeD( double x )`: Randomizes the raw dataset into training and test sets which are then normalized, preserving the frequency of outcomes in the training and test sets, takes the fraction  $x$  as a double value from 0 to 1 representing the portion of exemplars to be placed in the test set as argument ( *number of outputs must be only 1, and it must be discrete* ), returns true if successful
- Methods related to calculating accuracy
  - \* `void metricsReport( ostream& streamObj )`: Outputs all internal *TwoSet* metrics for a 1-output *Model*, takes ostream object *streamObj* as argument
- Methods related to logging to the history file:
  - \* `void setHistory( bool flag )`: Sets history logging, which *appends* all operations to the history file
  - \* `bool getHistory()`: Returns the current state of history logging
  - \* `void setHistoryFilename( string& filename )`: Sets the name of the history file to string *filename*
  - \* `bool addHistory( ostream& outputStream )`: appends the ostream object *outputStream* to the history file.
- *Private members*
  - `Matrix< double > Raw`: the raw dataset
  - `Matrix< double > TrainSetData`: the training set
  - `Matrix< double > TestSetData`: the test set

- TwoSet *TrainTwoSet*: the *TwoSet* object for the training set
  - TwoSet *TestTwoSet*: the *TwoSet* object for the test set
  - vector< double > *minima*: a vector containing the minimum values of the columns in *Raw*
  - vector< double > *maxima*: a vector containing the maximum values of the columns in *Raw*
  - unsigned *nInput*: the number of input nodes
  - unsigned *nOutput*: the number of output nodes
  - double *threshold*: the threshold for discrete output
  - double *inUpperLimit*: the upper limit for a normalized input variate
  - double *inLowerLimit*: the lower limit for a normalized input variate
  - double *outUpperLimit*: the upper limit for a normalized output variate
  - double *outLowerLimit*: the lower limit for a normalized output variate
  - bool *discreteFlag*: a flag to indicate if the output is to be considered discrete (held to 0,1)
  - bool *rawLoadedFlag*: a flag to indicate if *Raw* loaded
  - bool *trainLoadedFlag*: a flag to indicate if *TrainSetData* loaded
  - bool *testLoadedFlag*: a flag to indicate if *TestSetData* loaded
  - bool *trainTwoSetFlag*: a flag to indicate if the training set *TwoSet* object has been successfully constructed
  - bool *testTwoSetFlag*: a flag to indicate if the test set *TwoSet* object has been successfully constructed
  - bool *historyFlag*: a flag to indicate if operations will be logged to a history file, this file is *appended*
  - bool *minimaxFlag*: a flag to indicate if minima and maxima vectors have been set
  - string *historyFilename*: the name of history file
- *Private Utility functions*
    - void *minimax*( Matrix< double >& *data* ): computes the *minima* and *maxima* vectors from *Matrix data*
    - bool *checkDiscrete*( Matrix< double >& *data* ): returns true if the output columns of *Matrix data* are discrete, false if otherwise
    - void *normalize*( Matrix< double >& *data* ): normalizes input variates of *Matrix data*, and if nondiscrete, its output variates (*note: minima and maxima vectors must have been previously set*)
    - void *clear*(): clears *DataSet Matrix* members and flags

## 12.14 TwoSet

*TwoSet* is a specialized standalone object that is designed to calculate outcome metrics such as sensitivity, specificity, predictive value positive, predictive value negative and ROC area from a *one* output *Model*. The *TwoSet* data *Matrix* contains 2 columns, one for the known outcomes and one for the *Model*'s outputs. The *Model* output *must be* discrete. The format of this data *Matrix* is:

$$\begin{array}{cc} real_0 & test_0 \\ real_1 & test_1 \\ real_2 & test_2 \\ \dots & \dots \\ real_{n-1} & test_{n-1} \end{array}$$

where *real* represents a known outcome, and *test* is the *Model*'s output, for *n* examples in the data *Matrix*, e.g.:

```
0.000000 0.002851
0.000000 0.002978
0.000000 0.002857
1.000000 0.571710
```

For calculating ROC (receiver-operator characteristic) curve area, 2 methods are available. One is the traditional method using trapezoids and numerical calculation to calculate the area under the curve. By default, the number of trapezoids is set to the number of data points, but that may be changed by the *setTrapThresholds()* method. However, as should be evident, this method *underestimates* the ROC area. Wickens<sup>8</sup> provides a statistical method to calculate ROC area. Here,  $z = Z(A) = \Phi^{-1}(A)$  is linearly modeled for sensitivity and 1 - specificity, and this linear model is used to calculate ROC area. If enough data is available to bin and generate errors in the abscissa and ordinate of this line, then a p-value will be generated to reflect the probability of a least squares fit to that line, where a smaller p-value is *worse* (closer to 1 is more linear).

The method *ROCarea(ostringstream&)* may be used to automatically choose which method of calculating ROC curve area is used, depending on how many data points are available.

Two errors may be thrown from this class. The first is *DivisionByZero* where division by zero is encountered, generally in calculating positive and negative predictive value. The second is *twoSetErr*, which will contain errors generated in statistical calculation or linear modeling. *DivisionByZero* always contains the message "division by zero", whereas *twoSetErr*'s *what()* member will contain a message reflecting the type of error. An example of the latter would be:

---

<sup>8</sup>Wickens, Thomas D., Elementary Signal Detection Theory, Oxford University Press, New York, NY 2002, pp. 60–74

```

string reply;
cout << "What is the name of the file? ";
cin >> reply;
TwoSet data;
data.load( reply );
try {
    data.ROCarea( cout );
} catch ( TwoSet::twoSetErr& e ) { cout << e.what() << endl; }

```

The *TwoSet* object consists of:

- *Public* methods

The constructor `TwoSet( Matrix< double >& M )` constructs and loads *Matrix* *M* into a *TwoSet* object

- Methods for loading Matrices and files into a *TwoSet* object:

- \* `void setMatrix( Matrix< double >& M )`: Loads *Matrix* *M* into a *TwoSet* object, returns nothing
- \* `void load( string& filename )`: Takes *filename* as argument (string), and loads a file into the *TwoSet* data *Matrix*, returns nothing
- \* `bool loaded()`: Returns a flag to indicate if the *TwoSet* data *Matrix* is loaded

- *TwoSet* accessors:

- \* `Matrix< double >& getData()`: Returns the *TwoSet* data *Matrix*
- \* `bool save( string& filename )`: Takes *filename* as argument (string), and saves the *TwoSet* object into a file, returns flag indicating if save operation was successful
- \* `unsigned getNumElements()`: Returns the number of elements in the *TwoSet* data *Matrix*
- \* `TwoSet& insertReal( vector< double >& v )`: Insert vector *v* into the real column of the *TwoSet* data *Matrix*, the number of elements of *v* must equal the number of rows in the *TwoSet* data *Matrix*
- \* `vector< double >& real( vector< double >& v )`: Populate vector *v* with the real column of the *TwoSet* data *Matrix*, the number of elements of *v* must equal the number of rows in the *TwoSet* data *Matrix*
- \* `vector< double > real()`: Get a new vector from the real column of the *TwoSet* data *Matrix*
- \* `TwoSet& insertTest( vector< double >& v )`: Insert vector *v* into the test column of the *TwoSet* data *Matrix*, the number of elements of *v* must equal the number of rows in the *TwoSet* data *Matrix*

- \* `vector< double >& test( vector< double >& v )`: Populate vector *v* with the test column of the *TwoSet* data *Matrix*, the number of elements of *v* must equal the number of rows in the *TwoSet* data *Matrix*
- \* `vector< double > test()`: Get a new vector from the test column of the *TwoSet* data *Matrix*
- \* `double real( unsigned r )`: An accessor to set or retrieve the element in row *r* in the real column of the *TwoSet* data *Matrix*
- \* `double test( unsigned r )`: An accessor to set or retrieve the element in row *r* in the test column of the *TwoSet* data *Matrix*
- Accessors and methods for *TwoSet* calculated metrics
  - \* `void setThreshold( double x )`: Sets threshold value to *x*, any test element  $\geq x$  will be counted as 1, whereas any test element  $< x$  will be counted as 0
  - \* `double getClassAcc()`: Returns classification accuracy
  - \* `double getSens()`: Returns sensitivity, throws exception `TwoSet::DivisionByZero& e`, where `e.what()` is “division by zero”
  - \* `double getSpec()`: Returns specificity, throws exception `TwoSet::DivisionByZero& e`, where `e.what()` is “division by zero”
  - \* `double getPVP()`: Returns predictive value positive, throws exception `TwoSet::DivisionByZero& e`, where `e.what()` is “division by zero”
  - \* `double getPVN()`: Returns predictive value negative, throws exception `TwoSet::DivisionByZero& e`, where `e.what()` is “division by zero”
  - \* `void ClassTable( ostream& outputStream )`: Outputs to `ostream&` the classification table
- ROC area calculation by trapezoidal method
  - \* `void setTrapThresholds( unsigned n )`: Sets number of trapezoids to calculate ROC area to *n*
  - \* `unsigned getTrapThresholds()`: Returns number of trapezoids to calculate ROC area
  - \* `double getTrapROCArea()`: Returns ROC area calculated by trapezoidal method
- ROC area calculation by statistical method
  - \* `void setNbins( unsigned n )`: If data can be binned, sets the number of bins to *n*
  - \* `unsigned getNbins()`: Returns number of bins specified by user
  - \* `void setBinSize( unsigned n )`: If data can be binned, sets number of elements to be contained in a bin to *n*
  - \* `unsigned getBinSize()`: Return bin size specified by user

- \* void NbinThreshold( unsigned *n* ): Sets the number of data points below which data will not be binned
- \* unsigned getNBinThreshold(): Returns the number of data points below which data will not be binned
- \* bool getBinned(): Returns *true* if data was binned
- \* void setNbinsSetsSize( bool *flag* ): Set *flag* if number of bins sets bin size
- \* bool getNbinsSetsSize(): Returns *true* if number of bins sets bin size
- \* double getStatROCarea(): Returns statistical ROC area, throws exception TwoSet::twoSetErr& *e*, where *e.what()* contains a statistical or linear regression error message
- \* double getStatP(): Returns p-value for fitted line (smaller is worse)
- \* double getStatChi2(): Returns chi-squared value for fitted line
- Methods and accessors which output to ostream a ROC report which uses either or both the trapezoidal and/or statistical method
  - \* void ROCarea( ostream& *outputStream* ): Outputs to ostream& ROC area, and a report detailing method and results of calculation
  - \* void setROCReportFlag( *flag* ): Set *flag* true if both statistical and trapezoidal methods are to be reported, false to use either the trapezoidal method if number of data points < threshold (*calcThresh*) or statistical method if >= threshold
  - \* bool getROCReportFlag(): Returns flag indicating if either (false) or both (true) of the statistical and trapezoidal methods are used and reported, according to the above description of setROCReportFlag(...)
  - \* void setROCthresh( unsigned *n* ): Sets threshold for number of non-zero, non-one sensitivity and 1 - specificity data points under which statistics will not be used
  - \* unsigned getROCthresh(): Returns number of non-zero, non-one sensitivity and 1 - specificity data points under which statistics will not be used
  - \* void setROCSearchFlag( bool *flag* ): Set *flag* true if best *p* and ROC areas are searched and reported within a specified range of number of bins
  - \* bool getROCSearchFlag(): Returns flag indicating if best *p* and ROC areas are searched and reported within a specified range of bins
  - \* void setMinROCSearchBins( unsigned *n* ): Sets minimum number of bins to search for best *p* and ROC areas (must be > 2)
  - \* unsigned getMinROCSearchBins(): Returns minimum number of bins to search for best *p* and ROC areas

- \* void setMaxROCSearchBins( const unsigned  $n$  ): Sets maximum number of bins to search for best  $p$  and ROC areas
- \* unsigned getMaxROCSearchBins(): Returns maximum number of bins to search for best  $p$  and ROC areas
- Accessors for the Kolmogorov-Smirnov test<sup>9</sup>
  - \* double getKSD(): Returns Kolmogorov-Smirnov statistic D
  - \* double getKSP(): Returns **Kolmogorov-Smirnov  $p$ -value**
- *Private members*
  - Matrix< double > A: the *TwoSet* data *Matrix*
  - unsigned n: the number of elements in the *TwoSet* data *Matrix*
  - unsigned tp: the number of true positives
  - unsigned tn: the number of true negatives
  - unsigned fp: the number of false positives
  - unsigned fn: the number of false negatives
  - unsigned nThresholds: the number of thresholds for trapezoidal method
  - unsigned nBins: the number of bins for statistical ROC
  - unsigned binSize: the bin size for statistical ROC
  - unsigned binThresh: the number of data points under which data will not be binned
  - unsigned calcThresh: the number of data points under which statistics will not be used
  - double threshold: the threshold value
  - double statP:  $p$ -value for fitted line for statistical ROC calculation
  - double statChi2: chi-squared value for fitted line for statistical ROC calculation
  - double KSD: Kolmogorov-Smirnov test D value
  - double KSP: Kolmogorov-Smirnov test  $p$ -value
  - bool loadedFlag: a flag to indicate if the *TwoSet* data *Matrix* is loaded
  - bool thresholdFlag: a flag to indicate if the threshold is set
  - bool nBinFlag: flag indicates if number of bins determines bin size
  - bool binnedFlag: flag indicates if data was binned for statistical ROC calculation
  - bool statROCcalcFlag: flag indicates if statistical ROC area calculated

---

<sup>9</sup>Adapted from W. H. Press et. al, Numerical Recipes in C, 2nd edition, Cambridge University Press, Cambridge, 1992, pp. 623–626

- bool reportFlag: flag directs how to report statistical and/or trapezoidal ROC
- bool searchFlag: flag indicates if largest p, largest ROC is searched
- bool KScalcFlag: flag indicates if Kolmogorov-Smirnov test calculated
- *Private* Utility functions
  - void initialize(): Sets common initial values for constructors
  - void calculate(): Given a threshold, calculates tp, tn, fp, and fn
  - bool checkDiscrete( Matrix< double >& *M* ): Returns true if the first column (representing real outcomes) of *Matrix M* is discrete, false if otherwise
  - void statReport( ostream& *outputStream*, unsigned *goodData*, unsigned *nBins*, double *AUC*, double *chi2*, double *p* ): Utility method for ROCarea, outputs statistical report given ostream& *outputStream*, number of non-zero, non-one data points *goodData*, number of bins *nBins*, area under the curve *AUC*, chi-square value *chi2*, and p-value *p*
  - void KScalc(): Utility method to calculate Kolmogorov-Smirnov test<sup>10</sup>

## 12.15 What you really need to know about programming a Model

### 12.15.1 Object Type

All concrete derived objects of *Model* inherit its protected member, (string) *objType*, which is the name of the object. Typically, the class constructor defines this member, for example:

```
class BareProp : public Network {
public:
    BareProp() { objType = "BareProp"; } // default constructor
```

Abstract classes *do not* define *objType*. This member is accessed by the *Model* method *getType()*.

---

<sup>10</sup>Adapted from W. H. Press et. al, Numerical Recipes in C, 2nd edition, Cambridge University Press, Cambridge, 1992, pp. 623–626

### 12.15.2 Copy constructor and overloaded = operator

All derived objects of *Model* **must** have a copy constructor and overloaded = operator. Typically, both call a protected utility function so that both copy the same members. The protected utility function typically begins with a call to the copy method of its immediate base class so that it inherits all of the members of its base class(es). For example, suppose a hypothetical neural network object named "LittleNet" is derived from *Network*, and contains the following members:

```
unsigned nNodes; // number of nodes

Matrix< double > W; // weight Matrix

vector< double > I, // input vector
               y; // output vector
```

The following would be defined in the header file:

```
class LittleNet : public Network {
public:
    // Copy constructor
    LittleNet( const LittleNet& rhs );

    // Overloaded = operator
    LittleNet& operator= ( const LittleNet& rhs );

private: // or protected if the object is a base class
    // Copy utility
    void copy( const LittleNet& rhs );
```

Which would then be coded in the C++ file as:

```
// Copy constructor
LittleNet::LittleNet( const LittleNet& rhs )
{
    LittleNet::copy( rhs ); // use the copy utility
}

// Overloaded = operator
LittleNet& LittleNet::operator= ( const LittleNet& rhs )
{
    if ( &rhs != this ) // check for self-assignment
        LittleNet::copy( rhs ); // use the copy utility

    return *this; // enables A = B = C
}
```

```
// Copy utility
void LittleNet::copy( const LittleNet& rhs )
{
    Network::copy( rhs ); // call immediate base object copy
    nNodes = rhs.nNodes;
    W = rhs.W;
    I = rhs.I;
    y = rhs.y;
}
```

**Remember:** if you add any member to any existing class, you *must* insert a corresponding line of code to copy it in the copy utility function of that class.

### 12.15.3 Implementation of pure virtual methods in *Model*

Ultimately, all models map the inputs of each exemplar to the outputs. Thus, most of the *Model* base class methods pertain directly to this process. If you are writing a *Model*, you need to implement the following pure virtual methods:

#### **setDataSet**

The pure virtual method

```
virtual void setDataSet( DataSet& $dataObj$ )
```

loads a *DataSet* object into the derived *Model*. The protected member inherited from the base class *Model* is named *theData*, so the first line of

```
void YourModel::setDataSet( DataSet& dataObj )
```

where “YourModel” is the name of your derived *Model* object, should be

```
theData = dataObj; // set "theData" to incoming DataSet
```

The remainder of *YourModel::setDataSet* would then contain code that tailors your model to the incoming *DataSet* object. See *SimpleProp::setDataSet* and *BareProp::setDataSet* for examples.

#### **train**

The pure virtual method

```
virtual double train()
```

should train a *Model*, and return the final error (double) for that model. In the case of neural networks, training would involve finding the weight vectors and Matrices that minimize final error. See *SimpleProp::train* and *BareProp::train* for examples.

### **reportAccuracy**

The pure virtual method

```
virtual void reportAccuracy( ostream& outputStream )
```

should report the accuracy of the trained model to an ostream outputStream. *If you are programming an object that extends Network, e.g. a neural network, you **do not** need to implement this virtual method, as Network implements reportAccuracy for you.*

### **outputHeader**

The pure virtual method

```
virtual void outputHeader( ostream& outputStream )
```

should output a header to an ostream outputStream describing the architecture of the derived *Model*. See SimpleProp::outputHeader and BareProp::outputHeader for examples.

## **12.15.4 *Model* protected members**

The following are the protected members in *Model* that your object needs to set or get:

### **theData**

The protected member

```
DataSet theData
```

represents the primary dataset, in the form of a DataSet object. Your setDataSet method needs to set this member. See SimpleProp::setDataSet and BareProp::setDataSet for examples.

### **Train**

The protected member

```
Matrix< double > Train
```

represents a Matrix that contains the training set input columns only, typically constructed by a call within your setDataSet method to the *Model* protected utility function extractInputMatrices(). See SimpleProp::setDataSet and BareProp::setDataSet for examples.

## Test

The protected member

`Matrix< double > Test`

represents a Matrix that contains the test set input columns only, typically constructed by a call within your `setDataSet` method to the *Model* protected utility function `extractInputMatrices()`. See `SimpleProp::setDataSet` and `BareProp::setDataSet` for examples.

## TrainOutput

The protected member

`Matrix< double > TrainOutput`

represents a Matrix that contains the training set output columns only, typically constructed by a call within your `setDataSet` method to the *Model* protected utility function `extractOutputMatrices()`. See `BackProp::setDataSet` for an example.

## TestOutput

The protected member

`Matrix< double > TestOutput`

represents a Matrix that contains the test set output columns only, typically constructed by a call within your `setDataSet` method to the *Model* protected utility function `extractOutputMatrices()`. See `BackProp::setDataSet` for an example.

## builtFlag

Set the protected member

`bool builtFlag`

to *true* when the Model has been formally constructed, and its architecture set. See `SimpleProp::setHidden` and `BareProp::setHidden` for examples.

## historyFlag

Use the protected member

`bool historyFlag`

to decide if operations will be logged to a history file using the protected *Model* utility function `addHistory`. See `SimpleProp` and `BareProp` code for examples, as most methods use `addHistory`.

## **errorType**

Use the protected member

**bool errorType**

to determine if the type of error will be least mean squared (*errorType* = 0) or cross-entropy (*errorType* = 1). Typically, this member is used as an argument in *errorFunction* constructors (see [section 12.11.3](#), and `SimpleProp::trainSet()` and `BareProp::trainSet()` as examples.)

## **12.15.5 *Model* protected utility functions**

The following are the protected utility functions in *Model* that your object will want to call:

### **extractInputMatrices**

Call the utility function

**void extractInputMatrices()**

in your `setDataSet` method to extract the input columns of the training set into the protected *Model* Matrix *Train*, and the input columns of the test set into the protected *Model* Matrix *Test*.

### **extractOutputMatrices**

Call the utility function

**void extractOutputMatrices()**

in your `setDataSet` method to extract the output columns of the training set into the protected *Model* Matrix *TrainOutput*, and the output columns of the test set into the protected *Model* Matrix *TestOutput*.

### **addHistory**

Pass an ostream object to the utility function

**bool addHistory( ostream \$outputStream\$ )**

for that object to be added to the history file. This method should be called for every operation. See `SimpleProp` and `BareProp` code for examples, as most methods use `addHistory`.

### **copy**

See [section 12.15.2](#) for details.

## 12.16 Model

In the following specification, methods and members that pertain directly to coding a concrete derived *Model* implementation are shown in **red** .

**Model** is the base object. *Model* provides access to the dataset, and contains the virtual methods common to all modeling techniques. *Model* consists of:

- *Public* methods
  - Copy constructor and overloaded = operator, see section 12.15.2 for discussion:
    - \* `Model& operator= ( const Model& rhs )`
    - \* `Model( const Model& rhs )`
  - Methods related to accessing the dataset to model:
    - \* **virtual void setDataSet( DataSet& data )**: A pure virtual method that loads the *DataSet* object *data* into the *Model*
    - \* `DataSet& getDataSet()`: Returns a the *Model*'s dataset in the form of a *DataSet* object
  - Methods related to training the model:
    - \* **virtual double train()**: A pure virtual method that trains a *Model*, and returns the final error (double) for that model
    - \* **virtual void reportAccuracy( ostream& outputStream )**: A pure virtual method that reports the accuracy of the trained model to an ostream *outputStream*
  - Methods related to logging to files or reporting to user:
    - \* **virtual void outputHeader( ostream& outputStream )**: A pure virtual method which outputs a header to an ostream *outputStream* describing the architecture of the model, *Model* type dependant
    - \* `void setHistory( bool flag )`: Sets history logging, which *appends* all operations to the history file
    - \* `bool getHistory()`: Returns the current state of history logging
    - \* `void setHistoryFilename( string& filename )`: Sets the name of the history file to string *filename*
    - \* `string getHistoryFilename()`: Returns the name of the history file
    - \* `void setLastop( bool flag )`: Sets last operation logging, which *overwrites* a file to report the building (training in the case of *Iterative*) of a specific model
    - \* `bool getLastop()`: Returns the current state of last operation logging
    - \* `void setLastopFilename( string& filename )`: Sets the name of the last operation logging file to *filename*

- \* string getLastopFilename(): Returns the name of the last operation logging file
- Methods related to setting the output error function
  - \* void setLMSError(): Sets the output error function to least mean squared, which is

$$E = \frac{1}{2} \sum_{i=1}^m (y_i - o_i)^2 \quad (12.3)$$

where  $E$  is the error,  $m$  the number of outputs,  $y$  is the known value and  $o$  the Model's output.

- \* void setXError(): Sets the output error function to be **cross-entropy**, which is

$$E = \sum_{i=1}^m -y \log(o) - (1 - y) \log(1 - o) \quad (12.4)$$

*Note: for the cross-entropy error function, the output **must** be discrete.*

- *Protected members*

- *DataSet theData*: the primary dataset, in the form of a *DataSet* object
- *Matrix< double > Train*: a Matrix that contains the training set input columns *only*, constructed by a call to the protected utility function *extractInputMatrices()*
- *Matrix< double > Test*: a Matrix that contains the test set input columns *only*, constructed by a call to the protected utility function *extractInputMatrices()*
- *Matrix< double > TrainOutput*: a Matrix that contains the training set output columns *only*, constructed by a call to the protected utility function *extractOutputMatrices()*
- *Matrix< double > TestOutput*: a Matrix that contains the test set output columns *only*, constructed by a call to the protected utility function *extractOutputMatrices()*
- *bool builtFlag*: a flag to indicate if the *Model* has been formally constructed
- *bool historyFlag*: a flag to indicate if operations will be logged to a history file, this file is *appended*
- *bool lastopFlag*: a flag to indicate if model building (training in the case of *Iterative*) will be logged to a file describing the last operation, this file is *overwritten*
- *bool errorType*: flag indicates type of error function, 0 if least mean squared, 1 if cross-entropy

- *bool builtFlag*: a flag to indicate the type of error function, 0 if least mean squared, 1 if cross-entropy
  - *string objType*: the name of the object, see [section 12.15.1](#)
  - *string historyFilename*: the name of history file
  - *string lastopFilename*: the name of the file to log the last operation
  - *string errorLabel*: a string containing the name of the output error function
- *Protected* Utility functions
    - *void extractInputMatrices()*: Utility function which is intended to be called by the derived *Model* in its *setDataSet* method, and places the input columns of the training set in Matrix *Train*, and the input columns of the test set in Matrix *Test*
    - *void extractOutputMatrices()*: Utility function which is intended to be called by the derived *Model* in its *setDataSet* method, and places the output columns of the training set in Matrix *TrainOutput*, and the output columns of the test set in Matrix *TestOutput*
    - *bool addHistory( ostream& outputStream )*: Utility function to append the ostream object *outputStream* to the history file.
    - *void copy( const Model& rhs )*: Copy utility function, see [section 12.15.2](#) for discussion.

## 12.17 What you really need to know about programming an Iterative Model

*Iterative* inherits methods *setDataSet*, *train*, *reportAccuracy* and *outputHeader* from *Model*. If you are programming an *Iterative Model*, there are 2 pure virtual methods in *Iterative* that you must also implement:

### **trainSet**

*Iterative* handles models that require multiple passes (iterations) to train. Objects derived from *Iterative* must thus provide a method to adjust a model by a single pass through the training set, and *Iterative* will call this method iteratively to reach the final trained state. Thus the method:

```
virtual double trainSet()
```

changes the components of a *Model* (e.g. weight Matrices and vectors in a neural network), and returns the set error. See *SimpleProp::trainSet()* and *BareProp::trainSet()* for examples.

### **classAccuracy**

With each iteration, *Iterative* reports the set error and classification accuracy for the training and/or test sets if the outputs are discrete. Thus, the pure virtual method:

```
virtual void classAccuracy( ostream& outputStream )
```

must output to an ostream object the classification accuracy for the training and/or test sets. *If you are programming a Network object, you **do not** need to implement this virtual method, as Network implements classAccuracy for you.*

### **getGradMax**

Most iterative algorithms employ some form of gradient descent, and the iterative class can report the maximum absolute gradient and use it to determine cessation of training. The pure virtual method:

```
virtual double getGradMax()
```

must be coded to return the maximum absolute gradient of the derived object. *If you are programming a Network object, you **do not** need to implement this virtual method, as Network implements getGradMax for you.* However, you must code `Network::pack()` to create the *Network* protected member *stackG* so that `Network::getGradMax()` returns the proper value.

### **runHeader**

If you are adding parameters specific to an *Iterative* derived object that should be output to screen or log files prior to an *Iterative* run, you will need to implement a runHeader method in the derived class:

```
virtual void runHeader( ostream& outputStream)
```

which outputs to an ostream object the specific parameters to be displayed. *If you are programming a Network object, you will need to add code to this method in Network.*

As with any derivation of *Model*, a copy utility function, copy constructor and overloaded `=` operator must be implemented. See section 12.15.2 for details.

### **iteration**

You may access the iteration index in your derived object by the protected member unsigned *iteration*.

### **boundsErrorFlag**

If your object can generate a numerical bounds error (see [section 12.11.3](#) for an example of a method which generates a numerical bounds error), then use the protected member bool *boundsErrorFlag* to indicate if such an error has been thrown.

### **gradMaxFlag**

Use the protected member bool *gradMaxFlag* to determine if a gradient is to be constructed in your derived object.

## **12.18 Iterative**

**Iterative** extends *Model*. Neural computational algorithms are iterative, and thus use *Iterative* as a base. *Iterative* manages methods and parameters related to iterating through a dataset.

In the following specification, methods and members that pertain directly to coding a concrete derived *Iterative* implementation are shown in red .

- *Public* methods
  - Copy constructor and overloaded = operator, see [section 12.15.2](#) for discussion:
    - \* Iterative& operator= ( const Iterative& rhs )
    - \* Iterative( const Iterative& rhs )
  - Methods pertaining to stopping training
    - \* void setMaxIterations( unsigned *maxIterations* ): Sets the maximum number of iterations through the training set to *maxIterations*
    - \* unsigned getMaxIterations(): Returns the maximum number of iterations through the training set
    - \* void setMinStop( bool *minStopFlag* ): Flag sets if a minimum error threshold will be used to stop training
    - \* bool getMinStop(): Returns flag indicating if a minimum error threshold will be used to stop training
    - \* void setMinError( double *minError* ): Sets minimum error threshold value to *minError*
    - \* double getMinError(): Returns minimum error threshold value
    - \* void setChangeStop( bool *changeStopFlag* ): Flag sets if training stops after error increases over 1 iteration
    - \* bool getChangeStop(): Returns flag indicating if change over 1 iteration will be used to stop training
    - \* void setChange( double  $\Delta$  ): Sets limit of change value to  $\Delta$
    - \* double getChange(): Returns limit of change value

- \* void setWindowStop( bool *windowStopFlag* ): Flag sets if training stops if set error increases over a specified window width
- \* bool getWindowStop(): Returns flag indicating if set error increases over a window width will be used to stop training
- \* void setWindow( unsigned *window* ): Sets the value of the window width to *window*
- \* unsigned getWindow(): Returns specified window width value
- \* void setGradStop( bool *gradMaxFlag* ): Flag sets if training will stop when the maximum absolute gradient decreases below the limit
- \* bool getGradStop(): Returns flag indicating if training will stop when the maximum absolute gradient decreases below the limit
- \* void setGradMaxLimit( double *limit* ): Sets the value of the maximum absolute gradient limit to *limit*
- \* double getGradMaxLimit(): Returns the value of the maximum absolute gradient limit
- Methods pertaining to printing while training
  - \* void setLogPrint( bool *logPrintFlag* ): Sets if prints out by powers of 10 ( 1-10 by ones, 10-100 by 10s, 100-1000 by 100s, etc. )
  - \* bool getLogPrint(): Returns flag indicating if printing by powers of 10
  - \* void setPrintCount( unsigned *printCount* ): Sets the linear print counter to *printCount*
  - \* unsigned getPrintCount(): Returns the linear print counter
- General methods
  - \* virtual void setDataSet( DataSet& *data* ): A pure virtual method inherited from *Model* that loads the *DataSet* object *data* into the *Model*
  - \* virtual double train(): A pure virtual method inherited from *Model* that trains a *Model*, and returns the final error (double) for that model
  - \* virtual void reportAccuracy( ostream& *outputStream* ): A pure virtual method inherited from *Model* that reports the accuracy of the trained model to an ostream *outputStream*
  - \* virtual void outputHeader( ostream& *outputStream* ): A pure virtual method inherited from *Model* which outputs a header to an ostream *outputStream* describing the architecture of the model, *Model* type dependant
  - \* virtual double trainSet(): A pure virtual method which trains one iteration through the training set, *Iterative Model* type dependant, and returns the set error

\* `virtual void classAccuracy( ostream& outputStream )`: A pure virtual method which outputs classification accuracies to an ostream *outputStream* for an *Iterative* output table

- *Protected* members

- `unsigned iteration`: the iteration index
- `unsigned maxIterations`: the maximum number of iterations through the training set
- `bool logPrintFlag`: flag indicates if printing by powers of 10
- `unsigned printCount`: the value for the linear print counter
- `bool minStopFlag`: flag indicates if a minimum error threshold will be used to stop training
- `double minError`: minimum error threshold value
- `bool changeStopFlag`: flag indicates if change in error over 1 iteration will be used to stop training
- `double delta`: value of change over 1 iteration
- `bool gradMaxFlag`: flag indicates if training will stop when maximum absolute gradient decreases below *gradMaxLimit*
- `double gradMaxLimit`: value of the maximum absolute gradient limit
- `bool windowStopFlag`: flag indicates if set error increases over a window width will be used to stop training
- `unsigned window`: value of the window width
- `bool boundsErrorFlag`: flag indicates if out of numerical bounds error encountered (e.g.  $\log(0)$ )

- *Protected* Utility functions

- `void copy( const Iterative& rhs )`: Copy utility function, see section 12.15.2 for discussion.
- `virtual void runHeader( ostream& outputStream )`: A pure virtual utility function that outputs parameters specific to derived objects preceeding an *Iterative* run
- `virtual double getGradMax()`: A pure virtual utility function that returns the maximum absolute gradient value

## 12.19 What you really need to know about programming a Network Model

### 12.19.1 Implementation of pure virtual methods in *Network*

*Network* inherits methods `setDataSet`, `train`, `reportAccuracy` and `outputHeader` from *Model*, and `trainSet` from *Iterative*, so you must code all of these methods. (See the respective methods in *SimpleProp* and *BareProp* as examples.) If you are programming a *Network Model*, there are 7 additional pure virtual methods in *Network* that you must implement in the concrete derived object:

#### **df**

The method

```
virtual unsigned df()
```

must be coded to return the number of degrees of freedom of your *Network* object, which is equal to the number of nodal connections. See *SimpleProp::df* and *BareProp::df* as examples.

#### **randomize**

The method

```
virtual void randomize()
```

must be coded to randomize the weights in your *Network* object. See *SimpleProp::randomize* and *BareProp::randomize* as examples.

#### **load and save**

The methods

```
virtual bool load( string& filename )  
virtual bool save( string& filename )
```

must be coded to load and save your *Network* derived object from and to a file named “filename”. Typically the method `save` would use your coded method `outputHeader` to write the *Network* object architecture to a file, then the Matrices and vectors representing the *Network* model. As expected, `load` should do the opposite. See *SimpleProp::load*, *SimpleProp::save*, *BareProp::load* and *BareProp::save* as examples.

## innerTrainSet

The method

```
virtual double innerTrainSet()
```

must be coded in a *Network* object to calculate the gradient and error. The reason that another method in addition to the pure virtual method `trainSet` inherited from *Iterative* must be coded is to support automatic stepsize selection. The strategy proceeds as follows: `trainSet` is intended to call `innerTrainSet` to determine if the set error would increase, and set the learning rate (“step size”) accordingly lower if it would. Thus, `innerTrainSet` calculates the gradient and error, but does not necessarily update the weights. If automatic stepsize *is not* selected, `trainSet` simply calls `innerTrainSet` to calculate the gradient and error, and updates the weights. If automatic stepsize *is* selected, `trainSet` calls `innerTrainSet` to calculate the gradient and error, and reduces the stepsize before updating the weights if the error is found to increase. See `SimpleProp::innerTrainSet`, `BareProp::innerTrainSet` and `BackProp::innerTrainSet` as examples of the way `trainSet` and `innerTrainSet` are coded to work together.

The search itself is written once, as `Network::searchStepSize` — a protected member template. Each concrete model’s `trainSet` supplies two things and nothing else:

- its own **activation guard**, which is *not* the same in every model. *Logistic* searches whenever `automaticStepSizeFlag` is set; *SimpleProp*, *BareProp* and *BackProp* require `batchEpochFlag` as well, because the search compares the error of a whole pass against the previous pass and that only means something when a pass makes one weight update. The guard is written out at each call site rather than delegated to a policy method, so the asymmetry cannot be lost.
- its own **WeightSnapshot**, a private nested type holding just that model’s weights: `hW` and `oW` for *SimpleProp* and *BareProp*, `Weights` for *BackProp*, `W` for *Logistic*. `searchStepSize` constructs one as a local of the search, so no model carries a duplicate of its weights between calls and nothing is copied at all when the search is off.

*Network* restores `lastG` and `lastF` itself, since those are its own members and are the history conjugate gradient descent and Shanno carry between iterations. Nothing else is restored:  $\eta$ , the gradients, the accumulators and `currGradMax` are left as the trial passes leave them. Restoration is explicit rather than a destructor, so an exception out of `innerTrainSet` leaves trial weights installed exactly as it always has.

Richard Golden has suggested the following automatic stepsize selection algorithm, which he terms the “Crummy Back-tracking Line Search Algorithm”:

- Let `DELTAERROR` be a small non-negative number (e.g., `DELTAERROR = 1e-10` or `0`)

- Let MAXLOOPS = a small positive integer (e.g., MAXLOOPS = 2 or 3 or 4)
- Let GAMMA be some positive number less than 1 (e.g., GAMMA = 0.5 or 0.8)
- Let LoopCounter = 0
- Set “candidate” learning rate equal to 1.0
- While Exit Loop is FALSE
  - LoopCounter = LoopCounter + 1
  - Compute change to weights using current candidate learning rate
  - Evaluate the new ERROR at the weights you obtained to get a “candidate error change”
  - If (“candidate error change” < DELTAERROR) OR (LoopCounter > MAXLOOPS), then EXITLOOP is set to TRUE
  - Else set current candidate learning rate = GAMMA [candidate learning rate]
  - End While Loop

Don’t be fooled by the name—this algorithm is far from “crummy”.

**Note:** This algorithm is only to be used with off-line or batch/epoch learning.

## forward

The method

```
virtual void forward( Matrix< double >& M, const unsigned r )
```

must be coded to take a Matrix containing only the inputs of a training or test set as one argument (typically the *Train* or *Test* inherited protected members from *Model*), the position of the exemplar in the training or test set as a second argument, and derive the *Network*’s output. *This is not returned by the method. This is very important: in single output Networks, the inherited Network protected member o would be set by this method, and in multiple output Networks, the inherited Network protected member vector< double > output would be set by this method. Also very important, if sigmoidal output nodes are supported, the inherited Network protected member x would be set by this method, and in multiple output Networks, the inherited Network protected member vector< double > xs would be set by this method.*

## removeInputs

The method

```
virtual void removeInputs( const vector< unsigned >& v )
```

must be coded to take as its argument a vector of unsigned representing positions of input nodes, and to remove those nodes from the *Network* derived object. Typically, this would be accomplished first by a call to *DataSet::removeInputs*(vector< unsigned >& v ), followed by a call to *Model::extractInputMatrices*(), followed by code which removes input nodes in the structures representing weights in the *Network* object, typically *Matrix* columns. See *SimpleProp::removeInputs* and *BareProp::removeInputs* as examples.

### 12.19.2 Cloning and construction required for a new *Network*

A new concrete *Network* type must participate in the centralized *cloneNetwork*() dispatch in *netclone.\**; this is the clone used by stepwise regression and automatic optimizer probes. If the type may be created or loaded through a human interface it must also participate in *modelfactory.\**. *RegressNet* contains no type switch and must not be edited for each concrete network. See section 13.6.

### 12.19.3 *Network* protected members

The following are the protected members in *Network* that your object needs to set or get:

#### trainingType

The protected member

```
unsigned trainingType
```

determines the type of gradient descent algorithm of your neural network. *trainingType* = 0 is defined as canonical backpropagation. If you add other types of training to the *Network* class, you must define a unique *trainingType* for them. Currently coded algorithms include *trainingType* =

1. Conjugate gradient descent
2. Shanno algorithm

**NOTE:** Look closely at the code for *Network::getGradMax*(). You will find that this code assumes that if *trainingType* is not 0, then *stackG* will be constructed. If this is not true for your *trainingType*, then you must modify the if block in *Network::getGradMax*().

**o**

The protected member

**double o**

is the output for a single output network. *If your Network object supports single outputs, it must set this protected member*, typically in the method `forward`. Since the method `trainSet` will typically call the method `forward`, `trainSet` will generally access the member `o`. See `SimpleProp::forward` and `BareProp::forward` for examples in setting the protected member `o`, and `SimpleProp::trainSet` and `BareProp::trainset` for examples in accessing the protected member `o`.

**x**

The protected member

**double x**

contains the  $x$  term for the sigmoidal output  $o$  in a single output network:

$$o = 1/(1 + e^{-x})$$

*If your Network object supports single sigmoidal outputs, it must set this protected member*, typically in the method `forward`. See `SimpleProp::forward` and `BareProp::forward` for examples in setting the protected member `x`.

**output**

The protected member

**vector< double > output**

is the output for a multiple output network. *If your Network object supports multiple outputs, it must set this protected member*, typically in the method `forward`. Since the method `trainSet` will typically call the method `forward`, `trainSet` will generally access the member `output`.

**xs**

The protected member

**vector< double > xs**

contains the  $\vec{x}$  terms for the sigmoidal outputs  $\vec{o}$  in a multiple output network:

$$\vec{o} = 1/(1 + e^{-\vec{x}})$$

*If your Network object supports multiple sigmoidal outputs, it must set this protected member*, typically in the method `forward`. See `BackProp::forward` for an example in setting the protected member `xs`.

### **randomLimit**

The protected member

**double randomLimit**

is used to specify the range of randomized weights, which will be between  $-randomLimit$  and  $+randomLimit$ . Access this protected member in your `randomize` method. See `SimpleProp::randomize` and `BareProp::randomize` for examples.

### **weightsSetFlag**

Your derived *Network* object must set the protected member

**bool weightsSetFlag**

to *true* when the weights are set, either by randomization or loading weights from a file. See `SimpleProp::randomize`, `BareProp::randomize`, `SimpleProp::load` and `BareProp::load` for examples of setting *weightsSetFlag* to *true*, and `SimpleProp::setHidden` and `BareProp::setHidden` for examples of setting *weightsSetFlag* to *false*.

### **biasFlag**

Use the protected member

**bool biasFlag**

to determine if your *Network* object has biases.

### **eta**

Use the protected member

**double eta**

for the learning rate  $\eta$  of your *Network* object.

### **batchEpochFlag**

Use the protected member

**bool batchEpochFlag**

to determine if your *Network* object uses batch/epoch learning or not. (See the description of classic off-line (“batch”) backprop in [section 6.2.2](#).) *Pay special attention to how the learning rate is set*. It is expected that the driver will guide the user to the appropriate setting of the learning rate. Note also that “off- line” learning is synonymous with “ batch” or “epoch” learning, whereas “on-line” learning refers to updating the weights with each exemplar. Here, if `batchEpochFlag` is *true*, “offline” learning is invoked, and if `batchEpochFlag` is *false*, “online” learning is used.

### weightDecayFlag

Use the protected member

`bool weightDecayFlag`

to determine if your *Network* object uses weight decay. (See the discussion of [equation 6.1 in section 6.1.2](#) for an explanation of weight decay.) The learning rate equivalent of [equation 6.1](#) is

$$\begin{aligned}
\vec{w}_{t+1} &= \vec{w}_t - \eta \left. \frac{\partial E'}{\partial w} \right|_{w_t} \\
&= \vec{w}_t - \eta \left. \frac{\partial E}{\partial w} \right|_{w_t} - 2\eta\lambda\vec{w}_t \\
&= (1 - 2\eta\lambda)\vec{w}_t - \eta \left. \frac{\partial E}{\partial w} \right|_{w_t}
\end{aligned}$$

where  $(1 - 2\eta\lambda)$  is “weight decay”.  $\lambda$  may be set by either of 2 methods ( $N$  is the sample number size, see [equation 6.1 in section 6.1.2](#) for a discussion of  $\sigma_w$ ):

- The complexity of the population objective function  $\gg$  sample objective function:

$$\lambda = \frac{1}{N2\sigma_w^2} \quad (12.5)$$

- The complexity of the population objective function  $\simeq$  sample objective function:

$$\lambda = \frac{1}{2\sigma_w^2} \quad (12.6)$$

It is anticipated that the driver will guide the user to set the appropriate value of  $\lambda$ , and *insure that the user knows what he or she’s doing if they change it!*

### decay

You can use the protected member

`double decay`

for the weight decay term  $2\lambda$  of your *Network* object. Note the multiple of 2 so that calculation of decay with the gradient as in the right hand term of [equation 6.10](#),  $(1/\sigma_w^2)[\mathbf{W} , \mathbf{b}]$ , does not require the additional step of multiplication by 2.

### **regularizer**

You can use the protected member

```
double regularizer
```

if you need to access  $\lambda$  directly, e.g. for calculation of your error function in equation 6.1, which like *decayTerm* only needs to be calculated once prior to iterative calls to the *TrainSet* method, and is thus placed in the member function *runHeader*, where the *Iterative* class will call it once prior to iteratively calling the *TrainSet* method.

### **decayTerm**

In the protected member function *runHeader* in the *Network* class,

```
double decayTerm
```

is defined to be  $1 - 2\eta\lambda$  (or  $1 - \eta\text{decay}$ ), which only needs to be calculated once prior to iterative calls to the *TrainSet* method, and is thus placed in the member function *runHeader*, where the *Iterative* class will call it once prior to iteratively calling the *TrainSet* method. You may therefore use *decayTerm* as the weight multiple for weight decay in your *Network* object if the decay is not calculated with the gradient. If the decay is calculated with the gradient as in equation 6.10, then *decay* is used. See *BareProp::TrainSet* and *SimpleProp::TrainSet* for examples using *decayTerm*.

### **stackG**

The protected member

```
vector< double > stackG
```

is a single vector which contains the entire weight gradient of your *Network* derived object. It is used by the protected utility functions *pack* and *unpack* (see below).

## **12.19.4 Network protected utility functions**

### **Copy utility function, constructor and overloaded = operator**

As with any derivation of *Model*, a copy utility function, copy constructor and overloaded = operator must be implemented. See section 12.15.2 for details.

### **runHeader utility function**

If you are adding parameters specific to a *Network* object that should be output to screen or log files prior to an *Iterative* run, you will need to add code to the *runHeader* method in *Network*. See section 12.17.

## pack and unpack

The methods

```
virtual void pack()
virtual void unpack()
```

must be coded to convert the weight gradient structure of your *Network* derived object to the member vector *stackG* (pack), and vice versa (unpack). See SimpleProp::pack, SimpleProp::unpack, BareProp::pack and BareProp::unpack as examples.

### 12.19.5 Built-in gradient descent algorithms

One of the really neat things about *Networks* that construct the *stackG* vector is that they may use the built-in advanced gradient descent algorithms, Shanno's and conjugate gradient descent.<sup>11</sup> All the programmer needs to do is to construct the *stackG* vector, add the line of code

```
engine( trainingType, iteration );
```

after the *Network* object gradient structure is calculated and before the weights are updated, and these algorithms will automagically be available for the user. See SimpleProp::trainSet() and BareProp::trainSet() for examples.

#### Shanno algorithm

<sup>12</sup>

Briefly, Shanno's algorithm proceeds as follows: let the computed gradient be  $\vec{g}$  (*stackG*), and the iteration be  $t$  (**Note:** the iteration is *not* the exemplar index through the training set. It is the actual iteration number for each calculated gradient. It should also be noted that this algorithm, as well as conjugate gradient descent *is intended for off-line or batch/epoch learning.*) Let the learning rate be  $\gamma$ .

- Step 0: Let  $t = 0$ .
- Step 1: Set  $\vec{f}(t) = -\vec{g}_t$ . Go to step 3.
- Step 2: If  $t$  is a multiple of the degrees of freedom of the *Network*, then go to step 1. Otherwise, compute  $\vec{u}_t = \vec{g}_t - \vec{g}_{t-1}$  and the scalars  $a_t, b_t, c_t$  such that

$$- a_t = \frac{\vec{f}(t-1)^T \vec{g}_t}{\vec{f}(t-1)^T \vec{u}_t},$$

---

<sup>11</sup>Richard M. Golden, Mathematical Methods for Neural Network Analysis and Design, 1996, MIT Press, ISBN 0-262-07174-6, pp. 217–218, 221–222

<sup>12</sup>D. F. Shanno, "Conjugate gradient methods with inexact searches," *Mathematics of Operations Research* 3(3), 244–256, 1978, <https://doi.org/10.1287/moor.3.3.244>.

$$\begin{aligned}
- b_t &= \frac{\vec{u}_t^T \vec{g}_t}{\vec{f}(t-1)^T \vec{u}_t}, \\
- c_t &= \gamma + \frac{|\vec{u}_t|^2}{\vec{f}(t-1)^T \vec{u}_t},
\end{aligned}$$

and descent vector

$$\vec{f}(t) = -\vec{g}_t + a_t \vec{u}_t + (b_t - c_t a_t) \vec{f}(t-1)$$

- Step 3: Update the weights by  $\vec{w}(t+1) = \vec{w}(t) + \gamma \vec{f}(t)$ . Let  $t = t+1$ . Go to step 2.

Interested readers are directed to Richard Golden's book, pp. 217–218. It should be noted that in the highly optimized code in `Network::shanno(unsigned iteration)`, *lastF* refers to  $-\vec{f}(t-1)$ .

### Conjugate gradient descent

Conjugate gradient descent is a special case of the Shanno algorithm, and briefly proceeds as follows: let the computed gradient be  $\vec{g}$  (*stackG*), and the iteration be  $t$  (**Note:** again, the iteration is *not* the exemplar index through the training set. It is the actual iteration number for each calculated gradient. It should also be noted that this algorithm, as well as Shanno's, *is intended for off-line or batch/epoch learning.*) Let the learning rate be  $\gamma$ .

- Step 0: Let  $t = 0$ .
- Step 1: Set  $\vec{f}(t) = -\vec{g}_t$ . Go to step 3.
- Step 2: If  $t$  is a multiple of the degrees of freedom of the *Network*, then go to step 1. Otherwise, compute  $\vec{u}_t = \vec{g}_t - \vec{g}_{t-1}$  and the scalar  $b_t$  such that

$$b_t = \frac{\vec{u}_t^T \vec{g}_t}{\vec{u}_t^T \vec{f}(t-1)}$$

and descent vector

$$\vec{f}(t) = -\vec{g}_t + b_t \vec{f}(t-1)$$

- Step 3: Update the weights by  $\vec{w}(t+1) = \vec{w}(t) + \gamma \vec{f}(t)$ . Let  $t = t+1$ . Go to step 2.

Interested readers are directed to Richard Golden's book, pp. 221–222. It should be noted that in the highly optimized code in `Network::CGD(unsigned iteration)`, *lastF* refers to  $-\vec{f}(t-1)$ .

### 12.19.6 Calculation of the condition number

The condition number neuron reports is a *design* diagnostic: it answers whether the estimation problem is well posed, or whether the columns of the design are close enough to collinear that the coefficients are poorly identified. It is the ratio of the largest to the smallest absolute eigenvalue of the observed Fisher information of the *unpenalized* log likelihood,

$$\mathbf{X}^T \mathbf{V} \mathbf{X}, \quad \mathbf{V} = \text{diag}(\pi_i(1 - \pi_i))$$

where  $\pi_i$  is the fitted probability for exemplar  $i$ . This is the same matrix the Wald analysis inverts for the coefficient covariance (Hosmer & Lemeshow equation 2.8), and `Logistic::reportAccuracy` forms it exactly once and uses it for both. `Network::conditionOf( const Matrix<double>& )` takes that matrix and stores the two extreme absolute eigenvalues and their ratio; `Network::reportCondNum( ostream& )` prints what it stored and computes nothing. The *Gnu Scientific Library* symmetric eigenvalue routines are used.

The penalty is deliberately excluded. Weight decay adds  $\lambda \mathbf{I}$  to the curvature, which improves the conditioning of *any* design by construction, so a penalized condition number could be driven downward simply by regularizing more heavily — concealing exactly the collinearity the diagnostic exists to expose. If a penalized-curvature condition number is ever wanted, it belongs alongside this one under its own name, not in place of it.

**Historical note.** Through 2026 this quantity was instead the outer product  $\mathbf{B} = \mathbf{g} \cdot \mathbf{g}^T$  of per-exemplar gradient vectors, accumulated by the overloaded `storeGrads` methods under a `finalFlag` that marked a *final training iteration* run for the purpose. That construction was retired in August 2026 for four independent reasons: it trained the model as a side effect of reporting it; under conjugate gradient descent and Shanno’s algorithm it stored search *directions* rather than gradients, because the optimizer transformation had already been applied; under canonical backpropagation with gradient stopping disabled it read a matrix that had never been written; and it included the weight-decay penalty, so regularization improved the reported conditioning. `storeGrads`, `finalFlag` and the gradient accumulator are gone with it.

See `Logistic` for example code, and see `SimpleProp` for comments which indicate where code would be placed in a neural network style implemented `Network` object.

## 12.20 Network

In the following specification, methods and members that pertain directly to coding a concrete derived `Network` implementation are shown in **red**.

**Network** extends *Iterative*. Algorithms with hidden layers, weights, learning rates, transfer functions and other general properties of neural networks are encapsulated in *Network*.

**Note:** It is expected that in objects derived from *Network*, the method `setDataSet` inherited from *Model* be called *before* the derived methods that specify object architecture, so that the derived object may know the number of input nodes and size the Matrices related to the dataset prior to completing its architecture. This sequence of loading a dataset using the inherited *Model* method `setDataSet` prior to establishing network architecture will be invoked for all *Network* objects.

- *Public methods*

- `Network& operator= ( const Network& rhs )`: see section 12.15.2 for discussion
- `Network( const Network& rhs )`: see section 12.15.2 for discussion
- `virtual void setDataSet( DataSet& data )`: A pure virtual method inherited from *Model* that loads the *DataSet* object *data* into the *Model*
- `virtual void outputHeader( ostream& outputStream )`: A pure virtual method inherited from *Model* which outputs a header to an ostream *outputStream* describing the architecture of the model, *Model* type dependant
- `void setEta( double  $\eta$  )`: Sets the learning rate to  $\eta$
- `double getEta()`: Returns the learning rate  $\eta$
- `void setBias( bool flag )`: Sets bias nodes in *Network*
- `bool getBias()`: Gets boolean value indicating if bias nodes specified in *Network*
- `void setBatchEpoch( bool flag )`: Sets if batch/epoch learning is used in a *Network* object (See the description of classic off-line (“batch”) backprop in section 6.2.2)
- `bool getBatchEpoch()`: returns if batch/epoch used
- `void setWeightDecay( bool flag )`: Sets if weight decay is used in a *Network* object (See the discussion of *weight decay* in section 12.19.3)
- `bool getWeightDecay()`: returns if weight decay used
- `void setDecay( double  $2\lambda$  )`: Sets weight decay term to  $2\lambda$  in a *Network* object
- `double getDecay()`: returns weight decay term  $\lambda$
- `void setTrainingType( unsigned T )`: Sets training type to *T*
- `unsigned getTrainingType()`: returns training type
- `void setAutoStepSize( bool flag )`: Sets if automatic stepsize selection is used (See the discussion of *automatic stepsize selection* in section 12.19.1)

- `bool getAutoStepSize()`: returns if automatic stepsize selection is used
- `virtual unsigned df()`: Returns the number of degrees of freedom, equal to the number of nodal connections
- `void setRandomLimit( double  $L$  )`: Sets the limit for the random weights  $-L$  to  $+L$
- `virtual void randomize()`: A pure virtual method which randomizes initial weights in a *Network* object
- `getWeightsSet()`: Returns a flag to indicate if weights set
- `virtual bool save( string& filename )`: A pure virtual method which saves the *Network* object architecture and weights to a file named *filename*, returns true if successful
- `virtual bool load( string& filename )`: A pure virtual method which loads the *Network* object architecture and weights from a file named *filename*, returns true if successful
- `virtual double trainSet()`: A pure virtual method which trains one iteration through the training set, *Network* type dependant, and returns the set error
- `virtual double innerTrainSet()`: A pure virtual method which is called by `trainSet()` to calculate the gradient and return set error, used for automatic stepsize selection
- `virtual void forward( Matrix< double >&  $M$ , unsigned  $n$  )`: A pure virtual method which forward propagates for one input row  $n$  in a Matrix of inputs  $M$  (e.g. either the inherited *Model* Matrix *Train* or *Test*)
- `virtual void reportAccuracy( ostream& outputStream )`: Implementation of the pure virtual method inherited from *Model* that reports the accuracy of the trained model to an ostream *outputStream*
- `virtual void classAccuracy( ostream& outputStream )`: A pure virtual method inherited from *Iterative* which outputs classification accuracies to an ostream *outputStream* for an *Iterative* output table
- `virtual void removeInputs( vector< unsigned >&  $v$  )`: A pure virtual method which takes as its argument a vector of unsigned  $v$  representing positions of input nodes, and removes those nodes from the *Network* object

- *Protected* members

- `unsigned trainingType`: the type of gradient descent algorithm, *trainingType* = 0 is reserved for canonical backpropagation (*trainingType* = 1 has been coded for conjugate gradient descent, and *trainingType* = 2 for Shanno algorithm)

- double *o*: the output for a single-output *Network*
- double *x*: the *x* term for the sigmoidal output *o* in a single output network:

$$o = 1/(1 + e^{-x})$$

- vector < double > *output*: the outputs vector for a multiple output *Network*
- vector < double > *xs*: the  $\vec{x}$  terms for the sigmoidal outputs  $\vec{o}$  in a multiple output network:

$$\vec{o} = 1/(1 + e^{-\vec{x}})$$

- double *randomLimit*: the limit for the random weights
- bool *weightsSetFlag*: a flag to indicate if weights set
- bool *biasFlag*: a flag to indicate if bias nodes specified in *Network*
- bool *batchEpochFlag*: a flag to indicate if batch/ epoch training is used in the *Network* object ( See the description of classic off-line (“batch”) backprop in [section 6.2.2](#))
- bool *weightDecayFlag*: a flag to indicate if weight decay is used in the *Network* object (See the discussion of [weight decay](#) in [section 12.19.3](#))
- double *decay*: the weight decay term  $2\lambda$
- double *regularizer*: the weight decay term  $\lambda$
- double *decayTerm*: the weight decay multiple  $1 - 2\eta\lambda$  calculated in the protected member function `runHeader` (see [section 12.19.3](#) for an explanation)
- double *eta*: the learning rate  $\eta$
- bool *automaticStepSizeFlag*: a flag to indicate if automatic step size algorithm is used
- double *o\_errAccumulate*: stores average output error accumulation over the exemplars used in `innerTrainSet()` method
- double *oldErrorAccumulate*: stores average output error accumulation for previous run over the exemplars, used in `innerTrainSet()` method
- double *deltaError*: `deltaError` constant for automatic stepsize selection
- double *currGradMax*: current absolute maximum gradient for shanno/CGD
- double *gamma*: gamma constant for automatic stepsize selection
- unsigned *maxLoops*: `maxLoops` constant for automatic stepsize selection

- vector < double > *stackG*: a single vector which contains the entire weight gradient of the *Network* derived object
  - vector < double > *lastG*:  $\vec{g}(t-1)$  for shanno/CGD
  - vector < double > *lastF*:  $\vec{f}(t-1)$  for shanno/CGD
  - bool *finalFlag*: a flag to indicate that the train methods (inner-TrainSet() or trainSet()) are at the final iteration, for code to be run only during the final iteration
  - Matrix < double > *grads*: a matrix to hold exemplar gradients formerly used to compute the **B** matrix for the condition number
- *Protected Utility functions*
    - void copy( const Network& rhs ): Copy utility function, see section 12.15.2 for discussion.
    - virtual void runHeader( ostream& outputStream ): A pure virtual utility function inherited from the Iterative class that outputs Network specific parameters preceeding an Iterative run
    - virtual void pack(): A pure virtual utility function that converts the entire weight gradient structure of the *Network* derived object to the protected member *stackG*
    - virtual void unpack(): A pure virtual utility function that converts the protected member *stackG* back to the weight gradient structure of the *Network* derived object
    - virtual double getGradMax(): A pure virtual utility function inherited from *Iterative* that returns the maximum absolute gradient
    - void engine( unsigned type, unsigned iteration ): an utility function which calls the appropriate training algorithm depending on training type *type*, and passes it the current iteration
    - void engine( unsigned type, unsigned iteration ): chooses which kind of advanced gradient descent algorithm (CGD or shanno) based on *type*, and passes *iteration* to it
    - void CGD( unsigned iteration ): implements conjugate gradient descent (see Golden pp. 221–222)<sup>13</sup>
    - void shanno( unsigned iteration ): implements Shanno algorithm (see Golden pp. 217–218)
    - void conditionOf( const Matrix<double>& symmetric ): stores the largest and smallest absolute eigenvalues of the passed symmetric matrix and their ratio, the condition number. The caller supplies the matrix, because which matrix the condition number describes is

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<sup>13</sup>Richard M. Golden, Mathematical Methods for Neural Network Analysis and Design, 1996, MIT Press, ISBN 0-262-07174-6

a statistical decision belonging to the caller; *Logistic* passes  $\mathbf{X}^T \mathbf{V} \mathbf{X}$  (see above). Replaced the overloaded `storeGrads(...)` methods and the *grads* accumulator in August 2026

- `void reportCondNum( ostream& outputStream )`: utility method to report the stored condition number to *outputStream*. It prints; it does not compute

## 12.21 BareProp

The **BareProp** class is a *Network* object that serves to illustrate neural network design in neUROn2++. *BareProp* is a one output node, one hidden layer neural network without biases.

**Note:** It is expected that the method `setDataSet` inherited from *Model* be called *before* the *BareProp* method `setHidden`, so that the *BareProp* object may know the number of input nodes and size the Matrices related to the dataset prior to completing its architecture. This sequence of loading a dataset using the *Model* method `setDataSet` prior to establishing network architecture will be invoked for all *Network* objects.

- *Public methods*

- `BareProp& operator= ( const BareProp& rhs )`: see section 12.15.2 for discussion
- `BareProp( const BareProp& rhs )`: see section 12.15.2 for discussion
- `virtual void setDataSet( DataSet& data )`: Implementation of the pure virtual method inherited from *Model* that loads the *DataSet* object *data* into the *BareProp Model*
- `virtual void outputHeader( ostream& outputStream )`: Implementation of the pure virtual method inherited from *Model* which outputs a header to an ostream *outputStream* describing the architecture of the *Bareprop Model*
- `void setHidden( unsigned H )`: Sets number of hidden nodes to *H*, which is all that is required to specify the architecture for a *BareProp* object
- `virtual unsigned df()`: Implementation of the pure virtual method inherited from *Network* which returns the number of degrees of freedom, equal to the number of nodal connections
- `virtual void randomize()`: Implementation of the pure virtual method inherited from *Network* which randomizes initial weights
- `virtual bool save( string& filename )`: Implementation of the pure virtual method inherited from *Network* which saves the *BareProp* object architecture and weights to a file named *filename*, returns true if successful

- virtual bool load( string& *filename* ): Implementation of the pure virtual method inherited from *Network* which loads the *BareProp* object architecture and weights from a file named *filename*, returns true if successful
  - virtual double trainSet(): Implementation of the pure virtual method inherited from *Iterative* which trains one iteration through the training set, and returns set error
  - virtual double innerTrainSet(): Implementation of the pure virtual method inherited from *Network* which is called by trainSet() to calculate the gradient and return set error, used for automatic stepsize selection
  - virtual void forward( Matrix< double >& *M*, unsigned *n* ): Implementation of the pure virtual method inherited from *Network* which forward propagates for one input row *n* in a Matrix of inputs *M* (e.g. either the inherited *Model* Matrix *Train* or *Test*)
  - virtual void removeInputs( vector< unsigned >& *v* ): Implementation of the pure virtual method which takes as its argument a vector of unsigned *v* representing positions of input nodes, and removes those nodes from the *BareProp* object
- *Private* members
    - unsigned *nHidden*: number of hidden nodes
    - Matrix< double > *hW*: hidden weight Matrix
    - Matrix< double > *hWup*: hidden weight update Matrix
    - Matrix< double > *hG*: hidden weight gradient Matrix
    - vector< double > *y*: vector containing dataset outputs
    - vector< double > *I*: vector for single exemplar inputs
    - vector< double > *hO*: hidden output vector
    - vector< double > *oW*: output weight vector
    - vector< double > *oG*: output weight gradient vector
    - vector< double > *h\_err*: hidden error vector
    - double *o\_err*: output error term
  - *Private* Utility functions
    - void copy( const BareProp& rhs ): Copy utility function, see section [12.15.2](#) for discussion.
    - virtual void pack(): Implementation of the pure virtual utility function inherited from *Network* that converts the entire weight gradient structure to the *Network* inherited protected member *stackG*
    - virtual void unpack(): Implementation of the pure virtual utility function inherited from *Network* that converts the *Network* inherited protected member *stackG* back to the weight gradient structure

## 12.22 SimpleProp

The **SimpleProp** class is a *Network* object that serves to illustrate neural network design in neUROn2++. *SimpleProp* is a one output node, one hidden layer neural network with biases.

**Note:** It is expected that the method `setDataSet` inherited from *Model* be called *before* the *SimpleProp* method `setHidden`, so that the *SimpleProp* object may know the number of input nodes and size the Matrices related to the dataset prior to completing its architecture. This sequence of loading a dataset using the *Model* method `setDataSet` prior to establishing network architecture will be invoked for all *Network* objects.

- *Public* methods
  - `SimpleProp& operator= ( const SimpleProp& rhs )`: see section 12.15.2 for discussion
  - `SimpleProp( const SimpleProp& rhs )`: see section 12.15.2 for discussion
  - `virtual void setDataSet( DataSet& data )`: Implementation of the pure virtual method inherited from *Model* that loads the *DataSet* object *data* into the *SimpleProp Model*
  - `virtual void outputHeader( ostream& outputStream )`: Implementation of the pure virtual method inherited from *Model* which outputs a header to an ostream *outputStream* describing the architecture of the *Bareprop Model*
  - `void setHidden( unsigned H )`: Sets number of hidden nodes to *H*, which is all that is required to specify the architecture for a *SimpleProp* object
  - `virtual unsigned df()`: Implementation of the pure virtual method inherited from *Network* which returns the number of degrees of freedom, equal to the number of nodal connections
  - `virtual void randomize()`: Implementation of the pure virtual method inherited from *Network* which randomizes initial weights
  - `virtual bool save( string& filename )`: Implementation of the pure virtual method inherited from *Network* which saves the *SimpleProp* object architecture and weights to a file named *filename*, returns true if successful
  - `virtual bool load( string& filename )`: Implementation of the pure virtual method inherited from *Network* which loads the *SimpleProp* object architecture and weights from a file named *filename*, returns true if successful
  - `virtual double trainSet()`: Implementation of the pure virtual method inherited from *Iterative* which trains one iteration through the training set, and returns set error

- virtual double innerTrainSet(): Implementation of the pure virtual method inherited from *Network* which is called by trainSet() to calculate the gradient and return set error, used for automatic stepsize selection
  - virtual void forward( Matrix< double >& *M*, unsigned *n* ): Implementation of the pure virtual method inherited from *Network* which forward propagates for one input row *n* in a Matrix of inputs *M* (e.g. either the inherited *Model* Matrix *Train* or *Test*)
  - virtual void removeInputs( vector< unsigned >& *v* ): Implementation of the pure virtual method which takes as its argument a vector of unsigned *v* representing positions of input nodes, and removes those nodes from the *SimpleProp* object
- *Private* members
    - unsigned *nHidden*: number of hidden nodes
    - unsigned *nH*: *nHidden* – 1, used for indexing with biases
    - Matrix< double > *hW*: hidden weight Matrix
    - Matrix< double > *hWup*: hidden weight update Matrix
    - Matrix< double > *hG*: hidden weight gradient Matrix
    - vector< double > *in\_bias*: input bias vector to add to input Matrix
    - vector< double > *y*: vector containing dataset outputs
    - vector< double > *I*: vector for single exemplar inputs
    - vector< double > *hO*: hidden output vector
    - vector< double > *oW*: output weight vector
    - vector< double > *oG*: output weight gradient vector
    - vector< double > *h\_err*: hidden error vector
    - double *o\_err*: output error term
  - *Private* Utility functions
    - void copy( const SimpleProp& rhs ): Copy utility function, see section 12.15.2 for discussion.
    - virtual void pack(): Implementation of the pure virtual utility function inherited from *Network* that converts the entire weight gradient structure to the *Network* inherited protected member *stackG*
    - virtual void unpack(): Implementation of the pure virtual utility function inherited from *Network* that converts the *Network* inherited protected member *stackG* back to the weight gradient structure

## 12.23 BackProp

The **BackProp** class is a *Network* object that allows multiple output nodes and multiple hidden layers in a neural network, and biases may or may not be present.

**Note:** It is expected that the *Model* method `setDataSet` be called *before* the *BackProp* method `setHidden`, so that the *BackProp* object may know the number of input nodes and size the Matrices related to the dataset prior to completing its architecture. This sequence of loading a dataset using the *Model* method `setDataSet` prior to establishing network architecture will be invoked for all *Network* objects.

- *Public* methods
  - `BackProp& operator= ( const BackProp& rhs )`: see section 12.15.2 for discussion
  - `BackProp( const BackProp& rhs )`: see section 12.15.2 for discussion
  - `virtual void setDataSet( DataSet& data )`: Implementation of the pure virtual method inherited from *Model* that loads the *DataSet* object *data* into the *BackProp Model*
  - `virtual void outputHeader( ostream& outputStream )`: Implementation of the pure virtual method inherited from *Model* which outputs a header to an ostream *outputStream* describing the architecture of the *Backprop Model*
  - `void setHidden( const vector< unsigned >& v )`: Sets the hidden layers, with the number of hidden nodes on each layer indicated by the corresponding vector *v* element
  - `virtual unsigned df()`: Implementation of the pure virtual method inherited from *Network* which returns the number of degrees of freedom, equal to the number of nodal connections
  - `virtual void randomize()`: Implementation of the pure virtual method inherited from *Network* which randomizes initial weights
  - `virtual bool save( string& filename )`: Implementation of the pure virtual method inherited from *Network* which saves the *BackProp* object architecture and weights to a file named *filename*, returns true if successful
  - `virtual bool load( string& filename )`: Implementation of the pure virtual method inherited from *Network* which loads the *BackProp* object architecture and weights from a file named *filename*, returns true if successful
  - `virtual double trainSet()`: Implementation of the pure virtual method inherited from *Iterative* which trains one iteration through the training set, and returns set error

- virtual double innerTrainSet(): Implementation of the pure virtual method inherited from *Network* which is called by trainSet() to calculate the gradient and return set error, used for automatic stepsize selection
  - virtual void forward( Matrix< double >& *M*, unsigned *n* ): Implementation of the pure virtual method inherited from *Network* which forward propagates for one input row *n* in a Matrix of inputs *M* (e.g. either the inherited *Model* Matrix *Train* or *Test*)
  - virtual void removeInputs( vector< unsigned >& *v* ): Implementation of the pure virtual method which takes as its argument a vector of unsigned *v* representing positions of input nodes, and removes those nodes from the *BackProp* object
- *Private* members
    - vector< unsigned > *nLayer*: describes the number of hidden nodes on each layer indicated by the corresponding vector element in *nLayer*
    - unsigned *nLayers*: number of hidden layers
    - vector< double > *I*: vector for single exemplar inputs
    - vector< double > *y*: vector containing dataset outputs
    - vector< double > *o\_err*: vector containing output errors
    - vector< Matrix< double > > *Weights*: vector of weight Matrix for all hidden layers and output layer
    - vector< Matrix< double > > *WeightsUp*: vector of Matrices to hold updated weights one for each weights Matrix
    - vector< Matrix< double > > *WeightsAccumulate*: vector of Matrices holding the per-exemplar sum accumulated across a batch/epoch pass. Divided by the number of training exemplars it is the averaged *raw* batch gradient of Methodology equation 2.14. It is not the quantity the weight update consumes when a search algorithm is selected — see *Gradient* below.
    - vector< Matrix< double > > *Gradient*: vector of Matrices holding the gradient as a separate structure, which is required whenever an algorithm other than canonical backpropagation is chosen, or gradient stopping is armed. The averaged batch gradient is copied into it; *engine* then transforms it into the selected optimizer’s search direction, by way of *pack* and *unpack*. **The weight update must therefore be taken from *Gradient***, in batch/epoch mode as well as on-line. Taking it from *WeightsAccumulate* discards the search direction and silently reduces conjugate gradient descent and Shanno to plain gradient descent (corrected 2026-08-01; see tests/backprop/check.bpoptimizer.cpp).
    - vector< vector< double > > *vpack*: per-layer holding vectors used by *unpack* to rebuild *Gradient* from *stackG*

- vector< vector< double > > *HOutputs*: vector containing a vector of outputs for each hidden layer
- vector< vector< double > > *HErrors*: vector containing a vector of errors for each hidden layer
- *Private* Utility functions
  - void copy( const BackProp& rhs ): Copy utility function, see section 12.15.2 for discussion.
  - virtual void pack(): Implementation of the pure virtual utility function inherited from *Network* that converts the entire weight gradient structure to the *Network* inherited protected member *stackG*
  - virtual void unpack(): Implementation of the pure virtual utility function inherited from *Network* that converts the *Network* inherited protected member *stackG* back to the weight gradient structure

## 12.24 Logistic

The **Logistic** class is a *Network* object that implements logistic regression. This form of data modeling is mathematically equivalent to a single output node neural network with a bias, the sigmoidal transfer function, no hidden nodes, and the cross-entropy error function. As a result, logistic regression is implemented as a derived *Network* class. This strategy also allows the user to perform stepwise forward and reverse regression on the built logistic regression model using the *RegressNet* class.

Note that the error is *cross-entropy* by definition, that the gradient descent method is *batch-epoch* by definition, and a bias node (y-intercept) is present by definition. At the end of a run, the  $\beta$  weights and standard error are reported and *p*-values for the Wald test that the  $\beta$  weights are nonzero.<sup>14</sup>

**Note:** It is expected that the method *setDataSet* inherited from *Model* be called *before* the *Logistic* method *randomize* () (which sets initial weights to zero or random values, depending on the value of the flag *randInitCondFlag* which is set by the method *setInitCond*()), so that the *Logistic* object may know the number of input nodes and size the vectors and Matrices related to the dataset prior to completing its architecture. This sequence of loading a dataset using the *Model* method *setDataSet* prior to establishing network architecture will be invoked for all *Network* objects.

- *Public* methods
  - Logistic& operator= ( const Logistic& rhs ): see section 12.15.2 for discussion

---

<sup>14</sup>see Hosmer and Lemeshow, Applied Logistic Regression, 2nd edition, John Wiley & Sons, Inc., New York, 2000, pp 36–42

- `Logistic( const Logistic& rhs )`: see section 12.15.2 for discussion
- `virtual void setDataSet( DataSet& data )`: Implementation of the pure virtual method inherited from *Model* that loads the *DataSet* object *data* into the *Logistic Model*
- `virtual void outputHeader( ostream& outputStream )`: Implementation of the pure virtual method inherited from *Model* which outputs a header to an ostream *outputStream* describing the architecture of the *Logistic Model*
- `virtual unsigned df()`: Implementation of the pure virtual method inherited from *Network* which returns the number of degrees of freedom, equal to the number of inputs + 1 for the y- intercept
- `virtual void randomize()`: Implementation of the pure virtual method inherited from *Network* which randomizes initial weights
- `virtual void reportAccuracy( ostream& outputStream )`: overrides (calls base *Network* `reportAccuracy()` method first) *Network* method inherited from *Iterative*, adding binary logistic regression specific footer information, including reporting  $\beta$  weights, standard error, and Wald test ( $p$  for  $\beta$  weights nonzero)
- `virtual bool save( string& filename )`: Implementation of the pure virtual method inherited from *Network* which saves the *Logistic* object  $\beta$  weights and y-intercept to a file named *filename*, returns true if successful
- `virtual bool load( string& filename )`: Implementation of the pure virtual method inherited from *Network* which loads the *Logistic* object  $\beta$  weights and y-intercept from a file named *filename*, returns true if successful
- `virtual double trainSet()`: Implementation of the pure virtual method inherited from *Iterative* which trains one iteration through the training set, and returns set error
- `virtual double innerTrainSet()`: Implementation of the pure virtual method inherited from *Network* which is called by `trainSet()` to calculate the gradient and return set error, used for automatic stepsize selection
- `virtual void forward( Matrix< double >& M, unsigned n)`: Implementation of the pure virtual method inherited from *Network* which applies the sigmoidal function for one input row  $n$  in a Matrix of inputs  $M$  (e.g. either the inherited *Model* Matrix *Train* or *Test*)
- `virtual void removeInputs( vector< unsigned >& v )`: Implementation of the pure virtual method which takes as its argument a vector of unsigned  $v$  representing positions of input nodes, and removes those nodes from the *Logistic* object

- *Private* members

- vector< double > *y*: vector containing dataset outputs
  - vector< double > *I*: vector for single exemplar inputs
  - vector< double > *W*: vector for  $\beta$  weights (last is y-intercept)
  - vector< double > *G*: vector for  $\beta$  weight gradients
  - double *o\_err*: output error term
- *Private* Utility functions
    - void copy( const Logistic& rhs ): Copy utility function, see section 12.15.2 for discussion.
    - virtual void pack(): Implementation of the pure virtual utility function inherited from *Network* that converts the  $\beta$  weight gradient structure to the *Network* inherited protected member *stackG*, as the  $\beta$  weight structure is already a single vector, this simply copies it
    - virtual void unpack(): Implementation of the pure virtual utility function inherited from *Network* that converts the *Network* inherited protected member *stackG* back to the weight gradient structure, which, since it is a vector, just copies it

## 12.25 DFA

The **DFA** class is a *Model* object that encodes discriminant function analysis. It is an abstract class, whose derived concrete classes are *LDFA* for linear discriminant function analysis, and *QDFA* for quadratic discriminant function analysis.<sup>15</sup>

- *Public* methods
  - DFA& operator= ( const DFA& rhs ): see section 12.15.2 for discussion
  - DFA( const DFA& rhs ): see section 12.15.2 for discussion
  - virtual void setDataSet( DataSet& data ): Implementation of the pure virtual method inherited from *Model* that loads the *DataSet* object *data* into the *DFA Model*
  - virtual void outputHeader( ostream& outputStream ): Implementation of the pure virtual method inherited from *Model* which outputs a header to an ostream *outputStream* describing the architecture of the *DFA Model*
  - virtual double train(): A pure virtual method inherited from *Model* that trains a *Model*, and returns the final error (double) for that model

---

<sup>15</sup> An excellent treatment of discriminant function analysis can be found in Richard O. Duda and Peter E. Hart, *Pattern Classification and Scene Analysis*, 1973, John Wiley & Sons, New York, ISBN 0-471-22361-1, pp. 24-32

- virtual void `reportAccuracy( ostream& outputStream )`: A pure virtual method inherited from *Model* that reports the accuracy of the trained model to an ostream *outputStream*
- *Protected* members
  - Matrix< double > *D0*, *D1*: Matrices for inputs of 1-output datasets
  - vector< Matrix< double > > *D*: a vector of Matrices for inputs of multiple output datasets
  - vector< double > *X*: holds inputs for 1 exemplar
  - vector< double > *U0*, *U1*: mean vectors for 1-output datasets
  - vector< vector< double > > *U*: a vector of mean vectors for multiple output datasets
  - double *P0*, *P1*: a priori probabilities for 1-output datasets
  - vector< double > *P*: a priori probabilities for multiple output datasets
  - double *K0*, *K1*: constants for 1-output datasets
  - vector< double > *K*: constants for multiple output datasets
  - double *d0*, *d1*: discriminant function results for 1-output datasets
  - vector< double > *d*: discriminant function results for multiple output datasets
  - vector< unsigned > *trainClasses*: outputs for multiple output training sets
  - vector< unsigned > *testClasses*: outputs for multiple output test sets
- *Protected* Utility functions
  - void `copy( const DFA& rhs )`: Copy utility function, see section 12.15.2 for discussion.

## 12.26 LDFA

The **LDFA** class is a *DFA* object that encodes linear discriminant function analysis.

- *Public* methods
  - LDFA& operator= ( const LDFA& rhs ): see section 12.15.2 for discussion
  - LDFA( const LDFA& rhs ): see section 12.15.2 for discussion

- virtual double `train()`: Implementation of the pure virtual method inherited from *Model* that trains a *LDFA Model*
- virtual void `reportAccuracy( ostream& outputStream )`: Implementation of the pure virtual method inherited from *Model* that reports the accuracy of the trained *LDFA* model to an ostream *outputStream*
- *Private* members
  - Matrix< double > *C*: the common covariance Matrix
  - Matrix< double > *S*: inverse of the common covariance Matrix
- *Private* Utility functions
  - void `copy( const LDFA& rhs )`: Copy utility function, see section 12.15.2 for discussion.

## 12.27 QDFA

The **QDFA** class is a *DFA* object that encodes quadratic discriminant function analysis.

- *Public* methods
  - QDFA& `operator= ( const QDFA& rhs )`: see section 12.15.2 for discussion
  - QDFA( const QDFA& rhs ) : see section 12.15.2 for discussion
  - virtual double `train()`: Implementation of the pure virtual method inherited from *Model* that trains a *QDFA Model*
  - virtual void `reportAccuracy( ostream& outputStream )`: Implementation of the pure virtual method inherited from *Model* that reports the accuracy of the trained *QDFA* model to an ostream *outputStream*
- *Private* members
  - Matrix< double > *C0*, *C1*: covariance Matrices for 1-output datasets
  - vector< Matrix< double > > *C*: a vector of covariance Matrices for multiple output datasets
  - Matrix< double > *S0*, *S1*: covariance Matrix inverses for 1-output datasets
  - vector< Matrix< double > > *S*: a vector of covariance Matrices for multiple output datasets
  - double *Det0*, *Det1*: covariance Matrix determinants for 1-output datasets

- `vector< double > Det`: covariance Matrix determinants for multiple output datasets
- *Private* Utility functions
  - `void copy( const QDFA& rhs )`: Copy utility function, see section [12.15.2](#) for discussion.

## Chapter 13

# Stepwise regression

Statistical analysis of neural computational models is based on the observation that given 2 fully trained neural networks with final errors  $E(\mathbf{y}_{reduced})$  and  $E(\mathbf{y}_{full})$  that are sufficiently close may be discriminated by a chi-square probability distribution, where

$$G^2 = -2N[E(\mathbf{y}_{full}) - E(\mathbf{y}_{reduced})].$$

with  $G^2$  the chi-square probability  $p$  with degrees of freedom the difference in free parameters (connections between nodes) between the 2 neural models, and  $N$  the number of exemplars in the dataset (see section 6.3.)  $E$  is expressed as the cross-entropy error (see equation 6.3):

$$e(\mathbf{t}^k, \mathbf{a}^{(m)}) = -\mathbf{t}^k \cdot \log[\mathbf{a}_k^{(m)}] - (1 - \mathbf{t}^k) \cdot \log[1 - \mathbf{a}_k^{(m)}]. \quad (13.1)$$

In stepwise regression, the 2 networks  $\mathbf{y}_{reduced}$  and  $\mathbf{y}_{full}$  are chosen to differ by 1 input variable (which may be represented by 1 or more input nodes.)

### 13.1 Stepwise reverse regression

Stepwise reverse regression proceeds by training the full network, and recording that network's final error. Input variables are removed 1 at a time, the sub-networks so generated are trained to completion and their errors recorded, and the variable with the largest  $p$ -value (least significant) is then recorded with its  $p$ -value. Using the final error of the network which is deficient the one node with the largest  $p$ -value as a starting point, input variables are then removed 2 at a time, with each pair containing the 1 variable with the largest  $p$ -value and 1 of the remaining input variables. The pair of variables with the largest  $p$ -value (least significant) is then recorded with its  $p$ -value, and the process repeats until a chosen threshold  $p$ -value is reached.

Figure 13.1 demonstrates stepwise reverse regression.

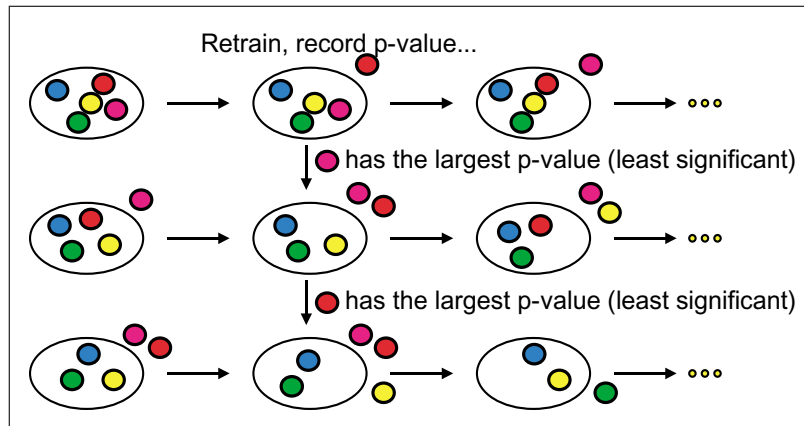


Figure 13.1: Stepwise reverse regression

## 13.2 Stepwise forward regression

Stepwise forward regression proceeds by training a baseline network with no input nodes, and recording that network's final error. Input variables are added 1 at a time, the subnetworks so generated are trained to completion and their errors recorded, and the variable with the smallest  $p$ -value (most significant) is then recorded with its  $p$ -value. Using the final error of the network which is the one node with the smallest  $p$ -value as a starting point, input variables are then added 2 at a time, with each pair containing the 1 variable with the smallest  $p$ -value and 1 of the remaining input variables. The pair of variables with the smallest  $p$ -value (most significant) is then recorded with its  $p$ -value, and the process repeats until a chosen threshold  $p$ -value is reached.

Figure 13.2 demonstrates stepwise forward regression.

## 13.3 The RegressNet class

**RegressNet** is a compound class based on the *Network* object that implements stepwise regression analysis on neural computational models.

- *Public methods*
  - `RegressNet& operator= ( const RegressNet& rhs )`: see section 12.15.2 for discussion
  - `RegressNet( const RegressNet& rhs )`: see section 12.15.2 for discussion
  - `void setInputStructure( const vector< vector< unsigned > >& v )`: Sets a vector of vector of unsigned  $v$  as the structure of input nodes

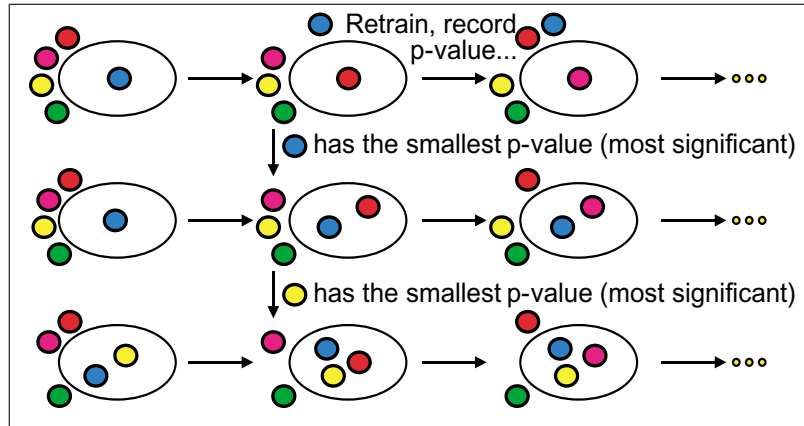


Figure 13.2: Stepwise forward regression

representing the variables for the stepwise regression. For example, in a 3 variable dataset, if variable 0 was represented by input nodes 0 and 1, variable 1 by input node 2, and variable 2 by input nodes 3, 4 and 5, vector[0] in  $v$  would contain (0,1), vector[1] in  $v$  would contain (2), and vector[3] in  $v$  would contain (3,4,5). All input nodes *must* be represented in  $v$ , and the groups must form an *exact partition* of them: *setNetwork* must have been called first, the structure and every group within it must be non-empty, every node index must belong to the network, and every input node must appear exactly once. Duplicates, overlaps, omissions — including an omission whose largest index still looks correct — and out-of-range nodes are each refused with their own message. The caller's order is preserved, between groups and within one: variable numbering in every report, progress announcement and audit-trail row is the caller's own, so normalizing here would renumber the answer. See section 10.8 for an utility method that parses a string description of input variables, and returns a vector of vector of unsigned.

- void setNetwork( Network\* *networkObj*, const double  $E$  ): sets the *Network* object *networkObj* in the *RegressNet* object, and the final error  $E$  of the trained *networkObj*
- void reverse\_regress(): performs stepwise reverse regression
- void forward\_regress(): performs stepwise forward regression

- *Private* members

- Network \**netPtr*: pointer to the incoming Network
- Network \**netCopyPtr*: pointer to a copy of the incoming Network to generate subnetworks

- bool *historyFlag*: indicates logging to history file
  - double *e\_in*: incoming Network's error
  - vector< vector< unsigned > > *variable\_defs*: the data structure representing input variables
  - typedef multimap< double, unsigned, less< double > > *ptable*: type definition of a table used to hold stepwise regression results, p-value first, followed by variable number
  - ostream *out*: to simplify writing to screen and history file
  - ostream *errorOut*: to build an exception message
  - string *errorString*: the exception message
- *Private Utility functions*
    - void report( ostream& o): a utility function which outputs an ostream to screen and history file, takes ostream o as its argument
    - void print\_input\_structure( const vector< vector< unsigned > > & v): a utility function which prints the input variable structure to screen and history file, takes data structure representing input variables v as its argument
    - void print\_regression\_table( const ptable& table ): a utility function which prints out a stepwise regression table, takes regression table multimap type *ptable* as argument
    - bool copy\_network(): a utility function which copies the incoming Network object *\*netPtr* into *\*netCopyPtr* so that the latter can be made feature deficient for subnetworks. **If you are creating a regressable *Network* object (and chances are that you are,) you must add code to this method.**
    - string network\_name() const: a utility function which returns the name of the type of *Network* object. **If you are creating a regressable *Network* object (and chances are that you are,) you must add code to this method.**
    - inline double Wilks( const double N, const double  $E_{full}$ , const double  $E_{sub}$  ) const: a utility function which calculates the chi-square parameter for 2 networks with N exemplars and which differ by their number of input nodes, the cross-entropy error of the fully trained network with all input nodes being  $E_{full}$ , and the fully trained feature deficient subnetwork's error being  $E_{sub}$
    - void copy( const RegressNet& rhs ): copy utility function, see section 12.15.2 for discussion.

- *Protected* contract and shared scaffold

The two selection procedures of sections 13.1 and 13.2 share their book-keeping and nothing else. Each keeps its own outer and inner loops, its own nesting of the two models, its own Wilks argument order, its own degrees-of-freedom expression, its own choice of the largest or smallest  $p$ -value, its own stopping test, and its own way of arriving at the final variable set. Those are the published mathematics of the two procedures, and a shared descriptor carrying a comparator and an argument order would conceal exactly the difference a reader of this class needs to see.

- static bool candidateFitEligible( Iterative::StopReason  $r$ , double  $E$  ): may a candidate fit's error enter a Wilks comparison? Two independent conditions, and the second is not implied by the first: a stopping *rule* must have ended the run, *and* the error must be a real number. A saturated fit satisfies the first and fails the second, its gradients having underflowed to zero while its cross-entropy overflowed
- void requireComparablePass( unsigned  $step$  ): refuses a pass in which no candidate's  $p$ -value could be calculated at all, rather than naming a variable that was never compared. This is *not* the case of a pass whose  $p$ -values are all zero: zero is a  $p$ -value, meaning maximally significant, and the first candidate to produce one wins its pass. An exact partition of the input nodes rules out *gammq*'s invalid-argument refusals, but not its refusal when *gcf* or *gser* exhausts *ITMAX*, which happens on entirely valid degrees of freedom and chi-square, so this is a live numerical defense rather than an unreachable one
- double beginAnalysis( const string& *banner*, const string& *question* ): the cross-entropy refusal, the structure check, the first clone, the banner and the input-structure print; returns the  $p$ -value threshold
- CandidateFit fitCandidate( ..., double  $E_{prior}$ , unsigned  $df$  ): trains one candidate subnetwork and records it, owning both progress announcements, the fit count, the record-*before*-judge ordering and the eligibility refusal. The caller has already cloned the network, removed the nodes and computed  $df$ , all three being direction-specific
- double reportComparison( const CandidateFit&  $f$ , double  $E_{prior}$ , double  $G^2$ , unsigned  $df$ , unsigned  $v$ , ptable&  $t$  ): reports one comparison and completes its audit-trail row. It is handed the statistic; it does not call Wilks, does not know which model was nested in which, and does not decide who wins. Returns the  $p$ -value, or not-a-number when the chi-square routine refused
- void endAnalysis( const string& *direction*, const string& *verb* ): the completion flag and the closing summary, reached only when a selection loop has decided

## 13.4 Exceptions

The *RegressNet* class throws exceptions of type *RegressNetErr*& *e*, where *e.what()* may contain:

- RegressNet error: maximum node value not *correct*, where *correct* is 1 less than the number of input nodes. This indicates that the vector of vector of unsigned representing the variable structure is incorrectly specified.
- RegressNet error: couldn't copy Network, unknown type. This indicates that the *Network* object has not been properly coded in *RegressNet::copy\_network()*.
- RegressNet error: stepwise regression not fully specified. This indicates that *Network* object and its final error have not been set in the *RegressNet* object.

## 13.5 Example

An example driver may thus appear as:

```
RegressNet regressionObj; // object to regress a Network

// Say you've already trained a Model pointed to by modelPtr, and
// it was a Network Model, and the last error was lastError
Network* netPtr = dynamic_cast< Network* >( modelPtr );
regressionObj.setNetwork( netPtr, lastError );

vector< vector< unsigned > > variable_defs; // variable definitions
string variable_string; // for entry of variable definitions

cout << "Please enter a string representing the variables: ";
cin >> variable_string;
try {
    variable_defs = util::variable_parse( variable_string );
}
catch ( util::utilErr& e ) {
    cout << e.what() << endl;
}

if ( !variable_defs.empty() ) // parser returned with a good result
{
    try { // try to set the input structure in the stepwise regression
        object
        regressionObj.setInputStructure( variable_defs );
    }
}
```

```

        catch ( RegressNet::RegressNetErr& e ) {
            cout << e.what() << endl;
        }
    }

    try {
        regressionObj.reverse_regress();
    }
    catch ( RegressNet::RegressNetErr& e ) {
        cout << e.what() << endl;
    }
}

```

## 13.6 Extending stepwise support today

RegressNet no longer contains a type-by-type copy switch, and no compiler-specific RTTI option is needed. Network cloning is centralized in `netclone.*`; construction from saved and requested model types is centralized in `modelfactory.*`. A new network type must participate in those shared interfaces and must preserve all architecture and training state when cloned.

Stepwise analysis works on clones and must never mutate the current trained model. Candidate fits use the same authoritative training loop as ordinary training, but run quietly and do not inspect the test set. Every candidate must converge before its likelihood may enter a Wilks comparison.

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